MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3110 FOUNDATION DESIGN OF CONVEYOR 3110-CNV-001

DESIGN CALCULATION

AS BUILT

MD-216-3110-EG-CV-CAL-1001

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyor Gallery 3110-CNV-001.
This conveyor is between Primary Crusher House and TP-1. Belt width of the conveyor is 1400 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3130-CNV-001. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The foundation of Ground Portal GR-1 to GR-4; TR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundation for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m depth below FGL (ie. RL 911.500m).
Refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3110-EG-ST-CAL-1001

5.1 LOAD COMBINATIONS

5.1.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)	

5.1.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)	

Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

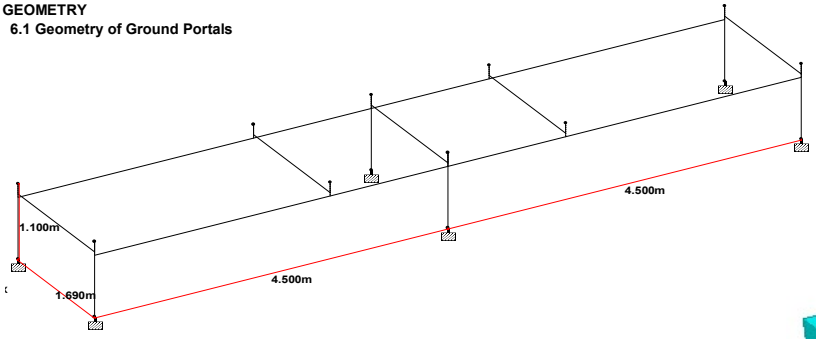
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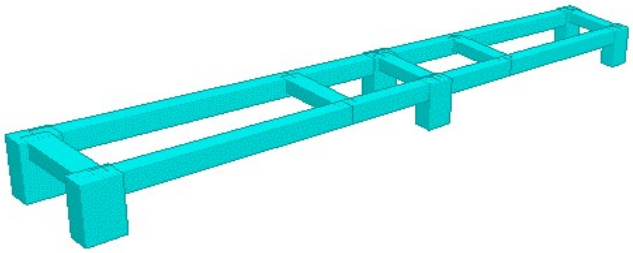
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6 GEOMETRY

6.1 Geometry of Ground Portals

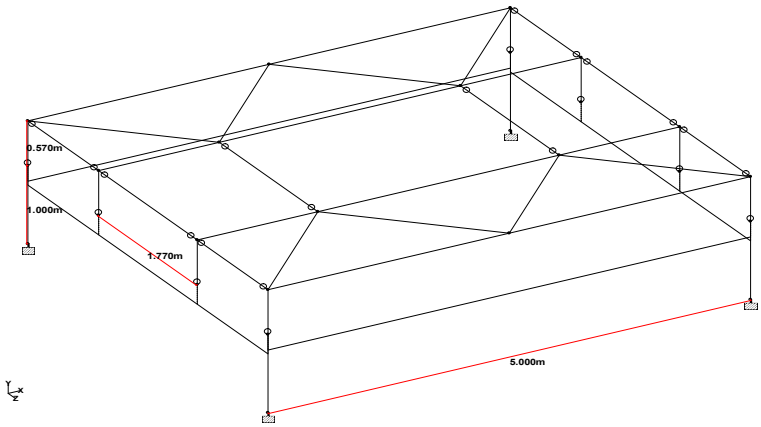


Staad model of On ground zone

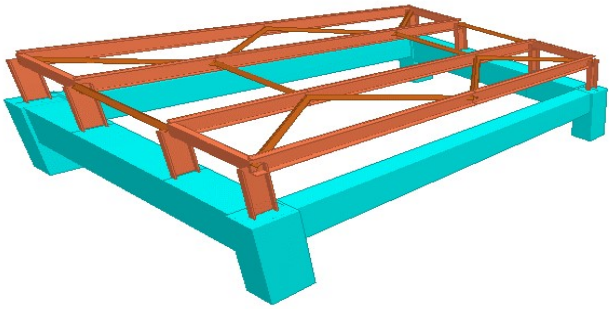


Isometric view of On ground zone (From Staad)

6.2 Geometry of Metal picking zone

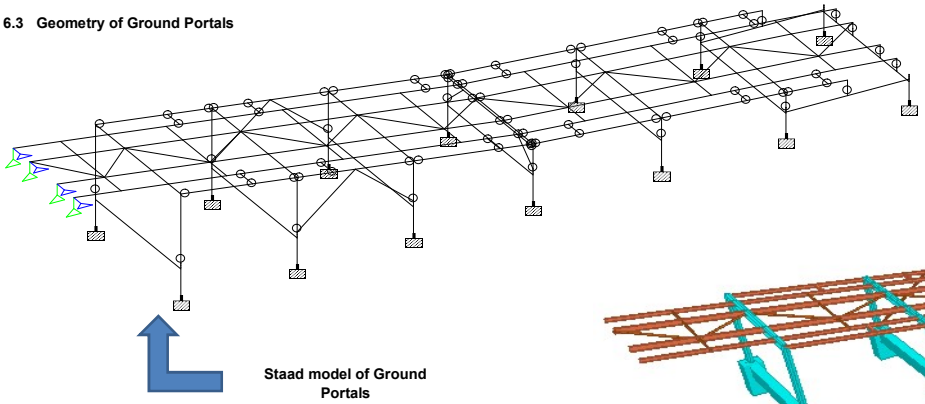


Staad Model of Metal picking zone



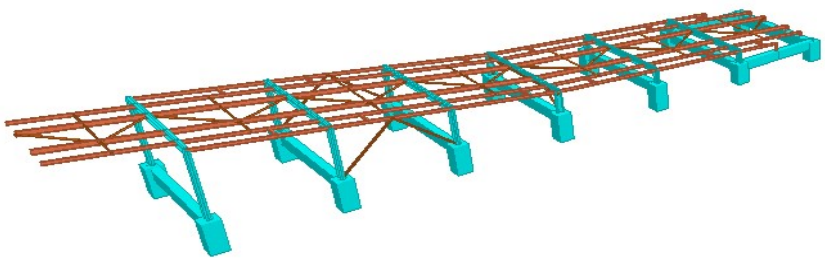
Isometric view of Metal picking zone

6.3 Geometry of Ground Portals



Staad model of Ground Portals

Isometric view of Ground Portals (From Staad)



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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 to 5 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1	400	400	8 No.- D16	2 Nos D8 @ 200c/c	On ground zone	8
P1A	400	400	8 No.- D16	2 Nos D8 @ 200c/c	On ground zone	2
P2	400	400	8 No.- D16	2 Nos D8 @ 200c/c	Metal picking zone	4
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1	2
P4	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP2	2
P5	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP3	2
P6	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP4 & GP7	4
P7	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP5 & GP6	4

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	250	300	3 Nos D12	3 Nos D12	2L D8 STRPS @ 200c/c	7
TB1A	300	400	3 Nos D12	3 Nos D12	2L D8 STRPS @ 200c/c	6
TB2	350	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2
TB2A	300	400	3 Nos D12	3 Nos D12	2L D8 STRPS @ 200c/c	2
TB3	350	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
TB3A	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2
TB4	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	6

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
On ground zone	**F1**	**1000**	**1000**	**300**	**96**	**D10 @ 200c/c B/W**	**D12 @ 200c/c B/W**	**0.19**	**10**
Metal picking zone	**F2**	**1000**	**1000**	**300**	**138**	**D10 @ 200c/c B/W**	**D12 @ 200c/c B/W**	**0.27**	**4**
GP1	**F3**	**1500**	**2000**	**300**	**72**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.14**	**2**
GP2	**F4**	**1500**	**2000**	**300**	**91**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.18**	**2**
GP3	**F5**	**1500**	**2000**	**300**	**93**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.18**	**10**
GP4	F5	1500	2000	300	47	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.09**	10
GP5	F5	1500	2000	300	76	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.15**	10
GP6	F5	1500	2000	300	60	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.12**	10
GP7	F5	1500	2000	300	51	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.10**	10

The design of the most critical foundation of the group is attached in the following pages.

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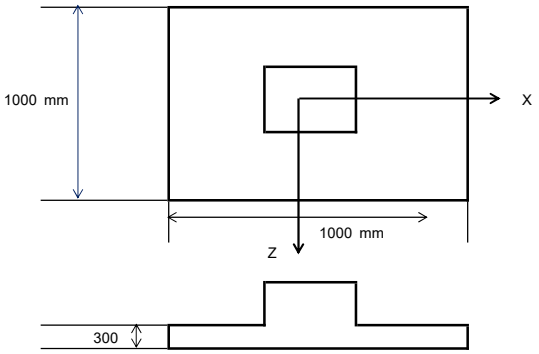
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7.3.2 Design of foundation mkd. F1 for On ground portion

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	1.00	m
Width of footing	Lx	=	1.00	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.40	m
Width of pedestal	Bz	=	0.40	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	0.17	m^3
Section Modulus of Pedestal	Zzz	=	0.17	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	518	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	10	mm
Bottom Reinforcement		=	12	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node nos	=	1, 2, 3,4, 5 & 6		*inputs
Weight of soil	=	10.58	KN	
Weight of foundation	=	7.20	KN	
Uplift	=	0.00	KN	
Surcharge	=	0	KN	
Pressure on top for tension	=	17.8	KN/m^2	



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
302 1.0 DL + 1.0 (L	-5.6	29.603	1.5	0.86	-0.1	-1.8	47.4	63.46	31.31	1	0.12
302 1.0 DL + 1.0 (L	9.2	30.091	1.5	0.86	0.098	-1.1	47.9	59.56	36.19	1	0.11
303 1.0 DL+ 0.75 W	2.3	45.56	5.2	3.94	0	-1.5	63.3	96.00	30.69	1	0.19
305 0.6 DL + 1.0 WL	-1.5	8.061	3.3	4.42	-0.311	-1.0	25.8	58.61	-6.92	1	0.11
306 0.6 DL + 1.0 WL	1.4	23.99	-3.9	-3.33	0	-0.9	41.8	67.14	16.41	1	0.13
305 0.6 DL + 1.0 WL	1.4	24.645	5.4	4.48	0.0	-0.9	42.4	74.68	10.18	1	0.14
306 0.6 DL + 1.0 WL	-2.7	14.315	-2.8	-3.58	0.21	-1.0	32.1	59.82	4.38	1	0.12
305 0.6 DL + 1.0 WL	-2.7	14.7	3.9	4.52	-0.319	-1.0	32.5	65.78	-0.81	1	0.13
305 0.6 DL + 1.0 WL	-2.7	14.7	3.9	4.52	-0.319	-1.0	32.5	65.78	-0.81	1	0.13
305 0.6 DL + 1.0 WL	4.9	14.992	3.9	4.52	0.319	-0.7	32.8	64.17	1.38	1	0.12
302 1.0 DL + 1.0 (L	-5.5	28.963	0.4	0.70	-0.1	-1.8	46.7	61.96	31.54	1	0.12
306 0.6 DL + 1.0 WL	4.8	14.608	-2.8	-3.58	-0.21	-0.7	32.4	58.12	6.66	1	0.11
0	0.0	0	0.0	0.00	0	0.0	17.8	17.78	17.78	1	0.03
0	0.0	0	0.0	0.00	0.0	0.0	17.8	17.78	17.78	1	0.03

Max P_{max}	=	96.00	KN/m^2
Max P_{min}	=	-6.92	KN/m^2
Resultant F.O.S	=	0.19	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
402 1.4 DL + 1.7 (L	-8.4	44.5	2.1	1.20	-0.1	-2.58	67.23	21.85
402 1.4 DL + 1.7 (L	13.5	45.2	2.1	1.20	0.137	-1.49	61.39	29.05
402 1.4 DL + 1.7 (L	3.3	66.2	2.8	1.49	0	-2.10	87.75	44.65
405 0.9 DL + 1.6 WL	-2.1	11.8	5.3	7.03	-0.493	-1.56	63.32	-39.76
406 0.9 DL + 1.6 WL	2.1	36.6	-6.2	-5.38	0	-1.35	76.92	-3.80
403 1.2 DL+ 1.6 WL	2.8	53.5	9.1	7.42	0	-1.80	108.79	-1.80
406 0.9 DL + 1.6 WL	-4.0	21.8	-4.5	-5.77	0.341	-1.56	65.77	-22.20
403 1.2 DL+ 1.6 WL	-6.2	33.3	6.7	7.44	-0.534	-2.11	90.59	-23.98
403 1.2 DL+ 1.6 WL	-6.2	33.3	6.7	7.44	-0.534	-2.11	90.59	-23.98
403 1.2 DL+ 1.6 WL	10.6	33.9	6.7	7.44	0.534	-1.38	86.77	-18.99
402 1.4 DL + 1.7 (L	-8.3	43.6	0.6	0.98	-0.118	-2.60	65.12	22.17
406 0.9 DL + 1.6 WL	7.3	22.2	-4.5	-5.77	-0.341	-1.06	63.21	-18.76
0	0.0	0.0	0.0	0.00	0	0.00	0.00	0.00
0	0.0	0.0	0.0	0.00	0	0.00	0.00	0.00

Max Net factored B.P = 108.79 KN/m^2

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENT DATED 20-05-2020 & AS MKD. (A03)	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T	0	1	2	3	4
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Approved	PRI	PRI	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 4.90 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 59.2 mm^2

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
greater of
or A_{stmin} = 0.0014bd

A_{stmin} = 394.2 mm^2
A_{req} = 59.2 mm^2
Required Spacing = **286.9** mm

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 4.90 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 62.6 mm^2

Effective Depth Along X = 207 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
greater of
or A_{stmin} = 0.0014bd

A_{stmin} = 372.6 mm^2
A_{req} = 62.6 mm^2
Required Spacing = **303.5** mm

Provide Bottom Reinforcement = ***12 mm dia @ 200 c/c***
Ast prov = 565.5 mm^2

Top Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 0.80 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 330.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 9.6 mm^2

Effective Depth Along X = 220 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
greater of
or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 198.0 mm^2
A_{req} = 9.6 mm^2
Required Spacing = **396.7** mm

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 0.80 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 304.10 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 10.1 mm^2

Effective Depth Along Z = 210 mm

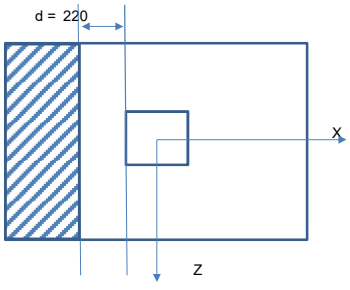
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
greater of
or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 189.0 mm^2
A_{req} = 10.1 mm^2
Required Spacing = **415.6** mm

Provide top Reinforcement = ***10 mm dia @ 200 c/c***
Ast prov = 392.7 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
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Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0026		
Effective Depth Along X	d	=	220	mm	
Shear Force at d distance	V_u	=	8.70	KN	
Allowable shear strength	φVc	=	163.39	KN	O.K
Utility ratio		=	0.05		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	0.35	KNm	
φ	=	0.75	for shear	
φVc	=	195.8	KN	

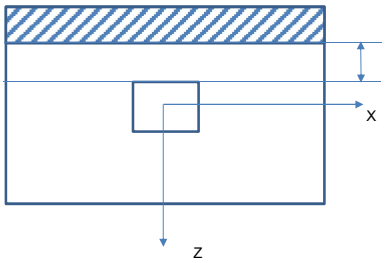
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ — *Table 22.5.5.1 SBC 304*

φVc	=	163.4	KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

	=	283.1	KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0027		
Effective Depth Along Z	d	=	210	mm	
Shear Force at d distance	V_u	=	9.79	KN	
Allowable shear strength	φVc	=	156.30	KN	O.K
Utility ratio		=	0.06		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	0.44	KNm	
φ	=	0.75	for shear	
φVc	=	182.7	KN	

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

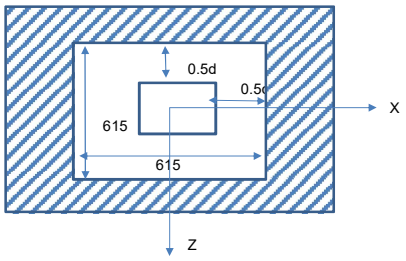
φVc	=	156.3	KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc	=	270.2	KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	1	m2
Punching area		=	0.38	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	2460.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.70 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	2.46	m	
Depth at section	d	=	215	mm	
Shear Force at section	Vu	=	67.6	KN	
Allowable shear strength	φVc	=	774.43	KN	O.K
Utility ratio			0.09		

Vc	=	$v_c b_0 d$		
	=	1032.57	KN	
φ	=	0.75	for shear	
φV_n	=	φVc		
φVc	=	774.4	KN	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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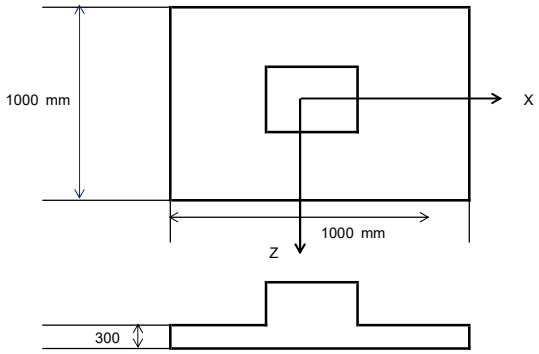
Outotec / L&T	0	1	2	3	4
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Approved	PRI	PRI	PRI	PRI	PRI

7.3.3 Design of foundation mkd. F2 for Metal pickup zone

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	1.00	m
Width of footing	Lx	=	1.00	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.40	m
Width of pedestal	Bz	=	0.40	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	0.17	m^3
Section Modulus of Pedestal	Zzz	=	0.17	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	518	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	10	mm
Bottom Reinforcement		=	12	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node nos		=	23, 24, 25 & 26	
Weight of soil		=	10.58	KN
Weight of foundation		=	7.20	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	17.8	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
303 1.0 (DL + TL2) +	-5.5	60.064	21.7	-1.46	10.7	4.1	77.8	110.97	44.73	1	0.21
303 1.0 (DL + TL2) +	8.5	61.213	21.9	-1.47	-11.268	-6.4	79.0	126.15	31.84	1	0.24
303 1.0 (DL + TL2) +	8.5	61.213	21.9	-1.47	-11.268	-6.4	79.0	126.15	31.84	1	0.24
306 0.6 DL + 1.0 WL	-2.7	15.729	-1.6	4.59	-2.577	0.1	33.5	61.81	5.22	1	0.12
305 1.0 DL+ 0.75 W	7.6	49.845	-21.1	-0.20	10.944	-5.3	67.6	100.55	34.71	1	0.19
304 1.0 DL+ 0.75 W	8.0	58.451	22.8	1.18	-10.9	-6.5	76.2	122.33	30.14	1	0.24
305 1.0 DL+ 0.75 W	-5.0	52.725	15.7	-3.76	10.269	4.1	70.5	117.87	23.15	1	0.23
304 1.0 DL+ 0.75 W	-4.6	44.119	-14.2	5.03	-10.327	2.9	61.9	109.52	14.29	1	0.21
302 1.0 (DL + TL1) +	7.6	61.213	21.8	-1.43	-14.446	-8.4	79.0	138.12	19.87	1	0.27
302 1.0 (DL + TL1) +	7.3	52.607	-20.2	2.57	14.505	-7.2	70.4	128.99	11.79	1	0.25
302 1.0 (DL + TL1) +	7.6	61.213	21.8	-1.43	-14.4	-8.4	79.0	138.12	19.87	1	0.27
302 1.0 (DL + TL1) +	-4.6	60.064	21.6	-1.43	13.871	6.1	77.8	122.94	32.76	1	0.24
0	0.0	0	0.0	0.00	0	0.0	17.8	17.78	17.78	1	0.03
0	0.0	0	0.0	0.00	0.0	0.0	17.8	17.78	17.78	1	0.03

Max P_{max}	=	138.12	KN/m^2
Max P_{min}	=	5.22	KN/m^2
Resultant F.O.S	=	0.27	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
403 1.4 (DL + TL2) +	-7.7	90.1	33.3	-2.30	17.4	6.73	144.25	35.99
403 1.4 (DL + TL2) +	11.9	91.7	33.6	-2.30	-18.185	-10.00	165.50	17.95
403 1.4 (DL + TL2) +	11.9	91.7	33.6	-2.30	-18.185	-10.00	165.50	17.95
406 0.9 DL + 1.6 WL	-4.1	23.3	-2.0	7.23	-3.864	0.19	67.80	-21.23
403 1.4 (DL + TL2) +	11.4	79.7	-31.3	3.89	18.268	-8.28	152.72	6.64
403 1.4 (DL + TL2) +	11.9	91.7	33.6	-2.30	-18.185	-10.00	165.50	17.95
405 1.2 DL+ 1.6 WL	-5.9	53.1	11.8	-6.53	9.105	3.51	113.31	-7.17
404 1.2 DL+ 1.6 WL	-5.5	42.7	-10.0	8.17	-9.173	2.04	104.00	-18.52
402 1.4 (DL + TL1) +	10.6	91.7	33.4	-2.25	-22.634	-12.84	182.26	1.19
402 1.4 (DL + TL1) +	10.2	79.7	-31.2	3.84	22.717	-11.13	169.49	-10.13
402 1.4 (DL + TL1) +	10.6	91.7	33.4	-2.25	-22.634	-12.84	182.26	1.19
402 1.4 (DL + TL1) +	-6.4	90.1	33.1	-2.25	21.829	9.57	161.01	19.23
0	0.0	0.0	0.0	0.00	0	0.00	0.00	0.00
0	0.0	0.0	0.0	0.00	0	0.00	0.00	0.00

Max Net factored B.P = 182.26 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T	0	1	2	3	4
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Approved	PRI	PRI	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.3 m

Moment about Z: Muz = 8.20 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 99.3 mm²

Length of cantilever along Z = 0.3 m

Moment about X: Mux = 8.20 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 105.1 mm²

Provide Bottom Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 394.2 mm²

A_{req} = 99.3 mm²

Required Spacing = **286.9** mm

Effective Depth Along X = 207 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 372.6 mm²

A_{req} = 105.1 mm²

Required Spacing = **303.5** mm

Ast prov = 565.5 mm²

Top Reinforcement

Length of cantilever along X = 0.3 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 330.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm²

Length of cantilever along Z = 0.3 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 304.10 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm²

Provide top Reinforcement = ***10 mm dia @ 200 c/c***

Effective Depth Along X = 220 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of $0.0014bd$ *Table 24.4.3.2 SBC 304*

A_{stmin} = 198.0 mm²

A_{req} = 0.0 mm²

Required Spacing = **396.7** mm

Effective Depth Along Z = 210 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of $0.0014bd$ *Table 24.4.3.2 SBC 304*

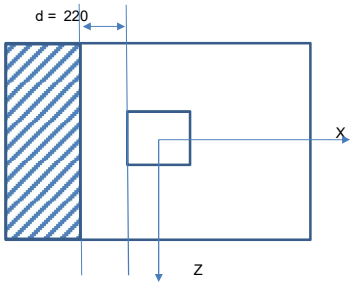
A_{stmin} = 189.0 mm²

A_{req} = 0.0 mm²

Required Spacing = **415.6** mm

Ast prov = 392.7 mm²

ONE WAY SHEAR CKECK



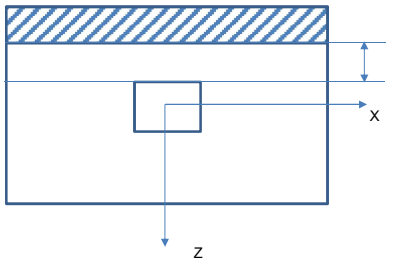
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Approved	PRI	PRI	PRI	PRI	PRI

Along X Direction Cantilever	L	=	0.08	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0026							*Table 22.5.5.1 SBY 304*
Effective Depth Along X	d	=	220	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	14.58	KN				Mu	=	0.58 KNm
Allowable shear strength	φVc	=	163.39	KN	O.K			φ	=	0.75 for shear
Utility ratio		=	0.09					φVc	=	195.8 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	163.4 KN *Table 22.5.5.1 SBY 304*
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									=	283.1 KN

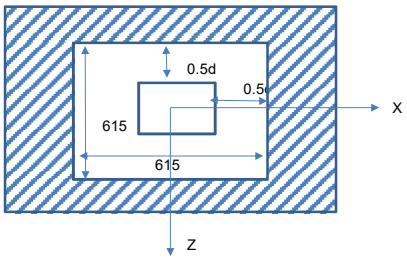
ONE WAY SHEAR CKECK



Along Z Direction Cantilever	L	=	0.09	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0027							*Table 22.5.5.1 SBY 304*
Effective Depth Along Z	d	=	210	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	16.40	KN				Mu	=	0.74 KNm
Allowable shear strength	φVc	=	156.30	KN	O.K			φ	=	0.75 for shear
Utility ratio		=	0.10					φVc	=	182.7 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	156.3 KN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	270.2 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	1	m2
Punching area		=	0.38	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	2460.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.70 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	2.46	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	215	mm			=	1032.57 KN
Shear Force at section	Vu	=	113.3	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	774.43	KN	O.K	φV_n	=	φVc
Utility ratio			0.15			φVc	=	774.4 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T	0	1	2	3	4
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

7.3.4 Design of foundation mkd. F3 for supporting trestle mkd. GP1

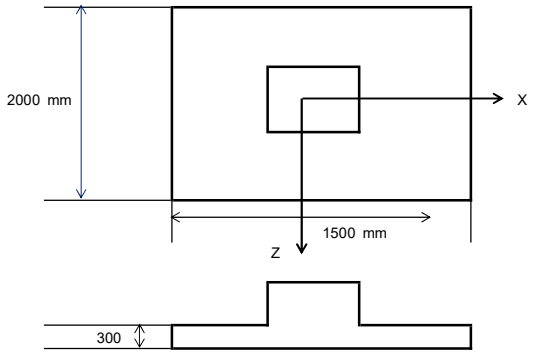
Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.00	m
Width of footing	Lx	=	1.50	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	1.00	m^3
Section Modulus of Pedestal	Zzz	=	0.75	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	518	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

Staad Node no	=	213 & 214	
Weight of soil	=	33.26	KN
Weight of foundation	=	21.60	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	18.3	KN/m^2

*inputs



CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	11.9	43.604	-13.9	9.71	10.6	-4.9	98.5	49.11	16.53	1	0.09
302 1.0 (DL + TL1) +	14.9	60.497	-24.5	16.47	17.313	-9.0	115.4	66.96	9.95	1	0.13
303 1.0 (DL + TL2) +	10.9	59.467	-23.1	16.53	7.315	-2.8	114.3	58.37	17.85	1	0.11
304 1.0 DL+ 0.75 W	16.4	56.95	-25.4	11.91	15.867	-9.4	111.8	61.67	12.88	1	0.12
305 1.0 DL+ 0.75 W	8.9	54.825	-17.3	17.69	7.907	-2.0	109.7	56.89	16.24	1	0.11
306 0.6 DL + 1.0 WL	12.1	27.579	-13.7	1.97	11.7	-7.9	82.4	39.97	14.99	1	0.08
307 0.6 DL + 1.0 WL	2.1	24.746	-2.9	9.68	1.057	2.0	79.6	38.84	14.24	1	0.07
301 1.0 DL	15.0	46.138	14.5	-9.39	-11.244	-5.5	101.0	50.43	16.90	1	0.10
302 1.0 (DL + TL1) +	20.5	63.598	24.7	-16.61	-19.774	-11.9	118.5	71.95	7.02	1	0.14
303 1.0 (DL + TL2) +	12.1	61.519	24.0	-15.79	-6.389	-1.5	116.4	56.54	21.05	1	0.11
304 1.0 DL+ 0.75 W	14.2	58.064	18.3	-16.12	-11.1	-4.5	112.9	59.71	15.58	1	0.12
305 1.0 DL+ 0.75 W	17.7	58.842	25.6	-12.87	-14.104	-8.3	113.7	61.87	13.93	1	0.12
306 0.6 DL + 1.0 WL	6.7	27.164	3.8	-7.80	-4.77	-0.7	82.0	36.13	18.56	1	0.07
307 0.6 DL + 1.0 WL	11.3	28.201	13.5	-3.46	-8.7	-5.9	83.1	39.02	16.36	1	0.08

Max P_{max}	=	71.95	KN/m^2
Max P_{min}	=	7.02	KN/m^2
Resultant F.O.S	=	0.14	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	16.6	61.0	-19.4	13.60	14.8	-6.91	43.16	-2.46
402 1.4 (DL + TL1) +	21.2	89.6	-37.3	25.10	24.75	-12.93	72.21	-12.47
403 1.4 (DL + TL2) +	15.6	88.2	-35.3	25.18	10.753	-4.21	60.18	-1.41
404 1.2 DL+ 1.6 WL	22.8	62.8	-30.3	8.88	22.073	-14.29	48.86	-7.01
405 1.2 DL+ 1.6 WL	6.8	58.2	-13.0	21.22	5.09	1.47	42.59	-3.76
406 0.9 DL + 1.6 WL	18.7	41.5	-21.2	2.57	18.037	-12.32	32.84	-5.17
407 0.9 DL + 1.6 WL	2.7	37.0	-3.8	14.91	1.055	3.44	31.82	-7.17
401 1.4 DL	21.1	64.6	20.3	-13.14	-15.741	-7.75	45.00	-1.94
402 1.4 (DL + TL1) +	29.1	94.0	37.6	-25.30	-28.235	-16.99	79.28	-16.64
403 1.4 (DL + TL2) +	17.2	91.1	36.6	-24.16	-9.497	-2.39	57.69	3.01
404 1.2 DL+ 1.6 WL	15.0	62.7	14.5	-18.14	-11.25	-3.09	43.17	-1.34
405 1.2 DL+ 1.6 WL	22.4	64.4	30.1	-11.20	-17.573	-11.34	47.79	-4.85
406 0.9 DL + 1.6 WL	9.8	40.7	5.2	-11.92	-6.958	-0.85	26.62	0.51
407 0.9 DL + 1.6 WL	17.2	42.4	20.8	-4.98	-13.281	-9.11	31.24	-3.00

Max Net factored B.P = 79.28 KN/m^2

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 8.03 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right) \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 98.0 mm²

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 19.42 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right) \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 257.7 mm²

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm²
A_{req} = 98.0 mm²
Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm²
A_{req} = 257.7 mm²
Required Spacing = **555.7** mm

Ast prov = 1005.3 mm²

Top Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right) \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm²

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right) \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm²

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm²
A_{req} = 0.0 mm²
Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

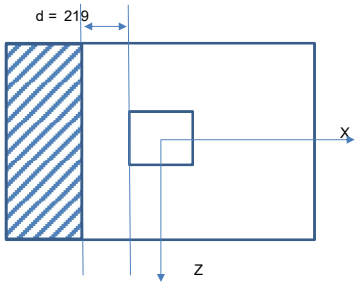
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 186.3 mm²
A_{req} = 0.0 mm²
Required Spacing = **607.1** mm

Ast prov = 565.5 mm²

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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Outotec / L&T	0	1	2	3	4
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Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0046		
Effective Depth Along X	d	=	219	mm	
Shear Force at d distance	V_u	=	18.31	KN	
Allowable shear strength	φVc	=	168.29	KN	O.K
Utility ratio		=	0.11		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 2.12 KNm
φ = 0.75 for shear
φVc = 179.8 KN

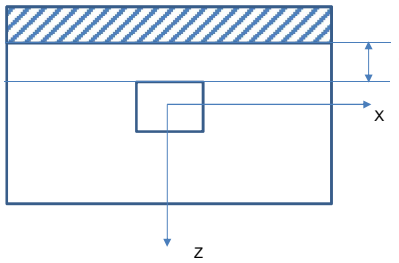
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ — *Table 22.5.5.1 SBC 304*

φVc = 168.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 281.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0049		
Effective Depth Along Z	d	=	207	mm	
Shear Force at d distance	V_u	=	39.08	KN	
Allowable shear strength	φVc	=	157.72	KN	O.K
Utility ratio		=	0.25		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 9.63 KNm
φ = 0.75 for shear
φVc = 157.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

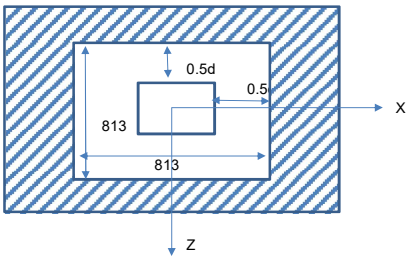
φVc = 159.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 3 m2
Punching area = 0.66 m2
λ = 1
β = 1
αs = 40
b0 = 3252.0 mm

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	
Depth at section	d	=	213	mm	
Shear Force at section	Vu	=	185.4	KN	
Allowable shear strength	φVc	=	1014.24	KN	O.K
Utility ratio			0.18		

Vc = $v_c b_0 d$
= 1352.32 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 1014.2 KN

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
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7.3.5 Design of foundation mkd. F4 for supporting trestle mkd. GP2

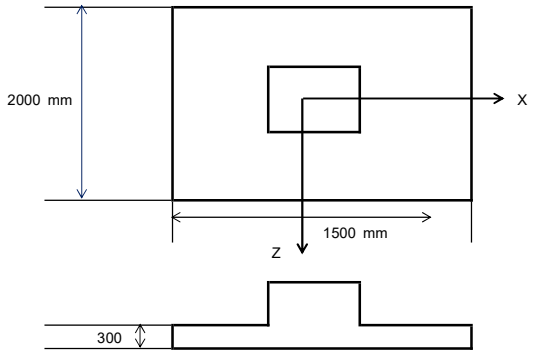
Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.00	m
Width of footing	Lx	=	1.50	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	1.00	m^3
Section Modulus of Pedestal	Zzz	=	0.75	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	518	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBY 304*

Staad Node no	=	211 & 212	
Weight of soil	=	33.26	KN
Weight of foundation	=	21.60	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	18.3	KN/m^2

*inputs



CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-0.9	49.896	-11.5	-6.77	-0.7	-5.2	104.8	48.60	21.24	1	0.09
302 1.0 (DL + TL1) +	0.7	82.326	-27.6	-32.64	-1.414	-5.8	137.2	86.11	5.35	1	0.17
303 1.0 (DL + TL2) +	-2.4	82.117	-20.5	-21.07	-0.754	-4.1	137.0	72.19	19.13	1	0.14
304 1.0 DL+ 0.75 W	1.5	69.066	-8.2	-9.40	-1.14	-6.8	123.9	59.74	22.88	1	0.12
305 1.0 DL+ 0.75 W	-3.2	79.214	-33.7	-34.27	-0.859	-3.2	134.1	83.29	6.09	1	0.16
306 0.6 DL + 1.0 WL	2.5	23.172	10.0	12.52	-0.6	-5.5	78.0	45.80	6.23	1	0.09
307 0.6 DL + 1.0 WL	-3.6	36.703	-23.9	-20.64	-0.26	-0.8	91.6	52.18	8.87	1	0.10
301 1.0 DL	-1.2	56.996	13.0	8.23	0.804	-6.8	111.9	54.60	19.97	1	0.11
302 1.0 (DL + TL1) +	1.9	88.22	26.3	31.37	1.569	-8.6	143.1	90.53	4.86	1	0.17
303 1.0 (DL + TL2) +	-4.0	90.239	24.4	24.84	0.723	-4.7	145.1	79.53	17.20	1	0.15
304 1.0 DL+ 0.75 W	-2.4	85.707	34.5	34.82	1.0	-6.0	140.6	89.70	4.02	1	0.17
305 1.0 DL+ 0.75 W	0.2	76.635	10.0	11.46	1.098	-7.4	131.5	65.15	22.51	1	0.13
306 0.6 DL + 1.0 WL	-2.5	40.246	24.1	20.51	0.433	-3.2	95.1	56.44	6.97	1	0.11
307 0.6 DL + 1.0 WL	1.0	28.15	-8.5	-10.64	0.5	-5.0	83.0	44.99	10.35	1	0.09

Max P_{max}	=	90.53	KN/m^2
Max P_{min}	=	4.02	KN/m^2
Resultant F.O.S	=	0.17	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-1.3	69.9	-16.2	-9.48	-1.0	-7.25	42.43	4.14
402 1.4 (DL + TL1) +	1.0	125.0	-42.4	-51.73	-2.08	-8.06	104.12	-20.82
403 1.4 (DL + TL2) +	-3.3	124.7	-32.5	-35.53	-1.156	-5.67	84.64	-1.53
404 1.2 DL+ 1.6 WL	3.9	65.2	7.0	8.36	-1.365	-9.85	43.23	0.24
405 1.2 DL+ 1.6 WL	-6.0	86.9	-47.2	-44.69	-0.764	-2.35	76.78	-18.87
406 0.9 DL + 1.6 WL	4.1	34.1	16.7	20.43	-0.973	-8.42	43.01	-20.29
407 0.9 DL + 1.6 WL	-5.8	55.7	-37.5	-32.62	-0.371	-0.91	52.41	-15.25
401 1.4 DL	-1.7	79.8	18.2	11.52	1.125	-9.54	50.84	2.35
402 1.4 (DL + TL1) +	2.7	133.2	40.5	49.89	2.299	-11.99	110.27	-21.49
403 1.4 (DL + TL2) +	-5.5	136.0	37.8	40.74	1.115	-6.60	94.87	-4.20
404 1.2 DL+ 1.6 WL	-4.1	94.2	47.8	44.73	1.057	-6.63	84.97	-22.18
405 1.2 DL+ 1.6 WL	1.3	74.8	-4.3	-5.10	1.214	-9.58	42.82	7.07
406 0.9 DL + 1.6 WL	-3.8	61.0	37.8	32.33	0.645	-4.66	58.86	-18.21
407 0.9 DL + 1.6 WL	1.6	41.6	-14.4	-17.51	0.802	-7.61	41.53	-13.78

Max Net factored B.P = 110.27 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 11.16 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 321.51 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 136.6 mm²

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 27.02 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 275.84 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 359.7 mm²

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm²
A_{req} = 136.6 mm²
Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm²
A_{req} = 359.7 mm²
Required Spacing = **555.7** mm

Ast prov = 1005.3 mm²

Top Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 327.46 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm²

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 292.56 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm²

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm²
A_{req} = 0.0 mm²
Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

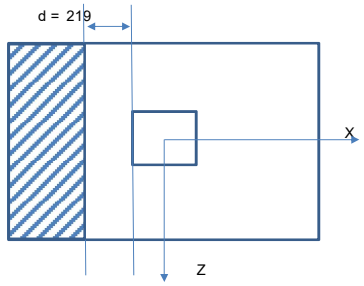
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 186.3 mm²
A_{req} = 0.0 mm²
Required Spacing = **607.1** mm

Ast prov = 565.5 mm²

ONE WAY SHEAR CKECK



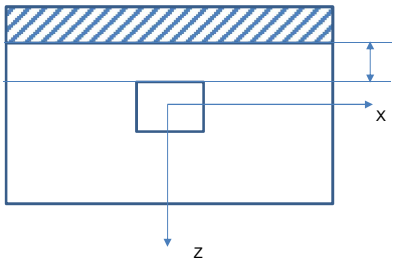
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Along X Direction	Cantilever	L	=	0.231	m		Least of	i) V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0046					Table 22.5.5.1 SBC 304
Effective Depth Along X		d	=	219	mm			λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	25.47	KN			Mu	= 2.94 KNm
Allowable shear strength		φVc	=	168.29	KN	O.K		φ	= 0.75 for shear
Utility ratio			=	0.15				φVc	= 179.8 KN
								ii) V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	= 168.3 KN Table 22.5.5.1 SBC 304
								iii) V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									= 281.8 KN

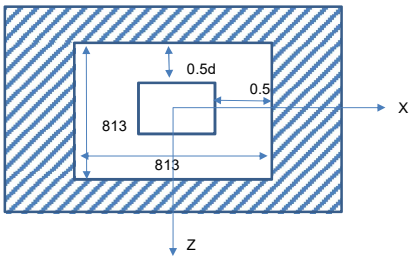
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.493	m		Least of	i) V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0049					Table 22.5.5.1 SBC 304
Effective Depth Along Z		d	=	207	mm			λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	54.36	KN			Mu	= 13.40 KNm
Allowable shear strength		φVc	=	157.72	KN	O.K		φ	= 0.75 for shear
Utility ratio			=	0.34				φVc	= 157.7 KN
								ii) V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	= 159.8 KN
								iii) V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
								φVc	= 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	3	m2
Punching area	=	0.66	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3252.0	mm

Allowable shear stress least of Table 22.6.5.2 SBC 304

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.27	N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face		u	=	3.25	m			Vc	= $v_c b_0 d$
Depth at section		d	=	213	mm				= 1352.32 KN
Shear Force at section		Vu	=	257.9	KN			φ	= 0.75 for shear
Allowable shear strength		φVc	=	1014.24	KN	O.K		φV_n	= φVc
Utility ratio				0.25				φVc	= 1014.2 KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	23-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
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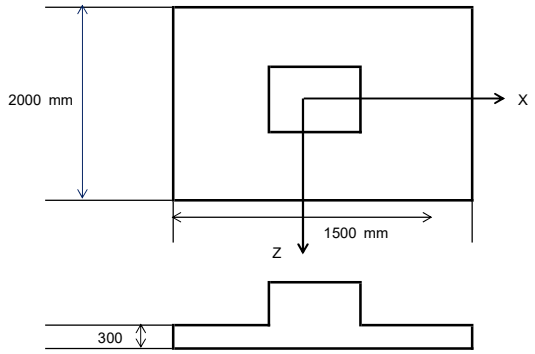
Outotec / L&T	0	1	2	3	4
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7.3.6 Design of foundation mkd. F5 for supporting trestle mkd. GP3 to GP7

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.00	m
Width of footing	Lx	=	1.50	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	1.00	m^3
Section Modulus of Pedestal	Zzz	=	0.75	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	518	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	209 & 210	
Weight of soil		=	33.26	KN
Weight of foundation		=	21.60	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	18.3	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-0.1	47.808	-3.5	-1.69	0.0	0.1	102.7	36.09	32.36	1	0.07
302 1.0 (DL + TL1) +	-0.5	82.815	-3.8	-14.21	-0.002	1.0	137.7	61.37	30.41	1	0.12
303 1.0 (DL + TL2) +	0.3	85.88	-12.4	-19.75	-0.001	-0.5	140.7	67.37	26.46	1	0.13
304 1.0 DL+ 0.75 W	-0.1	72.488	-2.2	-7.58	-0.002	0.2	127.4	50.26	34.64	1	0.10
305 1.0 DL+ 0.75 W	-0.1	77.937	-11.6	-18.74	-0.001	0.2	132.8	63.28	25.25	1	0.12
306 0.6 DL + 1.0 WL	0.0	25.052	4.2	6.43	0.0	0.1	79.9	33.15	20.13	1	0.06
307 0.6 DL + 1.0 WL	0.0	32.317	-8.4	-8.46	0	0.1	87.2	37.65	20.47	1	0.07
301 1.0 DL	-0.1	60.367	14.8	14.91	-0.002	0.3	115.2	53.69	23.13	1	0.10
302 1.0 (DL + TL1) +	-0.6	98.841	25.6	39.75	-0.001	1.1	153.7	92.41	10.06	1	0.18
303 1.0 (DL + TL2) +	0.2	95.489	14.7	22.50	-0.002	-0.4	150.4	73.09	27.15	1	0.14
304 1.0 DL+ 0.75 W	-0.2	90.708	23.4	32.32	0.0	0.4	145.6	81.34	15.71	1	0.16
305 1.0 DL+ 0.75 W	-0.1	85.222	14.2	21.82	-0.001	0.3	140.1	68.92	24.47	1	0.13
306 0.6 DL + 1.0 WL	-0.1	39.878	15.0	15.95	-0.002	0.2	94.7	47.81	15.35	1	0.09
307 0.6 DL + 1.0 WL	-0.1	32.563	2.8	1.95	0.0	0.1	87.4	31.25	27.03	1	0.06

Max P_{max} = 92.41 KN/m^2
Max P_{min} = 10.06 KN/m^2
Resultant F.O.S = 0.18

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-0.1	66.9	-4.9	-2.36	0.0	0.18	24.92	19.70
402 1.4 (DL + TL1) +	-0.7	126.9	-6.6	-24.49	-0.002	1.35	68.60	16.01
403 1.4 (DL + TL2) +	0.5	131.2	-18.7	-32.24	-0.002	-0.72	76.93	10.53
404 1.2 DL+ 1.6 WL	-0.1	69.8	3.5	2.24	-0.002	0.17	25.73	20.82
405 1.2 DL+ 1.6 WL	-0.1	81.5	-16.5	-21.58	-0.001	0.23	49.04	5.26
406 0.9 DL + 1.6 WL	0.0	37.2	6.9	10.39	-0.002	0.09	22.91	1.90
407 0.9 DL + 1.6 WL	-0.1	48.8	-13.2	-13.43	0	0.15	29.91	2.65
401 1.4 DL	-0.2	84.5	20.8	20.88	-0.002	0.39	49.57	6.77
402 1.4 (DL + TL1) +	-0.8	149.4	37.5	60.51	-0.001	1.52	112.34	-12.73
403 1.4 (DL + TL2) +	0.3	144.7	22.2	36.36	-0.002	-0.47	85.23	11.25
404 1.2 DL+ 1.6 WL	-0.2	96.7	30.3	37.20	-0.003	0.45	70.03	-5.56
405 1.2 DL+ 1.6 WL	-0.1	85.0	10.7	14.80	-0.001	0.31	43.53	13.13
406 0.9 DL + 1.6 WL	-0.2	60.2	23.1	24.62	-0.002	0.32	45.11	-4.99
407 0.9 DL + 1.6 WL	-0.1	48.5	3.5	2.22	-0.001	0.18	18.62	13.70

Max Net factored B.P = 112.34 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 11.37 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 139.1 mm^2

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 27.52 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 366.5 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm^2
A_{req} = 139.1 mm^2
Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm^2
A_{req} = 366.5 mm^2
Required Spacing = **548.6** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm^2

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

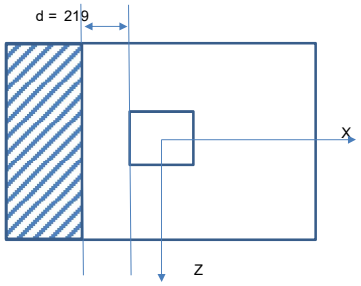
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 186.3 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **607.1** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Along X Direction Cantilever

L	=	0.231	m		
Reinforcement ratio ρ_w	=	0.0046			
Effective Depth Along X d	=	219	mm		
Shear Force at d distance V_u	=	25.95	KN		
Allowable shear strength φVc	=	168.29	KN	O.K	
Utility ratio	=	0.15			

Least of

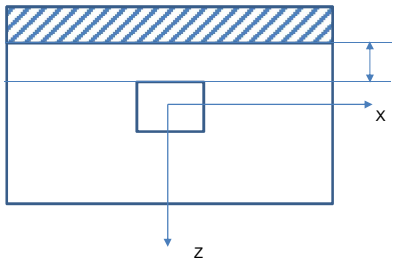
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

λ	=	1	Normal concrete
Mu	=	3.00	KNm
φ	=	0.75	for shear
φVc	=	179.8	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ = 168.3 KN — φVc = 168.3 KN — Table 22.5.5.1 SBC 304

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 281.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.493	m		
Reinforcement ratio ρ_w	=	0.0049			
Effective Depth Along Z d	=	207	mm		
Shear Force at d distance V_u	=	55.39	KN		
Allowable shear strength φVc	=	157.72	KN	O.K	
Utility ratio	=	0.35			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

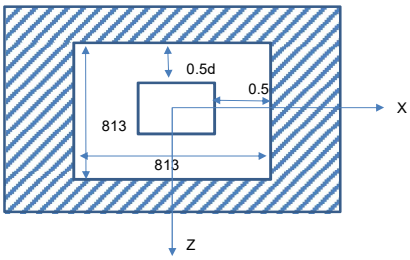
λ	=	1	Normal concrete
Mu	=	13.65	KNm
φ	=	0.75	for shear
φVc	=	157.7	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ = 159.8 KN — φVc = 159.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 266.4 KN — φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	3	m2
Punching area	=	0.66	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3252.0	mm

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.25	m		
Depth at section d	=	213	mm		
Shear Force at section Vu	=	262.8	KN		
Allowable shear strength φVc	=	1014.24	KN	O.K	
Utility ratio		0.26			

Vc	=	$v_c b_0 d$	
	=	1352.32	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	1014.2	KN

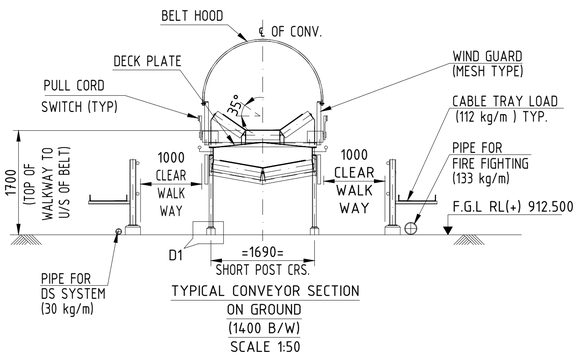
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Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

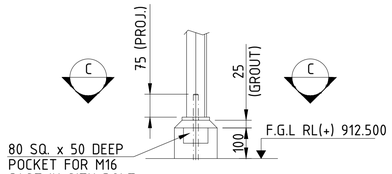
Outotec / L&T Revision No.:	0	1	2	3	4
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

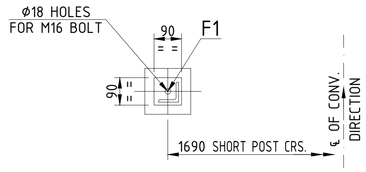
DRAWING NUMBER MD-216-3110-EG-ME-ELV-1001 REV D01



TYPICAL CONVEYOR SECTION ON GROUND (1400 B/W)
SCALE 1:50



DETAIL-D1
SCALE 1:10



VIEW C-C
SCALE 1:10

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3110-CNV-001

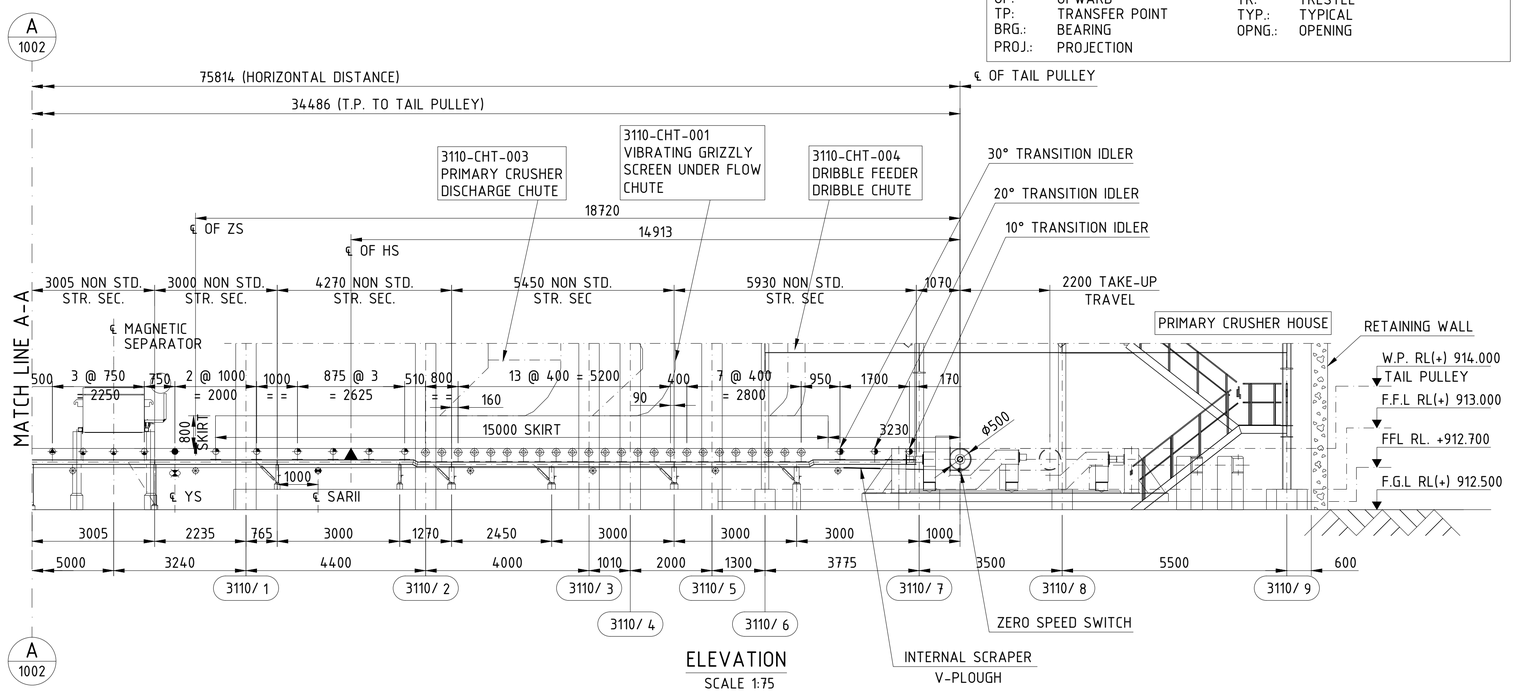
POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.65	(+)0.35	(±)0.11	(±)0.06

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QTY.
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	HGTU / 2200 mm	
	TAKE-UP LOAD	~2200 kg.(Apox.) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	N.A.	
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS) ▮	@ 50 M INTERVAL	2 PAIRS
	PULL CORD SWITCH (HS) ▲	@ 30 M INTERVAL	3 PAIRS
	ZERO SPEED SWAY SWITCH (SS) ◐	AT TAIL PULLEY	1 NO.
	BELT TRIP SWITCH (YS) ⋇	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm THK. CONTINUOUS	1 LOT.
	BELT HOOD	GI CORRUGATED SHEET	1 LOT.

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
T.P.:	TANGENT POINT	SYN:	SYNCHRONOUS
I.P.:	INTERSECTION POINT	thk.:	THICKNESS
W.P.:	WORKING POINT	RCC:	REINFORCED CEMENT CONCRETE
STD.:	STANDARD	B/W:	BELT WIDTH
STR.:	STRINGER	MAX:	MAXIMUM
SEC:	SECTION	deg.:	DEGREE
℄:	CENTER LINE	U/S.:	UNDERSIDE
DN:	DOWNWARD	B.O.M.R:	BOTTOM OF MONORAIL
UP:	UPWARD	T.O.W.W.:	TOP OF WALKWAY
TP:	TRANSFER POINT	TR:	TRESTLE
BRG.:	BEARING	TYP.:	TYPICAL
PROJ.:	PROJECTION	OPNG.:	OPENING



ELEVATION
SCALE 1:75

CONVEYOR CHARACTERISTICS

		PARTICULARS	SPECIFICATION	QTY.
CONVEYOR		EQUIPMENT TAG NO.	3110-CNV-001	1 NO.
		EQUIPMENT DESCRIPTION	PRIMARY CRUSHER SACRIFICIAL CONVEYOR	
		LOCATION	PRIMARY CRUSHER HOUSE TO TP#1	
		CAPACITY RATED / DESIGN (t/h)	702 / 772	
		BELT WIDTH (mm)	1400	
		BELT SPEED (m/sec)	1.498	
		TRIPPER FIXED / MOVABLE	N.A.	
		CONVEYOR LENGTH / LIFT (mm)	75814 / 3700	
MATERIAL		MATERIAL	GOLD ORE : FRESH ORE / OXIDE ORE	
		BULK DENSITY (t/m³)	1.6	
		MAX. LUMP SIZE (mm)	(-)340	
		MOISTURE CONTENT (%)	2	
		SURCHARGE ANGLE (deg.)	20	
BELTING		MAX. RUNNING TENSION (T1 kN)	26.87	
		SLACK SIDE TENSION (T2 kN)	10.23	
		TAIL TENSION (T0 kN)	15.18	
		BELT DESIGNATION	EP-630 / 4 (HEAVY DUTY)	
		TOP / BOTTOM COVER THICKNESS (mm)	8 / 3	
		COVER GRADE / FABRIC	DIN-X / EP	
PULLEYS		DISCHARGE WITH DRIVE (MKD. X DIA.)	D-3 x 630 mm	1 NO.
		DISCHARGE WITHOUT DRIVE (MKD. X DIA.)	N.A.	
		SNUB (MKD. X DIA.)	ND-4 x 500 mm	1 NO.
		TAIL (MKD. X DIA.)	ND-4 x 500 mm	1 NO.
		TAKE-UP (MKD. X DIA.)	N.A.	
		DEFLECTOR (MKD. X DIA.)	DR-2 x 273	1 NO.
		TRIPPER DISCHARGE (MKD. X DIA.)	N.A.	
		TRIPPER BEND (MKD. X DIA.)	N.A.	
		BEND (MKD. X DIA.)	N.A.	
		LAGGING (mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK		DISCHARGE WITH DRIVE	TBA	2 SETS
		DISCHARGE WITHOUT DRIVE	N.A.	
		SNUB	TBA	2 SETS
		TAIL	TBA	2 SETS
		TAKE-UP	N.A.	
		DEFLECTOR	TBA	2 SETS
		TRIPPER DISCHARGE	N.A.	
		TRIPPER BEND	N.A.	
IDLER		TROUGHING ⊕ (DIA. X deg.)	152.4 x 35°	39 NOS.
		TROUGHING RETRACTABLE ◔ (DIA. X deg.)	152.4 x 45°	6 NOS.
		TROUGHING TRAINER ● (DIA. X deg.)	152.4 x 35°	4 NOS.
		RETURN ⊕ (DIA. X deg.)	152.4 x 10°	20 NOS.
		SELF ALIGNING RETURN TRAINER INVERTED IDLER ⋈ (DIA. X deg.)	152.4 x 10°	3 NOS.
		IMPACT RETRACTABLE ⊕ (DIA. X deg.)	190 x 45°	24 NOS.
		TRANSITION (ADJUSTABLE) ◐ (DIA. X deg.)	152.4 x 10°/20°/30°	(3 + 2) NOS.
		RETURN SELF CLEANING ⊕ (DIA. X deg.)	190 x 0°	1 NO.
		NON MAGNETIC TROUGHING ⊕ (DIA. X deg.)	152.4 x 35°	5 NOS.
		WEIGHING IDLER ⊕ (DIA. X deg.)	152.4 x 35°	4 NOS.
		SPECIAL MACHINED IDLER ▲ (DIA. X deg.)	152.4 x 35°	6 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	55 / 1800	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	400V, 60Hz	
	REDUCER	TYPE	BEVEL HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	40	
		HOLD BACK	YES	
	BRAKE	TYPE	ELECTRO HYDRAULIC THRUSTER	
		MODEL NO.	TBA	
		DRAM DIAMETER	TBA	
	COUPLING HIGH SPEED	TYPE	FLUID COUPLING	1 NO.
		MODEL NO.	TBA	
	COUPLING LOW SPEED	TYPE	GEAR	1 NO.
		MODEL NO.	TBA	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	12/10/2023	AS BUILT	AFO	BRD	BD	BGR	
C01	16/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
		COMMENTS DT. 06.02.20 & REVISED AS MARKED (C01)					
A01	15/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3110-EG-ME-ELV-1002	CONV. 3110-CNV-001 - ELEVATION
MD-216-3110-EG-ME-LPX-1003	CONV. 3110-CNV-001 - PLAN VIEW

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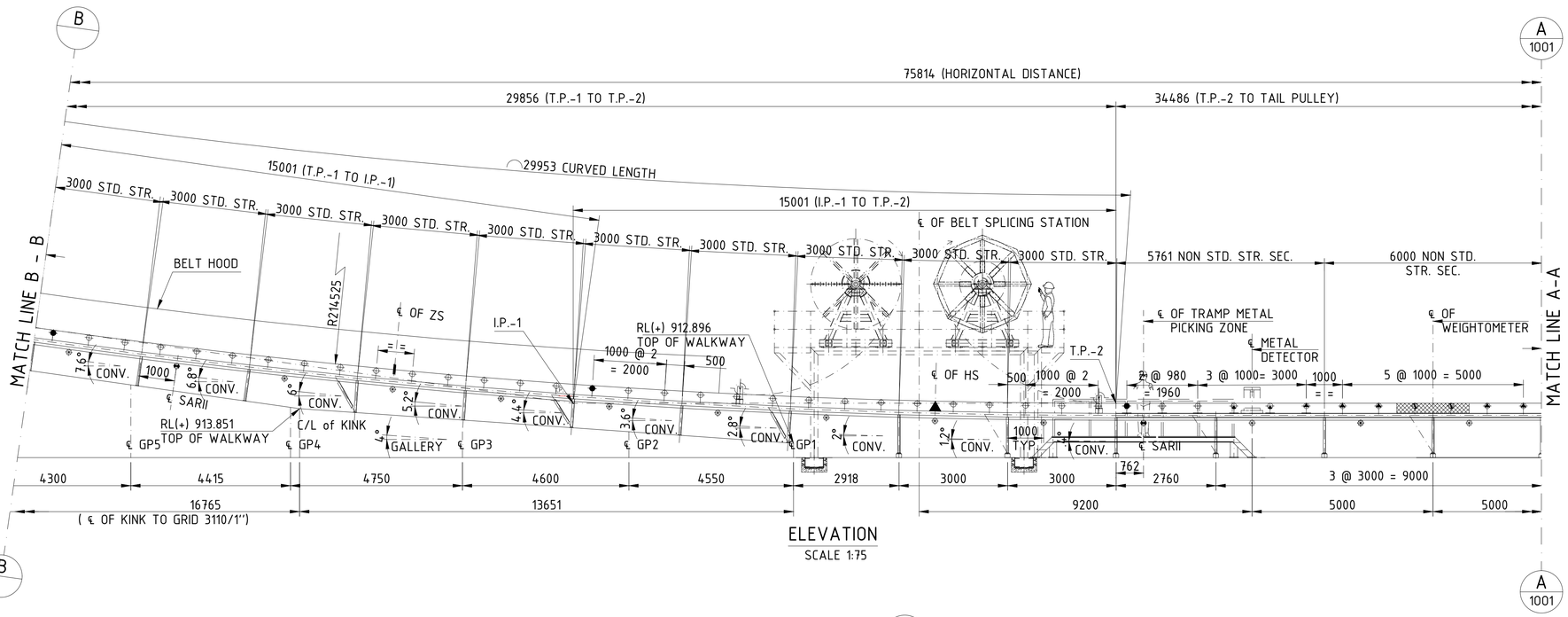
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APPROVED	BGR	15/01/2020
EMPLOYER		
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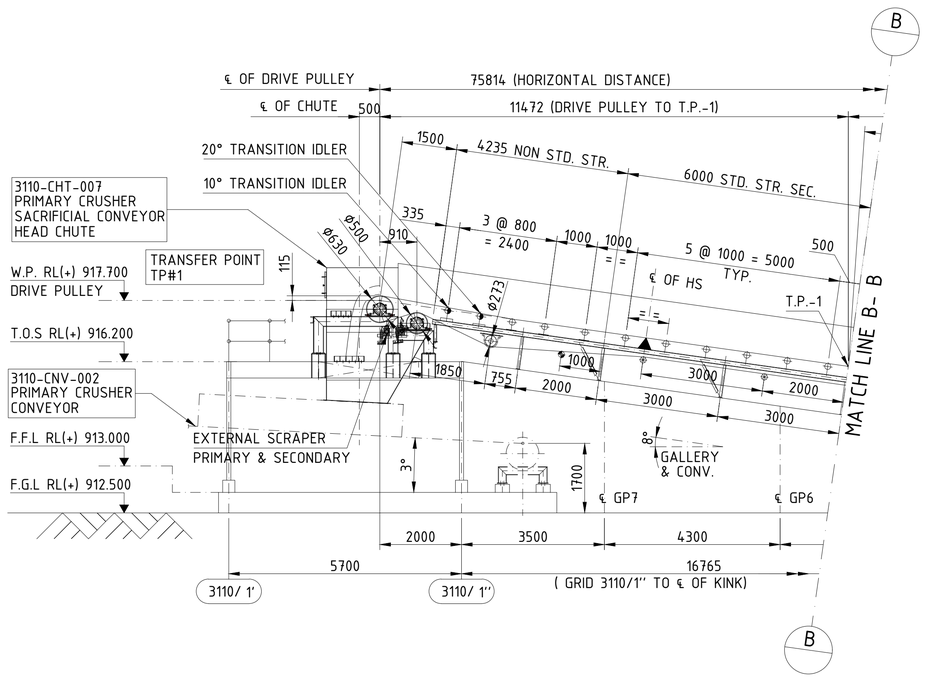
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-001 - ELEVATION

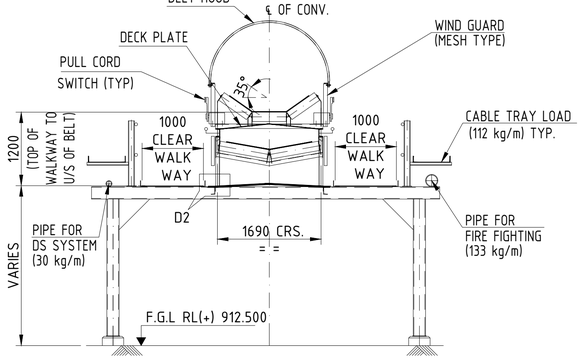
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REV: D01

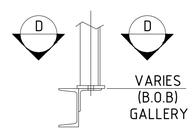
ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
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SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
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W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING
PROJ.:	PROJECTION		



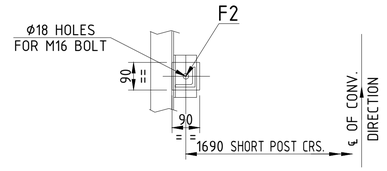




TYPICAL CONVEYOR SECTION OVER GROUND (1400 B/W)
SCALE 1:50



DETAIL-D2
SCALE 1:10



VIEW D-D
SCALE 1:10

TABLE-2 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3110-CNV-001

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F2	(+)0.65	(+)0.35	(±)0.11	(±)0.06

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	AFO	BRD	BD	BGR	
		C01	16/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1001	CONV. 3110-CNV-001 - ELEVATION			COMMENTS DT. 06.02.20 & REVISED AS MARKED /01					
MD-216-3110-EG-ME-LPX-1003	CONV. 3110-CNV-001 - PLAN VIEW	A01	15/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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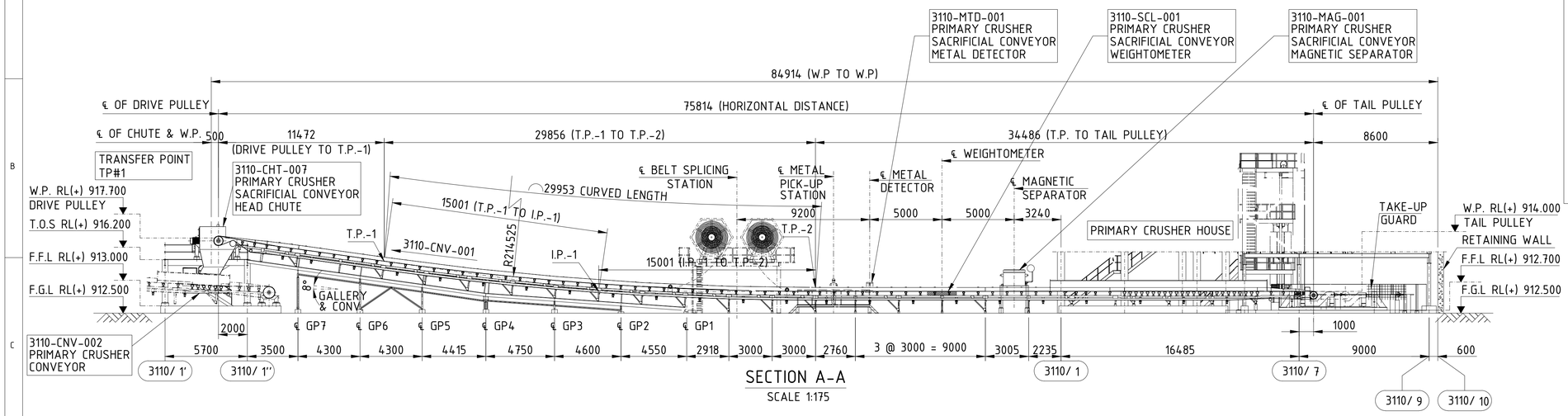
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31129	DOCUMENT ID OU100668878 / REVISION 2

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CHECKED	BRD	14/01/2020
ENGINEER	BD	14/01/2020
APPROVED	BGR	15/01/2020
EMPLOYER		
SCALE	1:10, 1:50, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-001 - ELEVATION

DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1002 REV D01

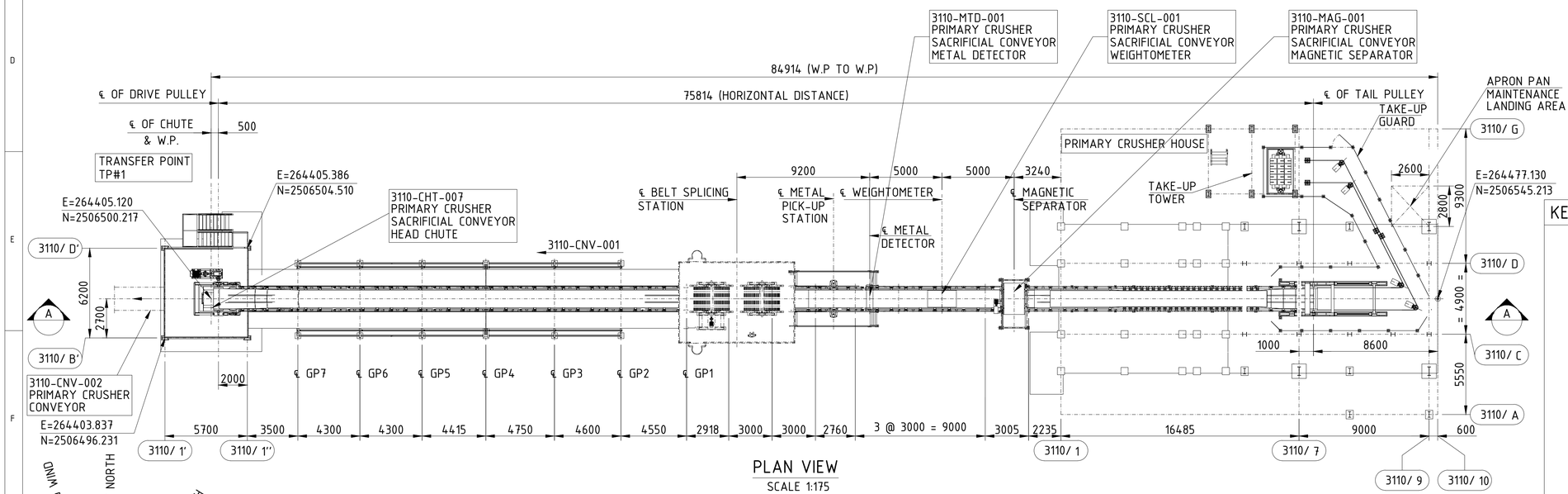
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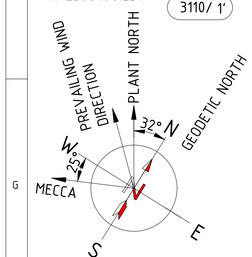
SECTION A-A
SCALE 1:175

ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING
PROJ.:	PROJECTION		

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

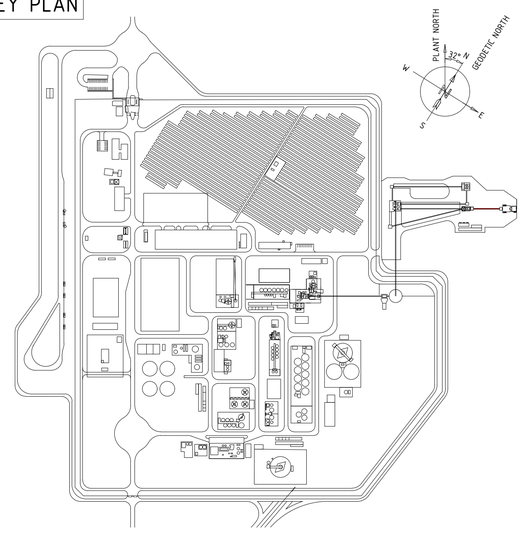


PLAN VIEW
SCALE 1:175



- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	AFO	BRD	BD	BGR	
		C01	16/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1002	CONV. 3110-CNV-001 - ELEVATION			COMMENTS DT. 06.02.20 & REVISED AS MARKED C01					
MD-216-3110-EG-ME-ELV-1001	CONV. 3110-CNV-001 - ELEVATION	A01	15/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_H_G_16111_001			2
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
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DRAWN	HKD	12/01/2020	
CHECKED	BRD	14/01/2020	
ENGINEER	BD	14/01/2020	
APPROVED	BGR	15/01/2020	
EMPLOYER			
SCALE	1:175	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-001 - PLAN VIEW

DRAWING NUMBER	REV
MD-216-3110-EG-ME-LPX-1003	D01

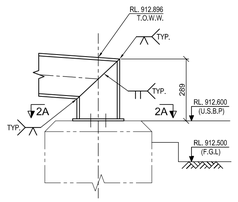
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED △A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

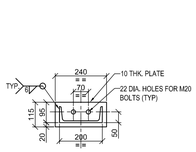
Outotec / L&T Revision No.:	0	1	2	3	4
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED △A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAIL**

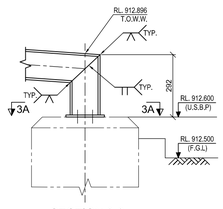
DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1001 | REV: D01



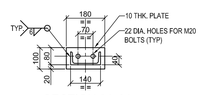
SECTION 2-2
(SCALE 1:10)



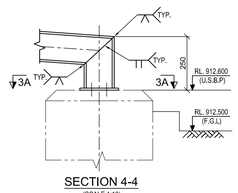
SECTION 2A-2A
(SCALE 1:10)



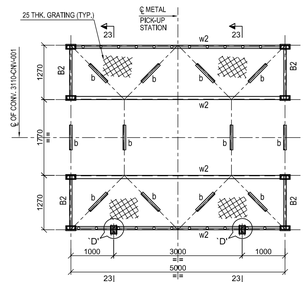
SECTION 3-3
(SCALE 1:10)



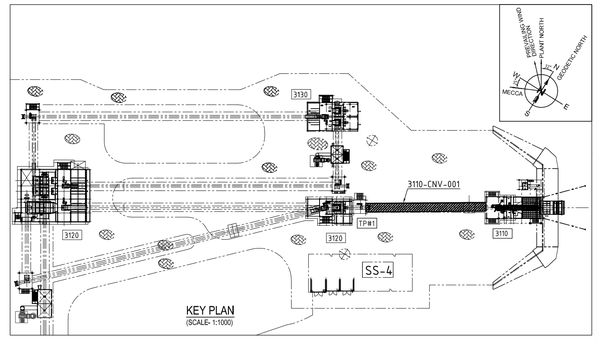
SECTION 3A-3A
(SCALE 1:10)



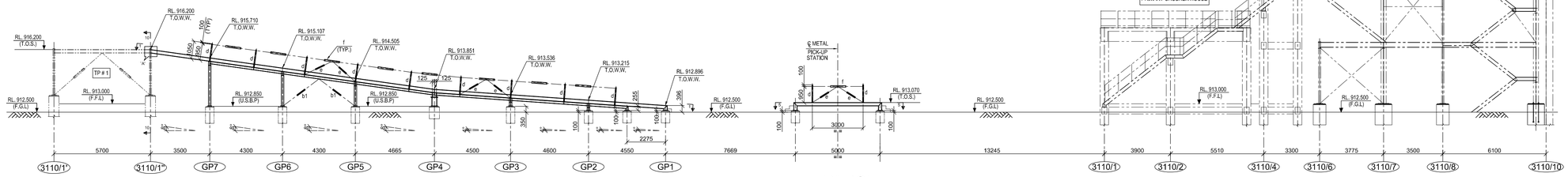
SECTION 4-4
(SCALE 1:10)



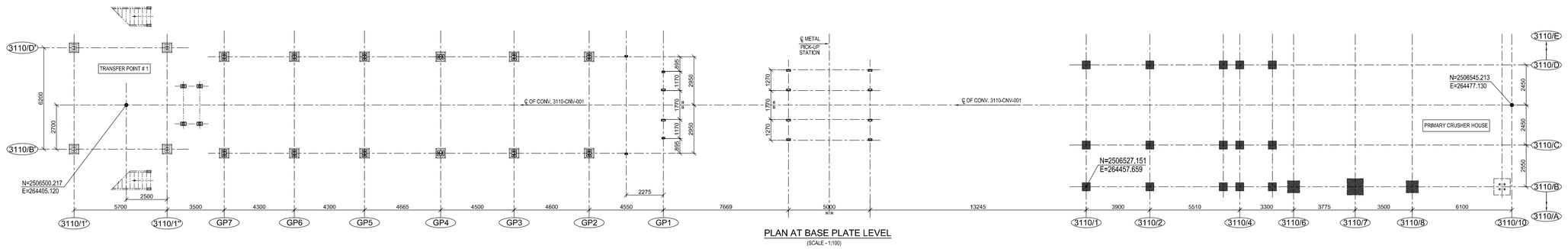
VIEW 5-5
(ALL PLAN BRACING ARE 75MM BELOW FROM T.O.S)
(SCALE - 1:50)



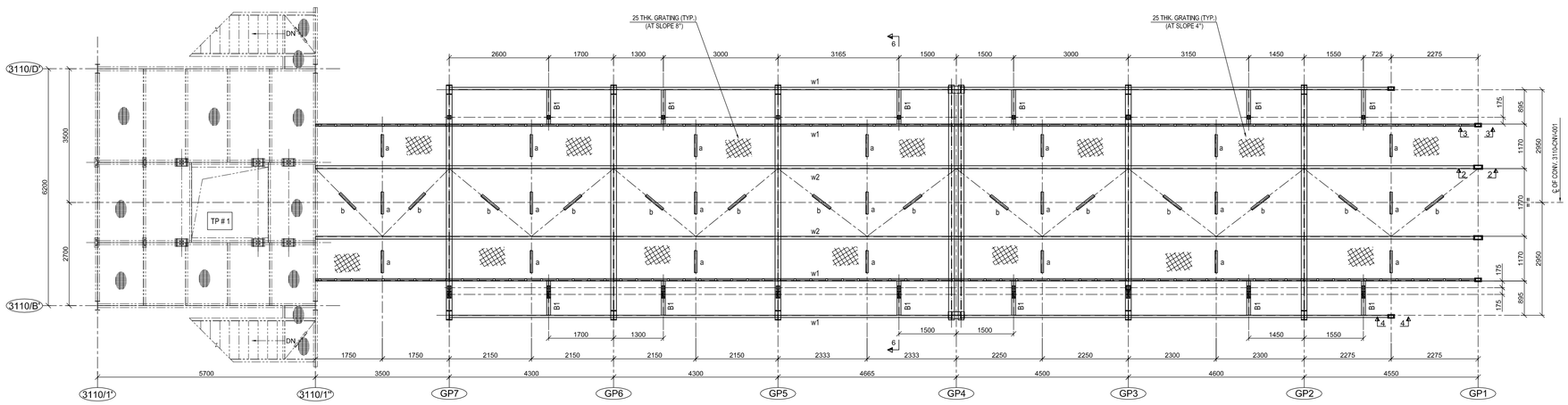
KEY PLAN
(SCALE- 1:1000)



ELEVATION
(SCALE - 1:150)



PLAN AT BASE PLATE LEVEL
(SCALE - 1:100)



VIEW 1-1
(ALL PLAN BRACING ARE 75MM BELOW FROM T.O.S)
(SCALE - 1:50)

LEGENDS :-

F.F.L - FINISHED FLOOR LEVEL
F.G.L - FINISHED GROUND LEVEL
T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
CHQD. PL. - CHEQUERED PLATE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W.W - TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT
T.O.W.W. - TOP OF WALKWAY
OPP. - OPPOSITE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3110-EG-ST-GAD-1002
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG-ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	HSK	SBY	APL	PRI	
MD-216-3110-EG-ME-ELV-1001	CONV. 3110-CNV-001 - ELEVATION	C01	30/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 29/04/2020 & REVISED AS MKD					
MD-216-3110-EG-ME-ELV-1002	CONV. 3110-CNV-001 - ELEVATION				HSK	SBY	APL	PRI	
MD-216-3110-EG-ME-LPX-1003	CONV. 3110-CNV-001 - PLAN VIEW	A01	31/03/2020	ISSUED FOR REFERENCE	HSK	SBY	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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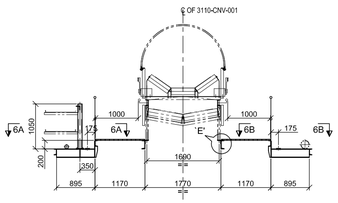
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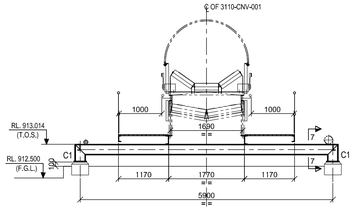
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_S_G_26013_001			2
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__3114 1		OU100670418	2
DRAWN	HSK	27/03/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	30/03/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APL	26/03/2020	STRUCTURAL GA FOR 3110-CNV-001, SH1
APPROVED	PRI	31/03/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE 1:100,1:50,1:10		A0	DRAWING NUMBER: MD - 216 - 3110 - EG - ST - GAD - 1001
			REV: D01

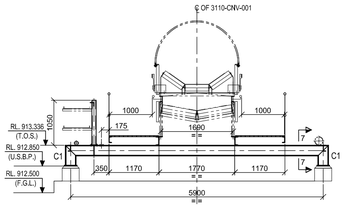
DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1002 | REV: D01



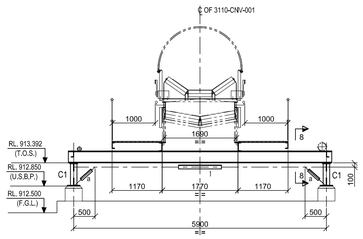
SECTION 6-6
(SCALE - 1:50)



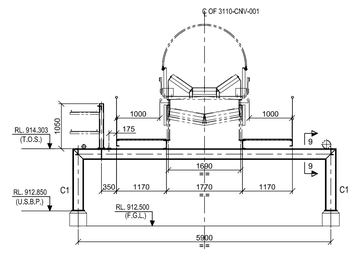
ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



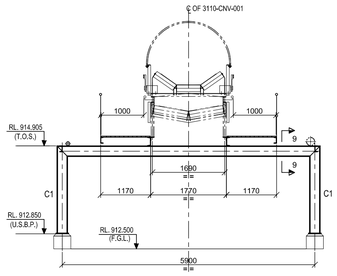
ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



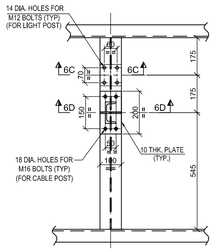
ELEVATION OF GROUND PORTAL MKD. GP4
(SCALE 1:50)



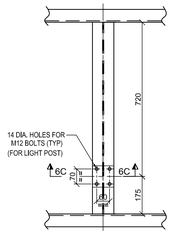
ELEVATION OF GROUND PORTAL MKD. GP5
(SCALE 1:50)



ELEVATION OF GROUND PORTAL MKD. GP6
(SCALE 1:50)



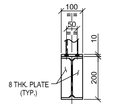
SECTION 6A-6A



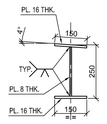
SECTION 6B-6B
(SCALE - 1:10)



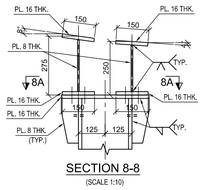
SECTION 6C-6C
(SCALE - 1:10)



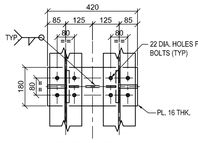
SECTION 6D-6D
(SCALE - 1:10)



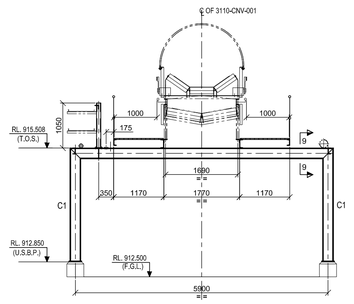
SECTION 7-7
(SCALE 1:10)



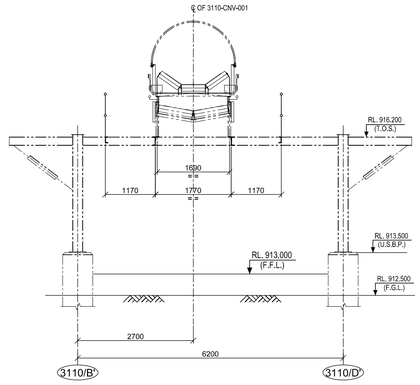
SECTION 8-8
(SCALE 1:10)



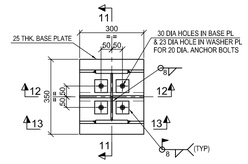
SECTION 8A-8A
(SCALE 1:10)



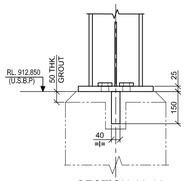
ELEVATION OF GROUND PORTAL MKD. GP7
(SCALE 1:50)



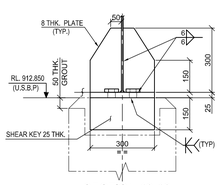
SECTION 10-10
(SCALE-1:50)



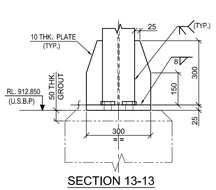
BASE PLATE DETAIL FOR GP2 & GP3
(SCALE - 1:10)



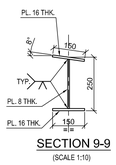
SECTION 11-11
(SCALE - 1:10)



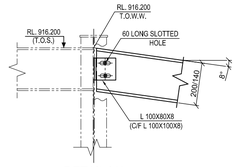
SECTION 12-12
(SCALE - 1:10)



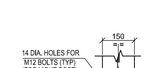
SECTION 13-13
(SCALE - 1:10)



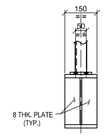
SECTION 9-9
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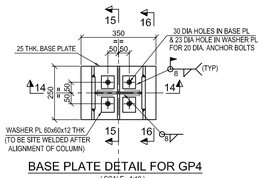
DETAIL `A'
(REF. ELEVATION OF DRG. NO. 1001)
(SCALE 1:10)



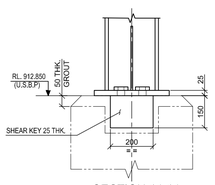
SECTION B1-B1
(SCALE 1:10)



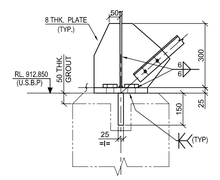
SECTION B2-B2
(SCALE 1:10)



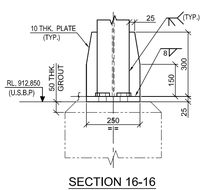
BASE PLATE DETAIL FOR GP4
(SCALE - 1:10)



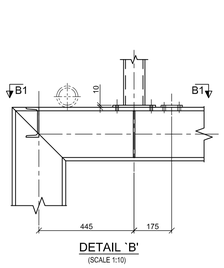
SECTION 14-14
(SCALE - 1:10)



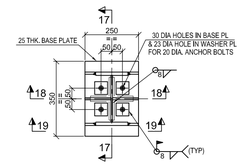
SECTION 15-15
(SCALE - 1:10)



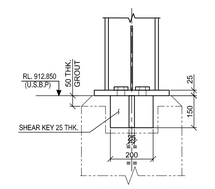
SECTION 16-16
(SCALE - 1:10)



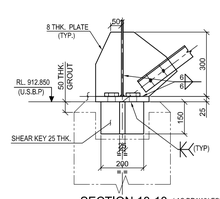
DETAIL `B'
(SCALE 1:10)



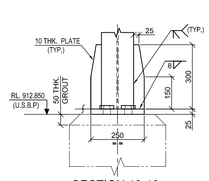
BASE PLATE DETAIL FOR GP5 & GP6
(SCALE - 1:10)



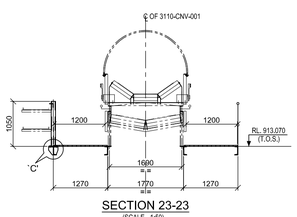
SECTION 17-17
(SCALE - 1:10)



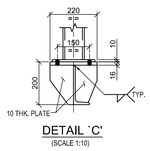
SECTION 18-18 (AS DRAWN FOR GP6)
SECTION 18-18 (OPP. HAND FOR GP5)
(SCALE - 1:10)



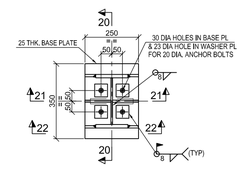
SECTION 19-19
(SCALE - 1:10)



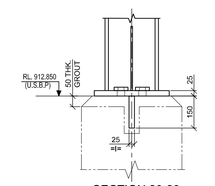
SECTION 23-23
(SCALE - 1:50)



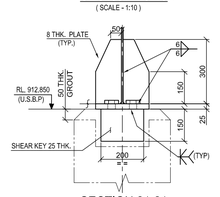
DETAIL `C'
(SCALE 1:10)



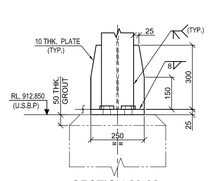
BASE PLATE DETAIL FOR GP7
(SCALE - 1:10)



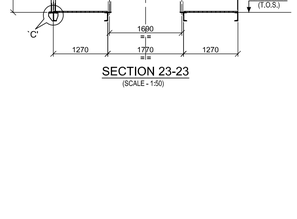
SECTION 20-20
(SCALE - 1:10)



SECTION 21-21
(SCALE - 1:10)



SECTION 22-22
(SCALE - 1:10)



SECTION 23-23
(SCALE - 1:50)

DETAIL `C'
(SCALE 1:10)

DETAIL `D'
(SCALE 1:10)

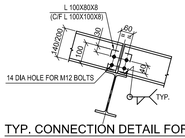
DETAIL `E'
(SCALE 1:10)

SECTION E1-E1
(SCALE 1:10)

SCHEDULE OF MEMBERS FOR GP-2 TO GP7

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	1. PL. 150X16 2. PL. 218X8	1 2 1	-	-	GUSSET PLATE 8 THK.
a	L 50X50X6	L	-	-	
b	L 65X65X6	L	-	-	
b1	2L 65X65X6	⊥	-	-	
c1	2L 75X75X8	⊥	-	-	
w1	UPN 140	[	-	-	
w2	UPN 200	[	-	-	
B1	IPE 200	I	-	-	
B2	UPN 140	[	-	-	
d	UPN 100	[	-	-	
e	L 50X50X6	L	-	-	
f	L 75X75X8	L	-	-	

GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



TYP. CONNECTION DETAIL FOR WALKWAY SUPPORTING CHANNEL FOR W1, W2
(SCALE 1:10)

LEGENDS :-

F.F.L - FINISHED FLOOR LEVEL
F.G.L - FINISHED GROUND LEVEL
T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE,
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
CHQD. PL. - CHEQUERED PLATE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W/W-TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT
T.O.W.W. - TOP OF WALKWAY
OPP. - OPPOSITE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3110-EG-ST-GAD-1001
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	HSK	SBY	APL	PRI	
MD-216-3110-EG-ME-ELV-1001	CONV. 3110-CNV-001 - ELEVATION	C01	30/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 27/04/2020 & REVISED AS MKD.	HSK	SBY	APL	PRI	
MD-216-3110-EG-ME-ELV-1002	CONV. 3110-CNV-001 - ELEVATION				HSK	SBY	APL	PRI	
MD-216-3110-EG-ME-LPX-1003	CONV. 3110-CNV-001 - PLAN VIEW	A01	31/03/2020	ISSUED FOR REFERENCE	HSK	SBY	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_S_G_26013_002		2
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31142	OU100670419	2

DRAWN	HSK	27/03/2020
CHECKED	SBY	30/03/2020
ENGINEER	APL	26/03/2020
APPROVED	PRI	31/03/2020
EMPLOYER		
SCALE	1:50,1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3110-CNV-001, SH2
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3110 - EG - ST - GAD - 1002 | REV: D01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3	4
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR ON GROUND PORTION**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 29-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART ON GROUND PORTION
JOB REF CONVEYOR 3110-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 -2.38419e-007 -1 0; 2 -2.38419e-007 -1 1.69; 3 4.5 -1 0; 4 4.5 -1 1.69;
5 9 -1 0; 6 9 -1 1.69; 7 -2.38419e-007 0.1 0; 8 -2.38419e-007 0.1 1.69;
9 3 0.1 0; 10 3 0.1 1.69; 11 4.5 0.1 0; 12 4.5 0.1 1.69; 13 6 0.1 0;
14 6 0.1 1.69; 15 9 0.1 0; 16 9 0.1 1.69;
MEMBER INCIDENCES
1 1 7; 2 2 8; 3 3 11; 4 4 12; 5 5 15; 6 6 16; 11 8 7; 12 9 10; 13 12 11;
14 13 14; 15 16 15; 21 7 9; 22 9 11; 23 11 13; 24 13 15; 25 8 10; 26 10 12;
27 12 14; 28 14 16;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
11 TO 15 PRIS YD 0.3 ZD 0.25
1 TO 6 PRIS YD 0.4 ZD 0.4
MEMBER PROPERTY EUROPEAN
21 TO 28 PRIS YD 0.4 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 FIXED
************************************************************
MEMBER OFFSET
21 TO 28 START 0 -0.2 0
```

```
21 TO 28 END 0 -0.2 0
11 TO 15 START 0 -0.15 0
11 TO 15 END 0 -0.15 0
LOAD 1 DL
SELFWEIGHT Y -1
JOINT LOAD
*CONVEYOR DEAD LOAD
7 TO 10 13 TO 16 FX 1.5 FY -6 FZ 0.8
************************************************************
LOAD 2 LL
JOINT LOAD
* Spillage Load @75kg/ Sq. m on seal plate
7 TO 10 13 TO 16 FY -1.9125
************************************************************
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
JOINT LOAD
7 TO 10 13 TO 16 FY -6.5
************************************************************
LOAD 4 WIND LOAD IN +Z DIRN
JOINT LOAD
7 TO 10 13 TO 16 FZ 2.58 MX 2.66
************************************************************
LOAD 5 WIND LOAD IN -Z DIRN
JOINT LOAD
7 TO 10 13 TO 16 FZ -2.58 MX -2.66
************************************************************
*LOAD 6 TL1
*TEMPERATURE LOAD
*1 TO 6 11 TO 15 21 TO 28 TEMP 25
************************************************************
*LOAD 7 TL2
*TEMPERATURE LOAD
*1 TO 6 11 TO 15 21 TO 28 TEMP -25
************************************************************
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
************************************************************
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
```

```
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 DL + 1.0  (LL+ML)
1 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 304 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 305 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 306 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 DL + 1.7  (LL+ML)
1 1.4 2 1.7 3 1.7
LOAD COMB 403 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 404 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 405 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 406 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
************************************************************
PERFORM ANALYSIS
************************************************************
LOAD LIST 401 TO 406
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 0 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 16 ALL
MINMAIN 12 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 6
DESIGN COLUMN 1 TO 6
DESIGN BEAM 11 TO 15 21 TO 28
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  1, 2020                *
*           Time=     9:25:30                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-001\Staad\Civil\3110-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 29-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART ON GROUND PORTION
    9. JOB REF CONVEYOR 3110-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 -2.38419E-007 -1 0; 2 -2.38419E-007 -1 1.69; 3 4.5 -1 0; 4 4.5 -1 1.69
   18. 5 9 -1 0; 6 9 -1 1.69; 7 -2.38419E-007 0.1 0; 8 -2.38419E-007 0.1 1.69
   19. 9 3 0.1 0; 10 3 0.1 1.69; 11 4.5 0.1 0; 12 4.5 0.1 1.69; 13 6 0.1 0
   20. 14 6 0.1 1.69; 15 9 0.1 0; 16 9 0.1 1.69
   21. MEMBER INCIDENCES
   22. 1 1 7; 2 2 8; 3 3 11; 4 4 12; 5 5 15; 6 6 16; 11 8 7; 12 9 10; 13 12 11
   23. 14 13 14; 15 16 15; 21 7 9; 22 9 11; 23 11 13; 24 13 15; 25 8 10; 26 10 12
   24. 27 12 14; 28 14 16
   25. DEFINE MATERIAL START
   26. ISOTROPIC STEEL
   27. E 2.05E+008
   28. POISSON 0.3
   29. DENSITY 76.8195
   30. ALPHA 1.2E-005
   31. DAMP 0.03
   32. TYPE STEEL
   33. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
   34. ISOTROPIC CONCRETE
   35. E 2.17185E+007
   36. POISSON 0.17
   37. DENSITY 23.5616
   38. ALPHA 1E-005
```

```
39. DAMP 0.05
40. TYPE CONCRETE
41. STRENGTH FCU 27579
42. END DEFINE MATERIAL
43. MEMBER PROPERTY EUROPEAN
44. 11 TO 15 PRIS YD 0.3 ZD 0.25
45. 1 TO 6 PRIS YD 0.4 ZD 0.4
46. MEMBER PROPERTY EUROPEAN
47. 21 TO 28 PRIS YD 0.4 ZD 0.3
48. CONSTANTS
49. MATERIAL CONCRETE ALL
50. SUPPORTS
51. 1 TO 6 FIXED
52. ******************************************************************
53. MEMBER OFFSET
54. 21 TO 28 START 0 -0.2 0
55. 21 TO 28 END 0 -0.2 0
56. 11 TO 15 START 0 -0.15 0
57. 11 TO 15 END 0 -0.15 0
58. LOAD 1 DL
59. SELFWEIGHT Y -1
60. JOINT LOAD
61. *CONVEYOR DEAD LOAD
62. 7 TO 10 13 TO 16 FX 1.5 FY -6 FZ 0.8
63. ******************************************************************
64. LOAD 2 LL
65. JOINT LOAD
66. * SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
67. 7 TO 10 13 TO 16 FY -1.9125
68. ******************************************************************
69. LOAD 3 ML
70. *CONVEYOR MATERIAL LOAD
71. JOINT LOAD
72. 7 TO 10 13 TO 16 FY -6.5
73. ******************************************************************
74. LOAD 4 WIND LOAD IN +Z DIRN
75. JOINT LOAD
76. 7 TO 10 13 TO 16 FZ 2.58 MX 2.66
77. ******************************************************************
78. LOAD 5 WIND LOAD IN -Z DIRN
79. JOINT LOAD
80. 7 TO 10 13 TO 16 FZ -2.58 MX -2.66
81. ******************************************************************
82. *LOAD 6 TL1
83. *TEMPERATURE LOAD
84. *1 TO 6 11 TO 15 21 TO 28 TEMP 25
85. ******************************************************************
86. *LOAD 7 TL2
87. *TEMPERATURE LOAD
88. *1 TO 6 11 TO 15 21 TO 28 TEMP -25
89. ******************************************************************
90. LOAD 11 G
91. REPEAT LOAD
92. 1 1.0
93. LOAD 12 G + QN + MN
94. REPEAT LOAD
```

```
 95. 1 1.0 2 1.0 3 1.0
 96. ************************************************************
 97. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
 98. *LOAD COMB 201 1.4 DL
 99. *1 1.4
100. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
101. *1 1.2 6 1.2 2 1.6 3 1.6
102. *LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
103. *1 1.2 7 1.2 2 1.6 3 1.6
104. *LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
105. *1 1.2 4 1.6 2 0.5 3 0.5
106. *LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
107. *1 1.2 5 1.6 2 0.5 3 0.5
108. *LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
109. *1 0.9 4 1.6
110. *LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
111. *1 0.9 5 1.6
112. ************************************************************
113. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
114. LOAD COMB 301 1.0 DL
115. 1 1.0
116. LOAD COMB 302 1.0 DL + 1.0  (LL+ML)
117. 1 1.0 2 1.0 3 1.0
118. LOAD COMB 303 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
119. 1 1.0 4 0.75 2 0.75 3 0.75
120. LOAD COMB 304 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
121. 1 1.0 5 0.75 2 0.75 3 0.75
122. LOAD COMB 305 0.6 DL + 1.0 WL(+Z)
123. 1 0.6 4 1.0
124. LOAD COMB 306 0.6 DL + 1.0 WL(-Z)
125. 1 0.6 5 1.0
126. ************************************************************
127. ****LOAD COMBINATION FOR FOUNDATION DESIGN
128. LOAD COMB 401 1.4 DL
129. 1 1.4
130. LOAD COMB 402 1.4 DL + 1.7  (LL+ML)
131. 1 1.4 2 1.7 3 1.7
132. LOAD COMB 403 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
133. 1 1.2 4 1.6 2 0.5 3 0.5
134. LOAD COMB 404 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
135. 1 1.2 5 1.6 2 0.5 3 0.5
136. LOAD COMB 405 0.9 DL + 1.6 WL(+Z)
137. 1 0.9 4 1.6
138. LOAD COMB 406 0.9 DL + 1.6 WL(-Z)
139. 1 0.9 5 1.6
140. ************************************************************
141. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**16**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 10/ 4/ 18 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 60
TOTAL LOAD COMBINATION CASES = 12 SO FAR.
SIZE OF STIFFNESS MATRIX = 2 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 50254.4 MB

```
142. ************************************************************
143. LOAD LIST 401 TO 406
144. START CONCRETE DESIGN
145. CODE ACI 2008
146. FC 35000 ALL
147. FYMAIN 420000 ALL
148. TRACK 0 ALL
149. CLB 0.05 ALL
150. CLT 0.05 ALL
151. CLS 0.05 ALL
152. MAXMAIN 16 ALL
153. MINMAIN 12 ALL
154. MINSEC 8 ALL
155. REINF 0 MEMB 1 TO 6
156. DESIGN COLUMN 1 TO 6
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1611.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 12 MM	1.131	404	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

157. DESIGN BEAM 11 TO 15 21 TO 28

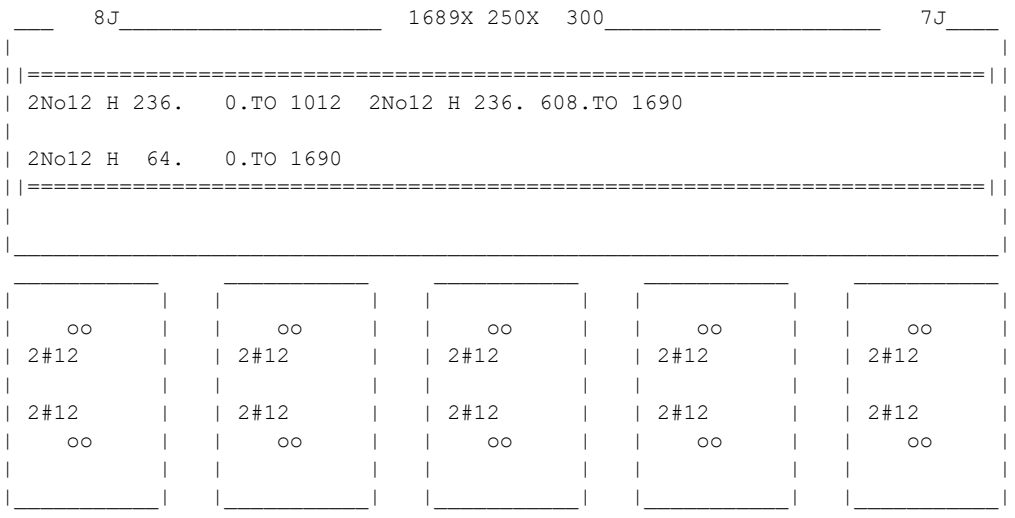
BEAM NO. 11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1690. MM FY - 420. FC - 35. MPA, SIZE - 250. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1690.	YES	YES
2	236.	2 - 12MM	0.	1012.	YES	NO
3	236.	2 - 12MM	608.	1690.	NO	YES

B E A M N O. 11 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.91 KNS Vc= 56.78 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.12 KNS Vc= 57.86 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



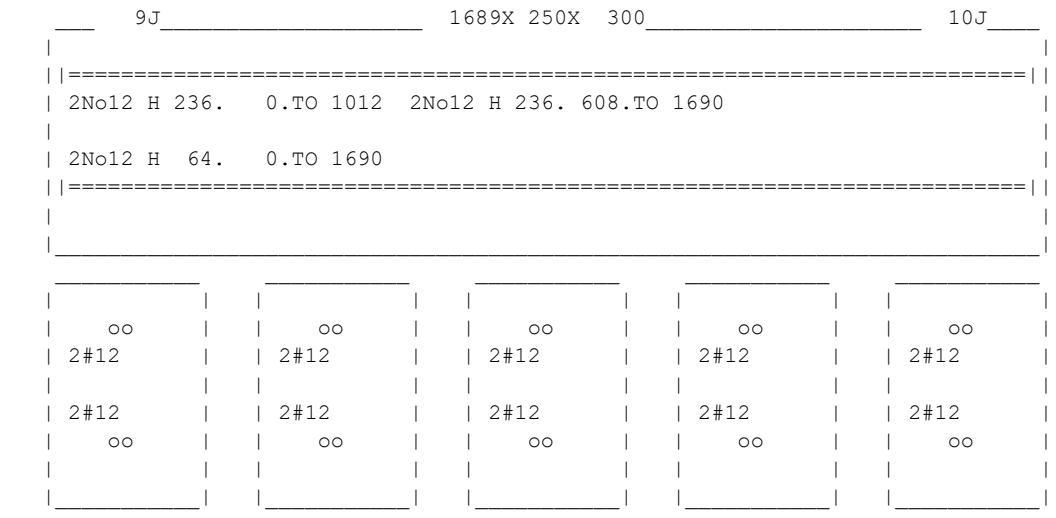
BEAM NO. 12 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1690. MM FY - 420. FC - 35. MPA, SIZE - 250. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1690.	YES	YES
2	236.	2 - 12MM	0.	1012.	YES	NO
3	236.	2 - 12MM	608.	1690.	NO	YES

B E A M N O. 12 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.26 KNS Vc= 55.97 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.76 KNS Vc= 101.45 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



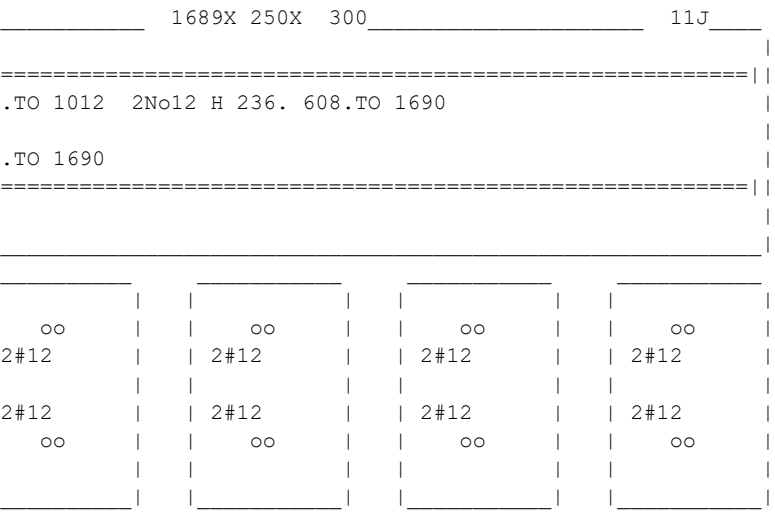
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1690. MM FY - 420. FC - 35. MPA, SIZE - 250. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1690.	YES	YES
2	236.	2 - 12MM	0.	1012.	YES	NO
3	236.	2 - 12MM	608.	1690.	NO	YES

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.88 KNS Vc= 56.58 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.14 KNS Vc= 56.91 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



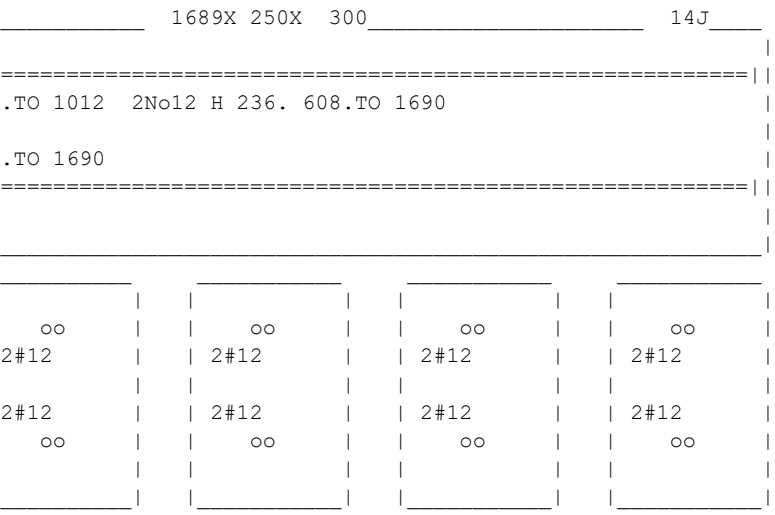
BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1690. MM FY - 420. FC - 35. MPA, SIZE - 250. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1690.	YES	YES
2	236.	2 - 12MM	0.	1012.	YES	NO
3	236.	2 - 12MM	608.	1690.	NO	YES

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.26 KNS Vc= 55.97 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.76 KNS Vc= 101.45 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



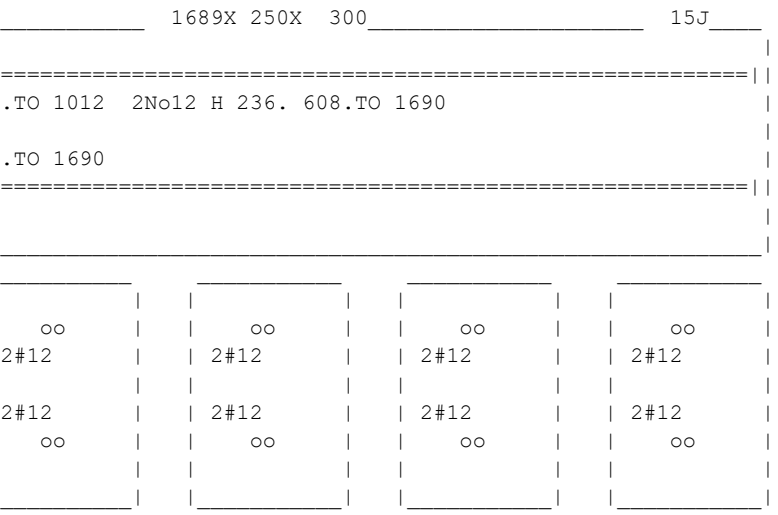
BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1690. MM FY - 420. FC - 35. MPA, SIZE - 250. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1690.	YES	YES
2	236.	2 - 12MM	0.	1012.	YES	NO
3	236.	2 - 12MM	608.	1690.	NO	YES

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.91 KNS Vc= 56.78 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.12 KNS Vc= 57.86 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 1.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



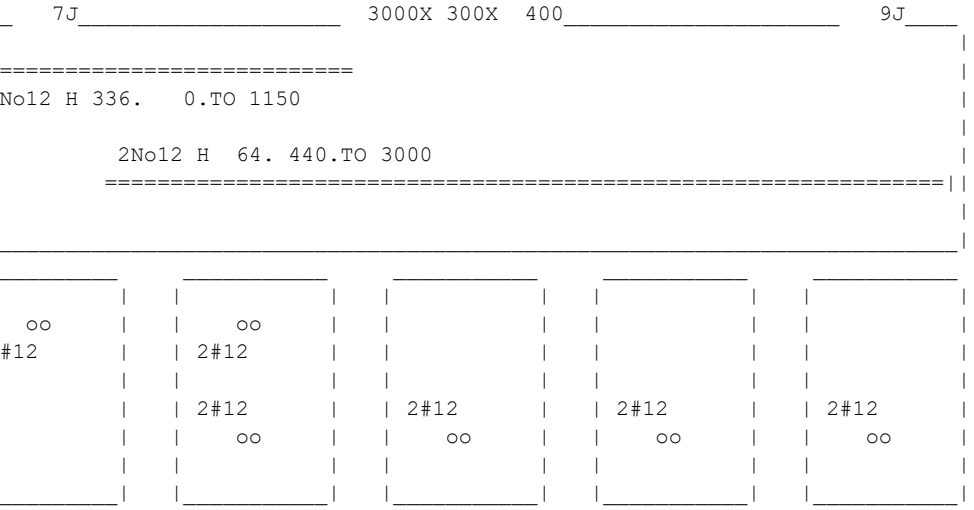
BEAM NO. 21 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	440.	3000.	NO	YES
2	336.	2 - 12MM	0.	1150.	YES	NO

B E A M N O. 21 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.88 KNS Vc= 97.53 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 0.35 KNS Vc= 94.13 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



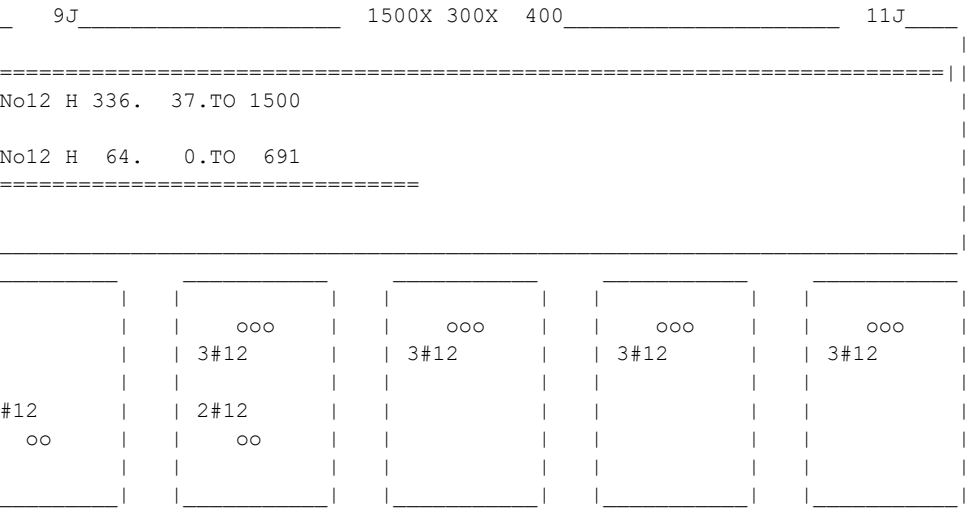
BEAM NO. 22 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	691.	YES	NO
2	336.	3 - 12MM	37.	1500.	NO	YES

B E A M N O. 22 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.23 KNS Vc= 102.42 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.52 KNS Vc= 95.70 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



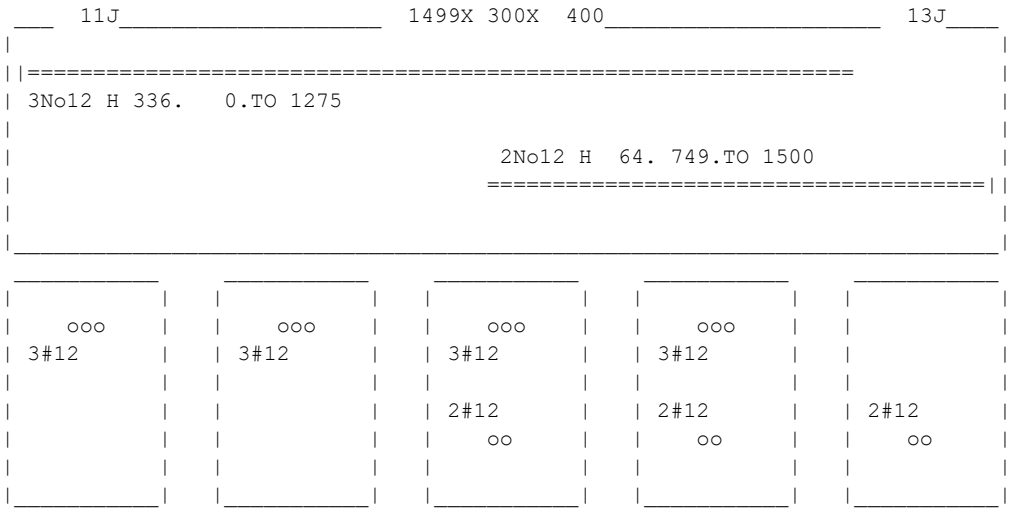
BEAM NO. 23 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	749.	1500.	NO	YES
2	336.	3 - 12MM	0.	1275.	YES	NO

B E A M N O. 23 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.84 KNS Vc= 95.64 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.55 KNS Vc= 99.48 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



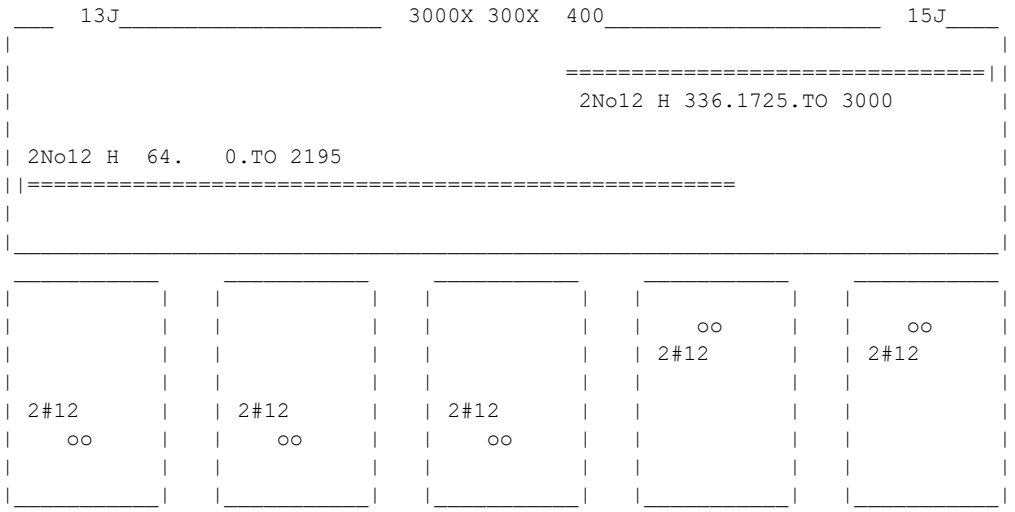
BEAM NO. 24 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	2195.	YES	NO
2	336.	2 - 12MM	1725.	3000.	NO	YES

B E A M N O. 24 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.33 KNS Vc= 94.13 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.57 KNS Vc= 96.69 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



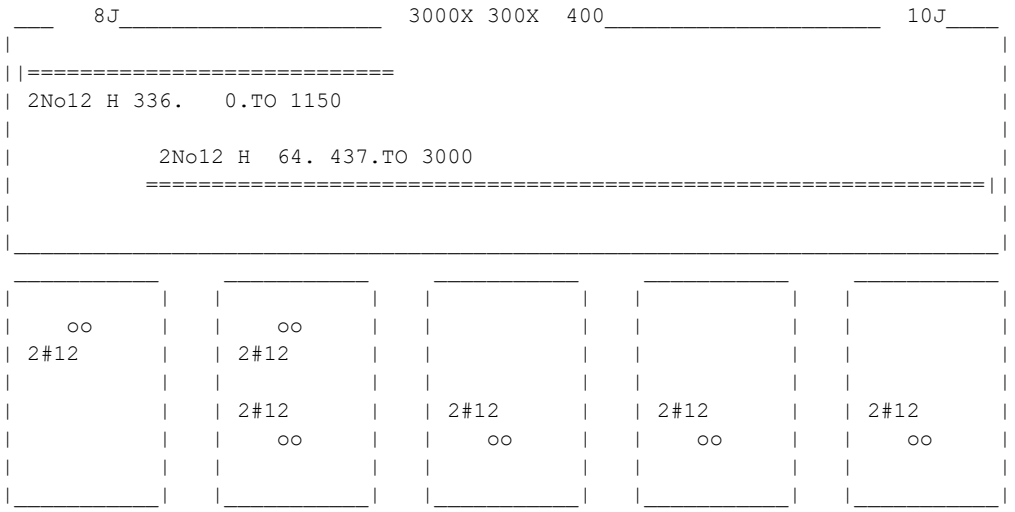
BEAM NO. 25 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	437.	3000.	NO	YES
2	336.	2 - 12MM	0.	1150.	YES	NO

B E A M N O. 25 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.99 KNS Vc= 97.44 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 0.24 KNS Vc= 94.11 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



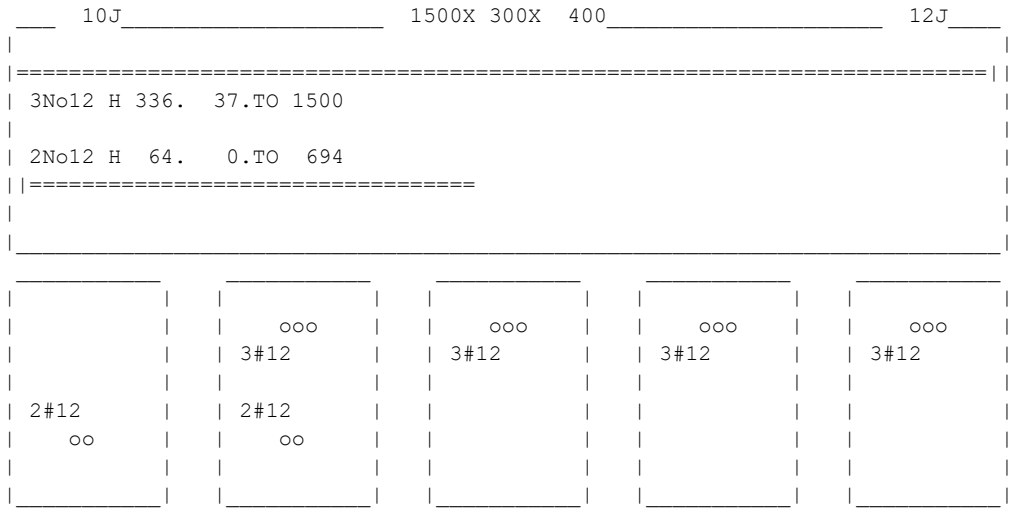
BEAM NO. 26 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	694.	YES	NO
2	336.	3 - 12MM	37.	1500.	NO	YES

B E A M N O. 26 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.63 KNS Vc= 102.42 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.92 KNS Vc= 95.74 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



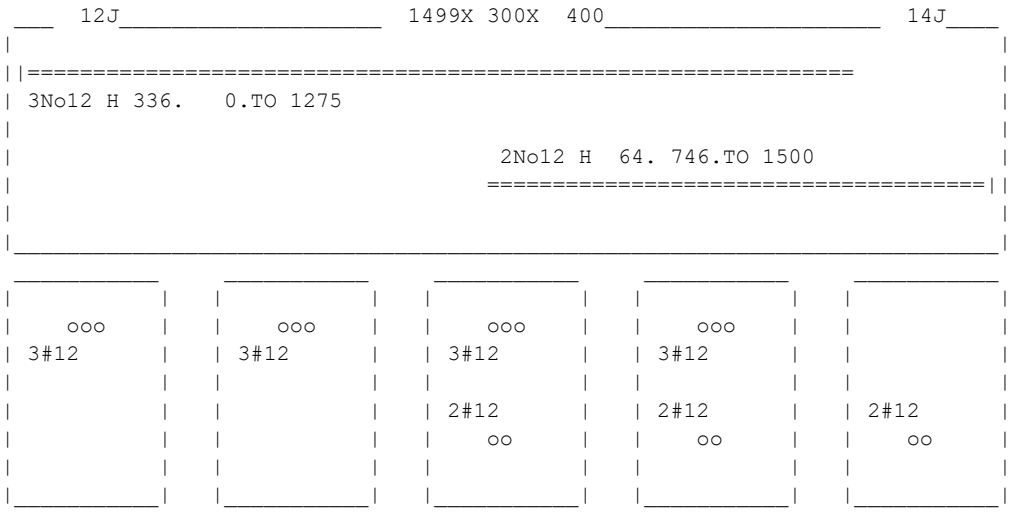
BEAM NO. 27 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	746.	1500.	NO	YES
2	336.	3 - 12MM	0.	1275.	YES	NO

B E A M N O. 27 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.24 KNS Vc= 95.67 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.94 KNS Vc= 99.57 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



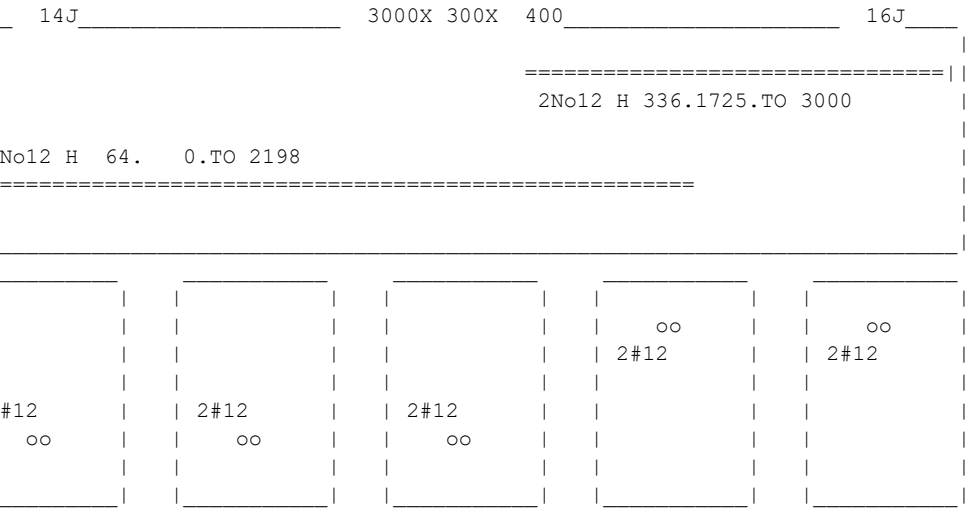
BEAM NO. 28 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	2198.	YES	NO
2	336.	2 - 12MM	1725.	3000.	NO	YES

B E A M N O. 28 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.44 KNS Vc= 94.14 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.67 KNS Vc= 96.67 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

158. END CONCRETE DESIGN
159. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 1,2020 TIME= 9:25:31 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED △A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3	4
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED △A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR METAL PICKING ZONE**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 29-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART METAL PICKING ZONE
JOB REF CONVEYOR 3110-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 -2.38419e-007 0 0; 2 -2.38419e-007 0 1.27; 3 -2.38419e-007 0 3.04;
4 -2.38419e-007 0 4.31; 5 5 0 0; 6 5 0 1.27; 7 5 0 3.04; 8 5 0 4.31;
9 -2.38419e-007 0.57 0; 10 -2.38419e-007 0.57 1.27; 11 -2.38419e-007 0.57 3.04;
12 -2.38419e-007 0.57 4.31; 13 1.25 0.57 1.27; 14 1.25 0.57 3.04;
15 3.75 0.57 1.27; 16 3.75 0.57 3.04; 17 5 0.57 0; 18 5 0.57 1.27;
19 5 0.57 3.04; 20 5 0.57 4.31; 21 2.49995 0.570159 0;
22 2.49995 0.570159 4.31; 23 -2.38419e-007 -1 0; 24 -2.38419e-007 -1 4.31;
25 5 -1 0; 26 5 -1 4.31;
MEMBER INCIDENCES
1 9 1; 2 10 2; 3 11 3; 4 12 4; 5 17 5; 6 18 6; 7 19 7; 8 20 8; 9 10 9;
10 11 10; 11 12 11; 12 9 13; 13 10 13; 14 11 14; 15 12 14; 16 13 14; 17 13 15;
18 14 16; 19 15 16; 20 15 17; 21 15 18; 22 16 19; 23 16 20; 24 18 17; 25 19 18;
26 20 19; 27 9 21; 28 12 22; 29 13 21; 30 14 22; 31 21 15; 32 22 16; 33 21 17;
34 22 20; 35 23 1; 36 24 4; 37 25 5; 38 26 8; 39 1 2; 40 2 3; 41 3 4; 42 5 6;
43 6 7; 44 7 8; 45 4 8; 46 1 5;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 8 13 14 17 18 21 22 27 28 33 34 TABLE ST UPN200
9 11 24 26 TABLE ST UPN140
12 15 20 23 29 TO 32 TABLE ST L65X65X6
10 16 19 25 TABLE ST L65X65X6
```

```
MEMBER PROPERTY EUROPEAN
39 TO 44 PRIS YD 0.5 ZD 0.35
35 TO 38 PRIS YD 0.4 ZD 0.4
MEMBER PROPERTY EUROPEAN
45 46 PRIS YD 0.4 ZD 0.3
CONSTANTS
MATERIAL STEEL MEMB 1 TO 34
MATERIAL CONCRETE MEMB 35 TO 46
SUPPORTS
23 TO 26 FIXED
MEMBER RELEASE
9 TO 11 16 19 24 TO 26 START MP 0.99
1 TO 11 16 19 24 TO 26 END MP 0.99
************************************************************
MEMBER TRUSS
12 15 20 23 29 TO 32
MEMBER OFFSET
39 TO 44 START 0 -0.25 0
39 TO 44 END 0 -0.25 0
45 46 START 0 -0.2 0
45 46 END 0 -0.2 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM & HANDRAIL LOAD @15KG/M
*WALKWAY WIDTH 1000MM & BELT WIDTH 1400MM
MEMBER LOAD
27 28 33 34 UNI GY -0.315
13 14 17 18 21 22 UNI GY -0.465
*CABLE TRAY LOAD @112KG/TRAY/M RUN * 2 NOS OF TRAY = 224KG/M RUN
*CONCENTRATED FORCE = 224*3=6.72kN & MOMENT= 6.72*0.4 = 2.69kNM
34 CON GY -6.72 1.5
34 CMOM GX 2.69 1.5
28 CON GY -6.72 1
28 CMOM GX 2.69 1
*LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
33 34 CON GY -1 1.75
27 28 CON GY -1 1.25
*PIPE DEAD LOAD
*CONVEYOR DEAD LOAD VERTICAL
22 CON GY -6 0.25
21 CON GY -6 0.25
13 14 CON GY -6 1
*CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
22 CON GX 1.5 0.25
21 CON GX 1.5 0.25
13 14 CON GX 1.5 1
*CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
22 CON GZ 0.8 0.25
21 CON GZ 0.8 0.25
13 14 CON GZ 0.8 1
************************************************************
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
```

```
27 28 33 34 UNI GY -2.4
13 14 17 18 21 22 UNI GY -2.4
************************************************************
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
22 CON GY -6.5 0.25
21 CON GY -6.5 0.25
13 14 CON GY -6.5 1
** Spillage Load @75kg/ Sq. m on seal plate
13 14 17 18 21 22 UNI GY -0.64
************************************************************
LOAD 4 WIND LOAD IN +Z DIRN
MEMBER LOAD
21 22 CON GZ 2.58 0.25
13 14 CON GZ 2.58 1
*****
21 22 CMOM GX 2.66 0.25
13 14 CMOM GX 2.66 1
*31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX 1.38
*TRESTLE BODY WIND
1 TO 8 UNI GZ 1.2
*WIND ON CABLE TRAY SUPPORT
28 CON GZ 1.11 1
28 CMOM GX 1
34 CON GZ 1.11 1.5
34 CMOM GX 1.5
************************************************************
LOAD 5 WIND LOAD IN -Z DIRN
MEMBER LOAD
21 22 CON GZ -2.58 0.25
13 14 CON GZ -2.58 1
*****
21 22 CMOM GX -2.66 0.25
13 14 CMOM GX -2.66 1
*31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX -1.38
*WIND ON TRESTLE BODY
1 TO 8 UNI GZ -1.2
*WIND ON CABLE TRAY SUPPORT
28 CON GZ -1.11 1
28 CMOM GX -1
34 CON GZ -1.11 1.5
34 CMOM GX -1.5
************************************************************
LOAD 6 TL1
TEMPERATURE LOAD
1 TO 34 TEMP 25
************************************************************
LOAD 7 TL2
TEMPERATURE LOAD
1 TO 34 TEMP -25
************************************************************
LOAD 11 G
REPEAT LOAD
```

```
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
************************************************************
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 6 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 7 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 6 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 7 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
************************************************************
PERFORM ANALYSIS
```

```
*STEEL TAKE OFF LIST 1 TO 11 13 14 17 18 21 22 24 TO 28 33 34
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 34
*FU 410000 MEMB 1 TO 34
*TRACK 0 MEMB 1 TO 34
*BEAM 1 MEMB 1 TO 34
*TMAIN 300 MEMB 1 TO 34
*MAIN 200 MEMB 1 TO 34
*NSF 0.8 MEMB 1 TO 34
*LZ 5 MEMB 13 14 17 18 21 22 27 28 33 34
*LY 2.5 MEMB 13 14 17 18 21 22 27 28 33 34
*UNT 2.5 MEMB 13 14 17 18 21 22 27 28 33 34
*UNB 5 MEMB 13 14 17 18 21 22 27 28 33 34
*CHECK CODE MEMB 1 TO 34
************************************************************
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 35 TO 46
FYMAIN 420000 MEMB 35 TO 46
TRACK 0 MEMB 35 TO 46
CLB 0.05 MEMB 35 TO 46
CLT 0.05 MEMB 35 TO 46
CLS 0.05 MEMB 35 TO 46
MAXMAIN 16 MEMB 35 TO 46
MINMAIN 12 MEMB 35 TO 46
MINSEC 8 MEMB 35 TO 46
REINF 0 MEMB 35 TO 38
DESIGN COLUMN 35 TO 38
DESIGN BEAM 39 TO 46
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  1, 2020                *
*           Time=    10:15:58                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-001\Staad\Civil\3110-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 29-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART METAL PICKING ZONE
    9. JOB REF CONVEYOR 3110-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 -2.38419E-007 0 0; 2 -2.38419E-007 0 1.27; 3 -2.38419E-007 0 3.04
   18. 4 -2.38419E-007 0 4.31; 5 5 0 0; 6 5 0 1.27; 7 5 0 3.04; 8 5 0 4.31
   19. 9 -2.38419E-007 0.57 0; 10 -2.38419E-007 0.57 1.27; 11 -2.38419E-007 0.57 3.04
   20. 12 -2.38419E-007 0.57 4.31; 13 1.25 0.57 1.27; 14 1.25 0.57 3.04
   21. 15 3.75 0.57 1.27; 16 3.75 0.57 3.04; 17 5 0.57 0; 18 5 0.57 1.27
   22. 19 5 0.57 3.04; 20 5 0.57 4.31; 21 2.49995 0.570159 0
   23. 22 2.49995 0.570159 4.31; 23 -2.38419E-007 -1 0; 24 -2.38419E-007 -1 4.31
   24. 25 5 -1 0; 26 5 -1 4.31
   25. MEMBER INCIDENCES
   26. 1 9 1; 2 10 2; 3 11 3; 4 12 4; 5 17 5; 6 18 6; 7 19 7; 8 20 8; 9 10 9
   27. 10 11 10; 11 12 11; 12 9 13; 13 10 13; 14 11 14; 15 12 14; 16 13 14; 17 13 15
   28. 18 14 16; 19 15 16; 20 15 17; 21 15 18; 22 16 19; 23 16 20; 24 18 17; 25 19 18
   29. 26 20 19; 27 9 21; 28 12 22; 29 13 21; 30 14 22; 31 21 15; 32 22 16; 33 21 17
   30. 34 22 20; 35 23 1; 36 24 4; 37 25 5; 38 26 8; 39 1 2; 40 2 3; 41 3 4; 42 5 6
   31. 43 6 7; 44 7 8; 45 4 8; 46 1 5
   32. DEFINE MATERIAL START
   33. ISOTROPIC STEEL
   34. E 2.05E+008
   35. POISSON 0.3
   36. DENSITY 76.8195
   37. ALPHA 1.2E-005
   38. DAMP 0.03
```

```
39. TYPE STEEL
40. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
41. ISOTROPIC CONCRETE
42. E 2.17185E+007
43. POISSON 0.17
44. DENSITY 23.5616
45. ALPHA 1E-005
46. DAMP 0.05
47. TYPE CONCRETE
48. STRENGTH FCU 27579
49. END DEFINE MATERIAL
50. MEMBER PROPERTY EUROPEAN
51. 1 TO 8 13 14 17 18 21 22 27 28 33 34 TABLE ST UPN200
52. 9 11 24 26 TABLE ST UPN140
53. 12 15 20 23 29 TO 32 TABLE ST L65X65X6
54. 10 16 19 25 TABLE ST L65X65X6
55. MEMBER PROPERTY EUROPEAN
56. 39 TO 44 PRIS YD 0.5 ZD 0.35
57. 35 TO 38 PRIS YD 0.4 ZD 0.4
58. MEMBER PROPERTY EUROPEAN
59. 45 46 PRIS YD 0.4 ZD 0.3
60. CONSTANTS
61. MATERIAL STEEL MEMB 1 TO 34
62. MATERIAL CONCRETE MEMB 35 TO 46
63. SUPPORTS
64. 23 TO 26 FIXED
65. MEMBER RELEASE
66. 9 TO 11 16 19 24 TO 26 START MP 0.99
67. 1 TO 11 16 19 24 TO 26 END MP 0.99
68. ************************************************************
69. MEMBER TRUSS
70. 12 15 20 23 29 TO 32
71. MEMBER OFFSET
72. 39 TO 44 START 0 -0.25 0
73. 39 TO 44 END 0 -0.25 0
74. 45 46 START 0 -0.2 0
75. 45 46 END 0 -0.2 0
76. LOAD 1 DL
77. SELFWEIGHT Y -1.1
78. *WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM _HANDRAIL LOAD @1
79. *WALKWAY WIDTH 1000MM _BELT WIDTH 1400MM
80. MEMBER LOAD
81. 27 28 33 34 UNI GY -0.315
82. 13 14 17 18 21 22 UNI GY -0.465
83. *CABLE TRAY LOAD @112KG/TRAY/M RUN * 2 NOS OF TRAY = 224KG/M RUN
84. *CONCENTRATED FORCE = 224*3=6.72KN _MOMENT= 6.72*0.4 = 2.69KNM
85. 34 CON GY -6.72 1.5
86. 34 CMOM GX 2.69 1.5
87. 28 CON GY -6.72 1
88. 28 CMOM GX 2.69 1
89. *LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
90. 33 34 CON GY -1 1.75
91. 27 28 CON GY -1 1.25
92. *PIPE DEAD LOAD
93. *CONVEYOR DEAD LOAD VERTICAL
94. 22 CON GY -6 0.25
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```
 95. 21 CON GY -6 0.25
 96. 13 14 CON GY -6 1
 97. *CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
 98. 22 CON GX 1.5 0.25
 99. 21 CON GX 1.5 0.25
100. 13 14 CON GX 1.5 1
101. *CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
102. 22 CON GZ 0.8 0.25
103. 21 CON GZ 0.8 0.25
104. 13 14 CON GZ 0.8 1
105. ************************************************************
106. LOAD 2 LL
107. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
108. MEMBER LOAD
109. 27 28 33 34 UNI GY -2.4
110. 13 14 17 18 21 22 UNI GY -2.4
111. ************************************************************
112. LOAD 3 ML
113. *CONVEYOR MATERIAL LOAD
114. MEMBER LOAD
115. 22 CON GY -6.5 0.25
116. 21 CON GY -6.5 0.25
117. 13 14 CON GY -6.5 1
118. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
119. 13 14 17 18 21 22 UNI GY -0.64
120. ************************************************************
121. LOAD 4 WIND LOAD IN +Z DIRN
122. MEMBER LOAD
123. 21 22 CON GZ 2.58 0.25
124. 13 14 CON GZ 2.58 1
125. *****
126. 21 22 CMOM GX 2.66 0.25
127. 13 14 CMOM GX 2.66 1
128. *31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX 1.38
129. *TRESTLE BODY WIND
130. 1 TO 8 UNI GZ 1.2
131. *WIND ON CABLE TRAY SUPPORT
132. 28 CON GZ 1.11 1
133. 28 CMOM GX 1
134. 34 CON GZ 1.11 1.5
135. 34 CMOM GX 1.5
136. ************************************************************
137. LOAD 5 WIND LOAD IN -Z DIRN
138. MEMBER LOAD
139. 21 22 CON GZ -2.58 0.25
140. 13 14 CON GZ -2.58 1
141. *****
142. 21 22 CMOM GX -2.66 0.25
143. 13 14 CMOM GX -2.66 1
144. *31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX -1.38
145. *WIND ON TRESTLE BODY
146. 1 TO 8 UNI GZ -1.2
147. *WIND ON CABLE TRAY SUPPORT
148. 28 CON GZ -1.11 1
149. 28 CMOM GX -1.
150. 34 CON GZ -1.11 1.5
```

```
151. 34 CMOM GX -1.5
152. ************************************************************
153. LOAD 6 TL1
154. TEMPERATURE LOAD
155. 1 TO 34 TEMP 25
156. ************************************************************
157. LOAD 7 TL2
158. TEMPERATURE LOAD
159. 1 TO 34 TEMP -25
160. ************************************************************
161. LOAD 11 G
162. REPEAT LOAD
163. 1 1.0
164. LOAD 12 G + QN + MN
165. REPEAT LOAD
166. 1 1.0 2 1.0 3 1.0
167. ************************************************************
168. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
169. *LOAD COMB 201 1.4 DL
170. *1 1.4
171. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
172. *1 1.2 6 1.2 2 1.6 3 1.6
173. *LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
174. *1 1.2 7 1.2 2 1.6 3 1.6
175. *LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
176. *1 1.2 4 1.6 2 0.5 3 0.5
177. *LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
178. *1 1.2 5 1.6 2 0.5 3 0.5
179. *LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
180. *1 0.9 4 1.6
181. *LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
182. *1 0.9 5 1.6
183. ************************************************************
184. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
185. LOAD COMB 301 1.0 DL
186. 1 1.0
187. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
188. 1 1.0 6 1.0 2 1.0 3 1.0
189. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
190. 1 1.0 7 1.0 2 1.0 3 1.0
191. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
192. 1 1.0 4 0.75 2 0.75 3 0.75
193. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
194. 1 1.0 5 0.75 2 0.75 3 0.75
195. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
196. 1 0.6 4 1.0
197. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
198. 1 0.6 5 1.0
199. ************************************************************
200. ****LOAD COMBINATION FOR FOUNDATION DESIGN
201. LOAD COMB 401 1.4 DL
202. 1 1.4
203. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
204. 1 1.4 6 1.4 2 1.7 3 1.7
205. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
206. 1 1.4 7 1.4 2 1.7 3 1.7
```

```
207. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
208. 1 1.2 4 1.6 2 0.5 3 0.5
209. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
210. 1 1.2 5 1.6 2 0.5 3 0.5
211. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
212. 1 0.9 4 1.6
213. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
214. 1 0.9 5 1.6
215. ************************************************************
216. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**26**	**NUMBER OF MEMBERS**	**46**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 22/ 8/ 48 DOF
TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 132
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 7 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.2/ 50254.7 MB

```
217. *STEEL TAKE OFF LIST 1 TO 11 13 14 17 18 21 22 24 TO 28 33 34
218. *LOAD LIST 201 TO 207
219. *PARAMETER 1
220. *CODE LRFD
221. *FYLD 275000 MEMB 1 TO 34
222. *FU 410000 MEMB 1 TO 34
223. *TRACK 0 MEMB 1 TO 34
224. *BEAM 1 MEMB 1 TO 34
225. *TMAIN 300 MEMB 1 TO 34
226. *MAIN 200 MEMB 1 TO 34
227. *NSF 0.8 MEMB 1 TO 34
228. *LZ 5 MEMB 13 14 17 18 21 22 27 28 33 34
229. *LY 2.5 MEMB 13 14 17 18 21 22 27 28 33 34
230. *UNT 2.5 MEMB 13 14 17 18 21 22 27 28 33 34
231. *UNB 5 MEMB 13 14 17 18 21 22 27 28 33 34
232. *CHECK CODE MEMB 1 TO 34
233. ************************************************************
234. LOAD LIST 401 TO 407
235. START CONCRETE DESIGN
236. CODE ACI 2008
237. FC 35000 MEMB 35 TO 46
```

```
238. FYMAIN 420000 MEMB 35 TO 46
239. TRACK 0 MEMB 35 TO 46
240. CLB 0.05 MEMB 35 TO 46
241. CLT 0.05 MEMB 35 TO 46
242. CLS 0.05 MEMB 35 TO 46
243. MAXMAIN 16 MEMB 35 TO 46
244. MINMAIN 12 MEMB 35 TO 46
245. MINSEC 8 MEMB 35 TO 46
246. REINF 0 MEMB 35 TO 38
247. DESIGN COLUMN 35 TO 38
```

==========

COLUMN NO. 35 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 36 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 37 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

==========

COLUMN NO. 38 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

248. DESIGN BEAM 39 TO 46

BEAM NO. 39 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

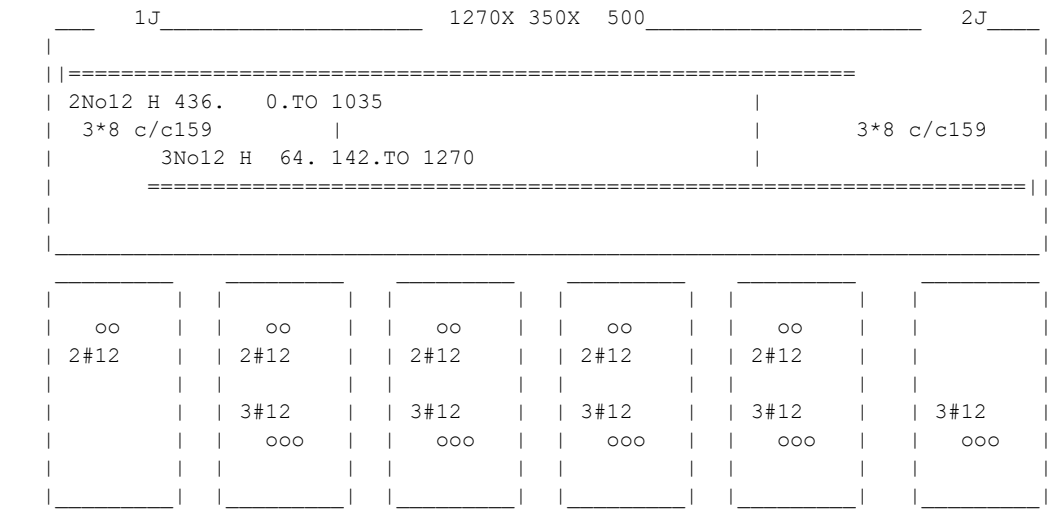
LEN - 1270. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	142.	1270.	NO	YES
2	436.	2 - 12MM	0.	1035.	YES	NO

B E A M N O. 39 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 46.13 KNS Vc= 269.41 KNS Vs= 0.00 KNS
Tu= 10.88 KN-MET Tc= 6.9 KN-MET Ts= 14.5 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.71 SQ.CM.

AT END SUPPORT - Vu= 43.58 KNS Vc= 155.89 KNS Vs= 0.00 KNS
Tu= 10.88 KN-MET Tc= 6.9 KN-MET Ts= 14.5 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.71 SQ.CM.



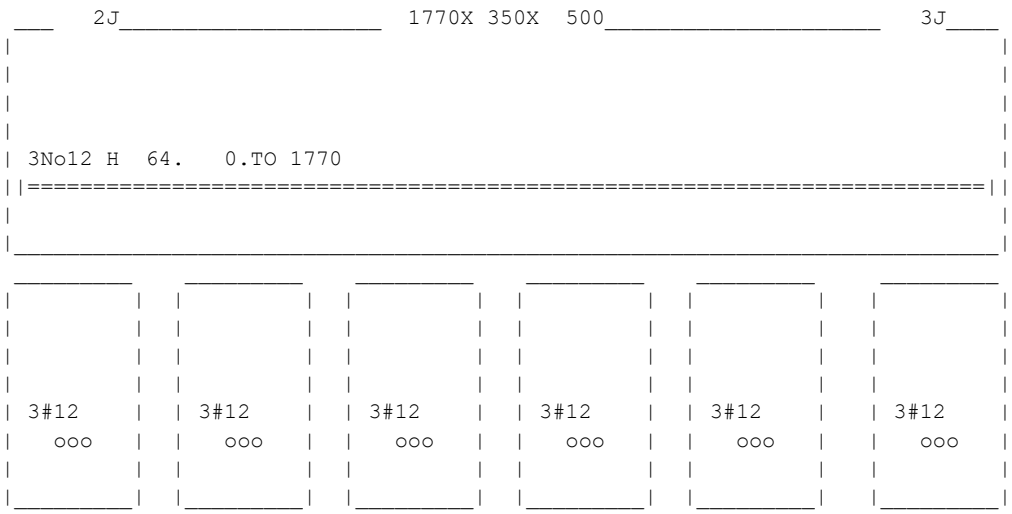
BEAM NO. 40 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1770. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1770.	YES	YES

B E A M N O. 40 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.57 KNS Vc= 142.68 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 4.16 KNS Vc= 143.16 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 41 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

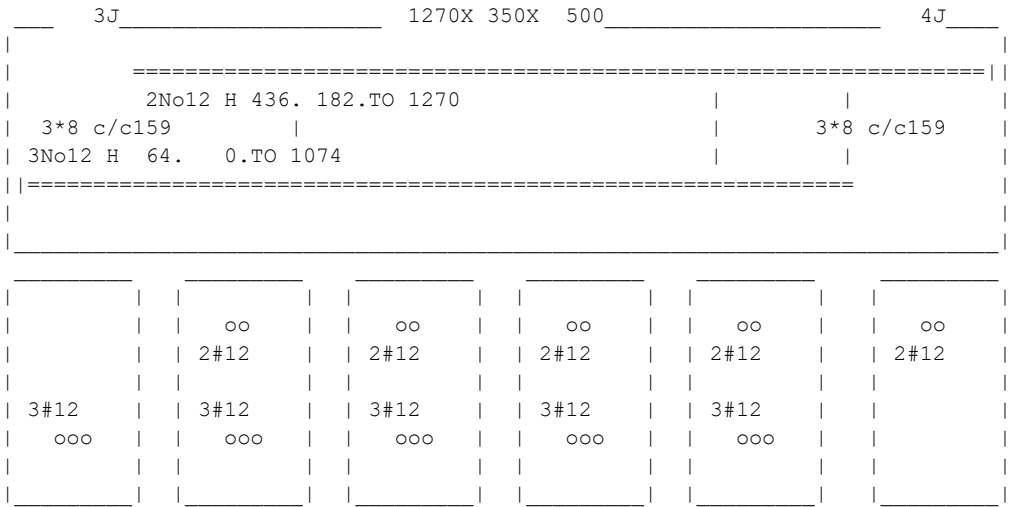
LEN - 1270. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1074.	YES	NO
2	436.	2 - 12MM	182.	1270.	NO	YES

B E A M N O. 41 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 43.75 KNS Vc= 159.05 KNS Vs= 0.00 KNS
Tu= 10.31 KN-MET Tc= 7.0 KN-MET Ts= 13.8 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.57 SQ.CM.

AT END SUPPORT - Vu= 46.30 KNS Vc= 270.34 KNS Vs= 0.00 KNS
Tu= 10.31 KN-MET Tc= 7.0 KN-MET Ts= 13.8 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.57 SQ.CM.



BEAM NO. 42 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

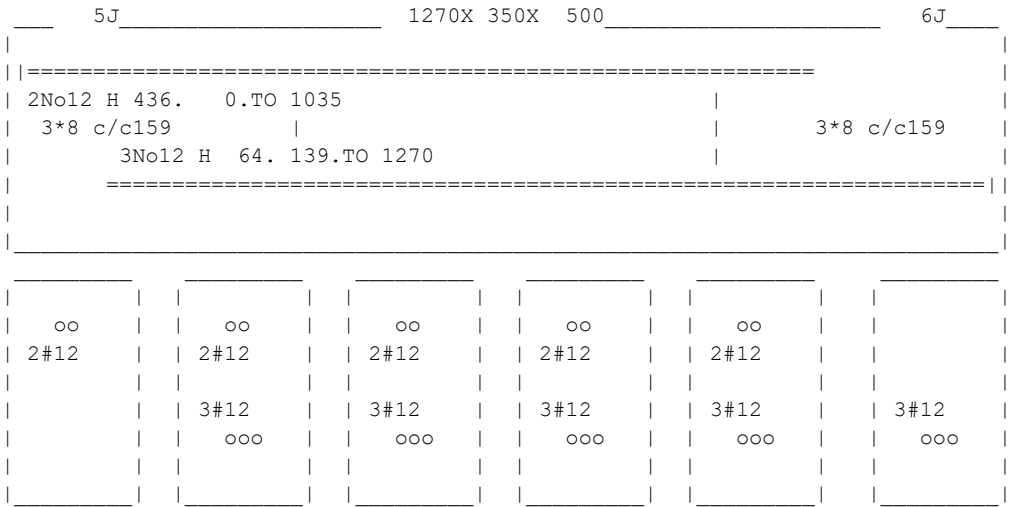
LEN - 1270. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	139.	1270.	NO	YES
2	436.	2 - 12MM	0.	1035.	YES	NO

B E A M N O. 42 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 46.57 KNS Vc= 269.43 KNS Vs= 0.00 KNS
Tu= 11.42 KN-MET Tc= 6.9 KN-MET Ts= 15.2 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.85 SQ.CM.

AT END SUPPORT - Vu= 44.02 KNS Vc= 155.97 KNS Vs= 0.00 KNS
Tu= 11.42 KN-MET Tc= 6.9 KN-MET Ts= 15.2 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.85 SQ.CM.



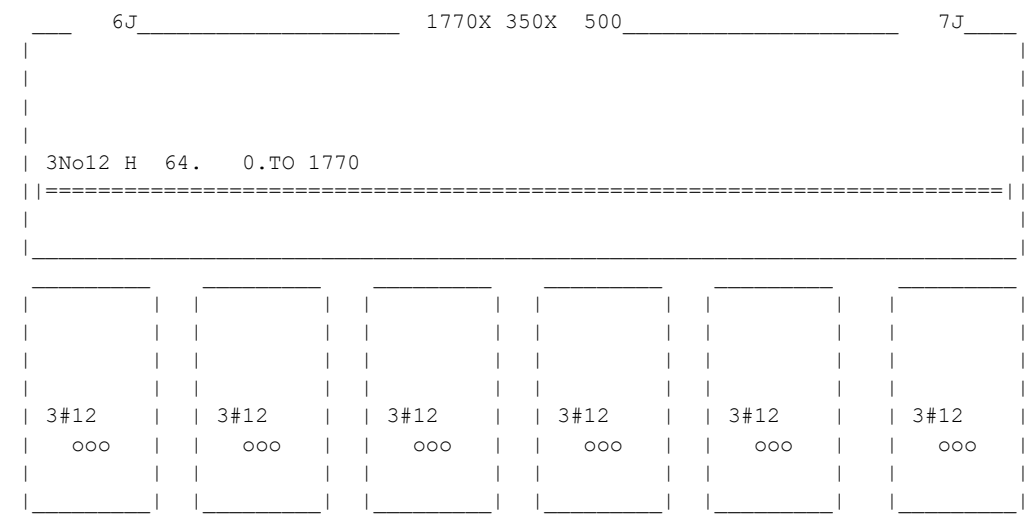
BEAM NO. 43 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1770. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1770.	YES	YES

B E A M N O. 43 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.57 KNS Vc= 142.68 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 4.16 KNS Vc= 143.15 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 44 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

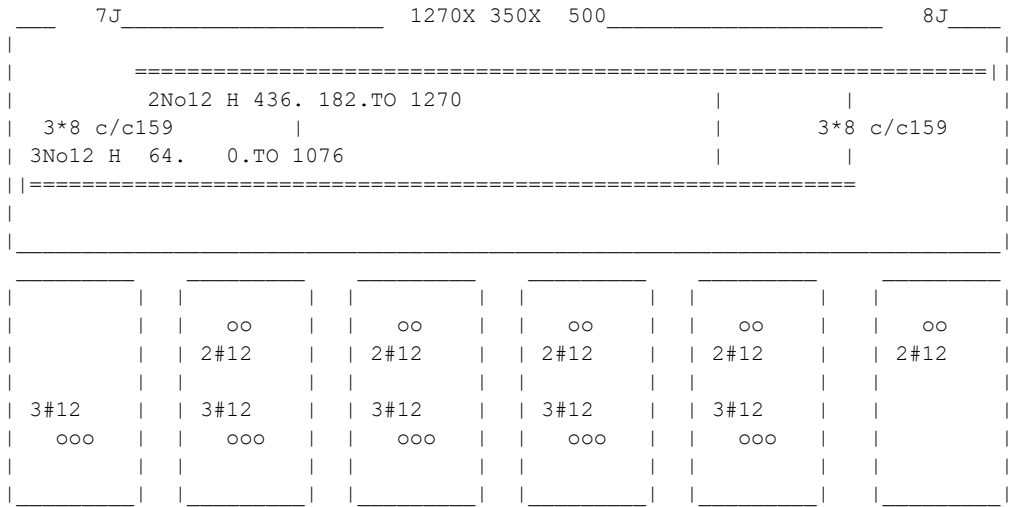
LEN - 1270. MM FY - 420. FC - 35. MPA, SIZE - 350. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1076.	YES	NO
2	436.	2 - 12MM	182.	1270.	NO	YES

B E A M N O. 44 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 44.19 KNS Vc= 159.10 KNS Vs= 0.00 KNS
Tu= 10.86 KN-MET Tc= 7.0 KN-MET Ts= 14.5 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.71 SQ.CM.

AT END SUPPORT - Vu= 46.74 KNS Vc= 270.35 KNS Vs= 0.00 KNS
Tu= 10.86 KN-MET Tc= 7.0 KN-MET Ts= 14.5 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 159. MM C/C FOR 201. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.71 SQ.CM.



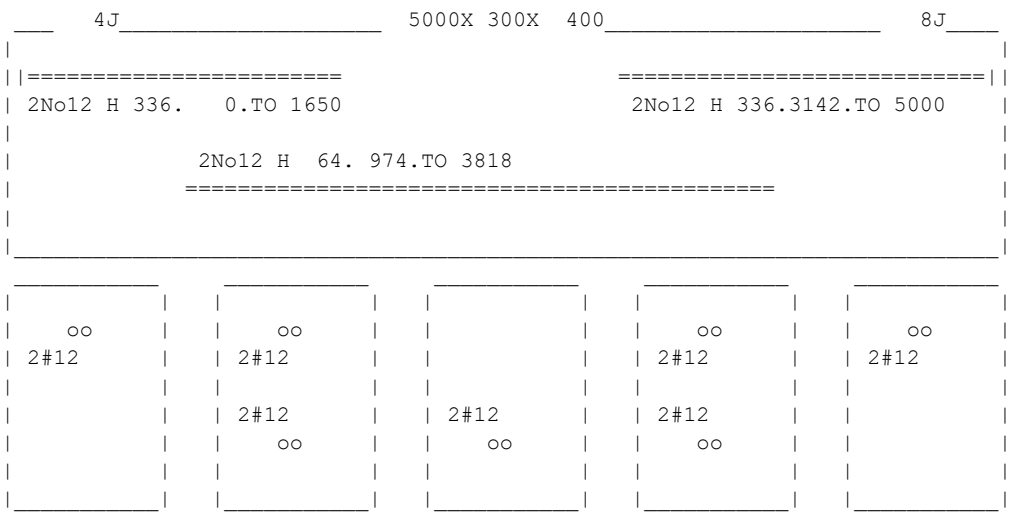
BEAM NO. 45 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	974.	3818.	NO	NO
2	336.	2 - 12MM	0.	1650.	YES	NO
3	336.	2 - 12MM	3142.	5000.	NO	YES

B E A M N O. 45 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.23 KNS Vc= 94.28 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.64 KNS Vc= 94.28 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



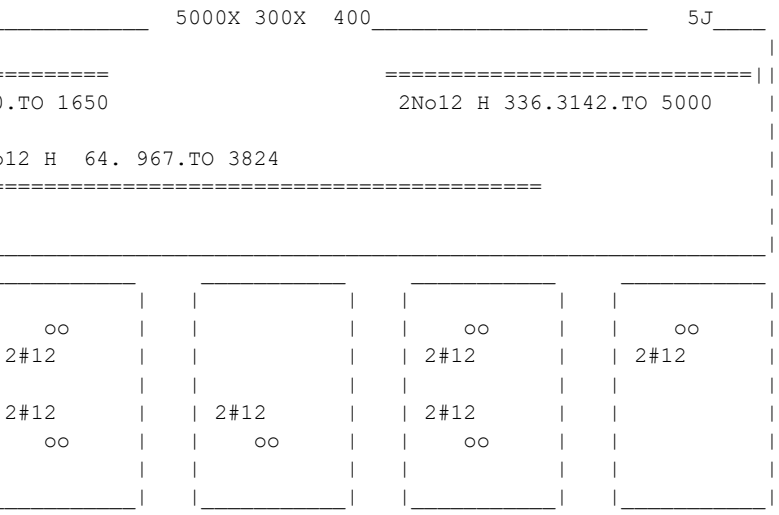
BEAM NO. 46 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	967.	3824.	NO	NO
2	336.	2 - 12MM	0.	1650.	YES	NO
3	336.	2 - 12MM	3142.	5000.	NO	YES

B E A M N O. 46 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.23 KNS Vc= 96.98 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.64 KNS Vc= 96.98 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

249. END CONCRETE DESIGN

250. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 1,2020 TIME= 10:15:59 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	A03	C01	D01
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3	4
Date	31-03-2020	27-04-2020	02-06-2020	04-07-2020	12-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 23-04-2020	REVISED AS PER EMPLOYER COMMENTS DATED 20-05-2020 & AS MARKED A03	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	APL	APL
Checked	SBY	SBY	SBY	SBY	SBY
Approved	PRI	PRI	PRI	PRI	PRI

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR GROUND PORTALS**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 30-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL
JOB REF CONVEYOR 3110-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0.35 -1.865; 2 0 0.35 3.635; 3 4.3 0.35 -1.865; 4 4.3 0.35 3.635;
5 8.6 0.35 -1.865; 6 8.6 0.35 3.635; 7 13.015 0.35 -1.865; 8 13.015 0.35 3.635;
9 17.765 0.35 -1.865; 10 17.765 0.35 3.635; 11 22.365 0.1 -1.865;
12 22.365 0.1 3.635; 13 26.915 0.1 -1.865; 14 26.915 0.35 -1.07;
15 26.915 0.35 0; 16 26.915 0.35 1.77; 17 26.915 0.35 2.84;
18 26.915 0.1 3.635; 19 -3.5 3.7 -1.07; 20 -3.5 3.7 0; 21 -3.5 3.7 1.77;
22 -3.5 3.7 2.84; 31 0 3.215 -1.865; 32 0 3.215 -1.07; 33 0 3.215 0;
34 0 3.215 1.77; 35 0 3.215 2.84; 36 0 3.215 3.635; 37 4.3 2.611 -1.865;
38 4.3 2.611 -1.07; 39 4.3 2.611 0; 40 4.3 2.611 1.77; 41 4.3 2.611 2.84;
42 4.3 2.611 3.635; 43 8.6 2.006 -1.865; 44 8.6 2.006 -1.07; 45 8.6 2.006 0;
46 8.6 2.006 1.77; 47 8.6 2.006 2.84; 48 8.6 2.006 3.635;
49 13.015 1.386 -1.865; 50 13.015 1.386 -1.07; 51 13.015 1.386 0;
52 13.015 1.386 1.77; 53 13.015 1.386 2.84; 54 13.015 1.386 3.635;
55 17.765 1.035 -1.865; 56 17.765 1.035 -1.07; 57 17.765 1.035 0;
58 17.765 1.035 1.77; 59 17.765 1.035 2.84; 60 17.765 1.035 3.635;
61 22.365 0.715 -1.865; 62 22.365 0.715 -1.07; 63 22.365 0.715 0;
64 22.365 0.715 1.77; 65 22.365 0.715 2.84; 66 22.365 0.715 3.635;
67 13.015 1.261 -1.865; 68 13.015 1.261 -0.865; 69 13.015 1.261 2.635;
70 13.015 1.261 3.635; 101 -1.75 3.4575 -1.07; 102 -1.75 3.4575 0;
103 -1.75 3.4575 1.77; 104 -1.75 3.4575 2.84; 105 2.15 2.913 -1.07;
106 2.15 2.913 0; 107 2.15 2.913 1.77; 108 2.15 2.913 2.84;
109 2.65395 2.84221 -1.865; 110 2.65702 2.84178 -1.07;
111 2.65702 2.84178 2.84; 112 2.65395 2.84221 3.635; 113 5.62693 2.4243 -1.865;
114 5.62693 2.4243 -1.07; 115 5.62693 2.4243 2.84; 116 5.62693 2.4243 3.635;
117 6.45 2.3085 -1.865; 118 6.45 2.3085 -1.07; 119 6.45 2.3085 0;
120 6.45 2.3085 1.77; 121 6.45 2.3085 2.84; 122 6.45 2.3085 3.635;
127 10.8075 1.696 -1.07; 128 10.8075 1.696 0; 129 10.8075 1.696 1.77;
130 10.8075 1.696 2.84; 131 11.7879 1.55832 -1.865; 132 11.7879 1.55832 -1.07;
133 11.7879 1.55832 2.84; 134 11.7879 1.55832 3.635; 135 14.7722 1.2563 -1.865;
136 14.7722 1.2563 -1.07; 137 14.7722 1.2563 2.84; 138 14.7722 1.2563 3.635;
139 15.39 1.2105 -1.07; 140 15.39 1.2105 0; 141 15.39 1.2105 1.77;
142 15.39 1.2105 2.84; 143 20.065 0.875 -1.07; 144 20.065 0.875 0;
145 20.065 0.875 1.77; 146 20.065 0.875 2.84; 147 20.9229 0.815318 -1.865;
148 20.9229 0.815318 -1.07; 149 20.9229 0.815318 2.84;
150 20.9229 0.815318 3.635; 151 23.9212 0.605896 -1.865;
152 23.9212 0.605896 -1.07; 153 23.9212 0.605896 2.84;
154 23.9212 0.605896 3.635; 155 24.64 0.5555 -1.07; 156 24.64 0.5555 0;
157 24.64 0.5555 1.77; 158 24.64 0.5555 2.84; 201 0 -1 -1.865; 202 0 -1 3.635;
```

```
203 4.3 -1 -1.865; 204 4.3 -1 3.635; 205 8.6 -1 -1.865; 206 8.6 -1 3.635;
207 13.015 -1 -1.865; 208 13.015 -1 3.635; 209 17.765 -1 -1.865;
210 17.765 -1 3.635; 211 22.365 -1 -1.865; 212 22.365 -1 3.635;
213 26.915 -1 -1.865; 214 26.915 -1 3.635; 215 24.64 0.1 3.635;
216 24.64 0.1 -1.865; 219 24.64 0.544456 -1.865; 220 24.64 0.544456 3.635;
221 26.915 0.1 -1.07; 222 26.915 0.1 0; 223 26.915 0.1 1.77;
224 26.915 0.1 2.84;
MEMBER INCIDENCES
1 31 1; 2 36 2; 3 37 3; 4 42 4; 5 43 5; 6 48 6; 7 67 7; 8 49 67; 9 70 8;
10 54 70; 11 55 9; 12 60 10; 13 61 11; 14 66 12; 31 20 102; 32 102 33;
33 33 106; 34 106 39; 35 39 119; 36 119 45; 37 45 128; 38 128 51; 39 51 140;
40 140 57; 41 57 144; 42 144 63; 43 63 156; 44 156 15; 45 21 103; 46 103 34;
47 34 107; 48 107 40; 49 40 120; 50 120 46; 51 46 129; 52 129 52; 53 52 141;
54 141 58; 55 58 145; 56 145 64; 57 64 157; 58 157 16; 71 19 101; 72 101 32;
73 32 105; 74 105 110; 75 110 38; 76 38 114; 77 114 118; 78 118 44; 80 44 127;
81 127 132; 82 132 50; 83 50 136; 84 136 139; 85 139 56; 86 56 143; 87 143 148;
88 148 62; 89 62 152; 90 152 155; 91 155 14; 92 22 104; 93 104 35; 94 35 108;
95 108 111; 96 111 41; 97 41 115; 98 115 121; 99 121 47; 101 47 130;
102 130 133; 103 133 53; 104 53 137; 105 137 142; 106 142 59; 107 59 146;
108 146 149; 109 149 65; 110 65 153; 111 153 158; 112 158 17; 131 31 109;
132 109 37; 133 37 113; 134 113 117; 135 117 43; 137 43 131; 138 131 49;
139 49 135; 140 135 55; 141 55 147; 142 147 61; 143 61 151; 144 151 219;
145 36 112; 146 112 42; 147 42 116; 148 116 122; 149 122 48; 151 48 134;
152 134 54; 153 54 138; 154 138 60; 155 60 150; 156 150 66; 157 66 154;
158 154 220; 171 110 109; 172 114 113; 174 132 131; 175 136 135; 176 148 147;
177 152 151; 178 112 111; 179 116 115; 181 134 133; 182 138 137; 183 150 149;
184 154 153; 191 3 117; 192 5 117; 193 4 122; 194 6 122; 196 68 67; 197 69 68;
198 70 69; 199 7 68; 200 8 69; 201 103 20; 202 103 33; 203 107 33; 204 107 39;
205 120 39; 206 120 45; 207 129 45; 208 129 51; 209 141 51; 210 141 57;
211 145 57; 212 145 63; 213 157 63; 214 157 15; 221 102 101; 222 103 102;
223 104 103; 224 106 105; 225 107 106; 226 108 107; 227 119 118; 228 120 119;
229 121 120; 230 128 127; 231 129 128; 232 130 129; 233 140 139; 234 141 140;
235 142 141; 236 144 143; 237 145 144; 238 146 145; 239 156 155; 240 157 156;
241 158 157; 251 32 31; 252 33 32; 253 34 33; 254 35 34; 255 36 35; 256 38 37;
257 39 38; 258 40 39; 259 41 40; 260 42 41; 261 44 43; 262 45 44; 263 46 45;
264 47 46; 265 48 47; 266 50 49; 267 51 50; 268 52 51; 269 53 52; 270 54 53;
271 56 55; 272 57 56; 273 58 57; 274 59 58; 275 60 59; 276 62 61; 277 63 62;
278 64 63; 279 65 64; 280 66 65; 301 201 1; 302 202 2; 303 203 3; 304 204 4;
305 205 5; 306 206 6; 307 207 7; 308 208 8; 309 209 9; 310 210 10; 311 211 11;
312 212 12; 313 213 13; 314 214 18; 321 1 2; 322 3 4; 323 5 6; 324 7 8;
325 9 10; 326 11 12; 341 13 221; 342 221 222; 343 222 223; 344 223 224;
345 224 18; 346 12 215; 347 11 216; 348 215 18; 349 216 13; 350 216 219;
351 215 220; 352 221 14; 353 222 15; 354 223 16; 355 224 17;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
```

```
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 6 11 TO 14 251 TO 280 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
7 TO 10 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
31 TO 58 353 354 TABLE ST UPN200
71 TO 78 80 TO 99 101 TO 112 352 355 TABLE ST UPN140
131 TO 135 137 TO 149 151 TO 158 350 351 TABLE ST UPN140
171 172 174 TO 179 181 TO 184 TABLE ST UPN140
196 TO 198 TABLE LD L75X75X8 SP 0.01
191 TO 194 TABLE LD L65X65X6 SP 0.01
201 TO 214 TABLE ST L65X65X6
221 TO 241 TABLE ST L50X50X6
199 200 TABLE ST L50X50X6
MEMBER PROPERTY EUROPEAN
341 TO 345 PRIS YD 0.6 ZD 0.35
301 TO 314 PRIS YD 0.6 ZD 0.6
321 TO 326 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY EUROPEAN
346 TO 349 PRIS YD 0.5 ZD 0.3
CONSTANTS
BETA 90 MEMB 1 TO 6 11 TO 14
MATERIAL STEEL MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
221 TO 241 251 TO 280 350 TO 355
MATERIAL CONCRETE MEMB 301 TO 314 321 TO 326 341 TO 349
SUPPORTS
19 TO 22 FIXED BUT FX MX MY MZ
201 TO 214 FIXED
MEMBER TRUSS
191 TO 194 199 TO 214 221 TO 241
MEMBER RELEASE
131 133 137 141 143 145 147 151 155 157 171 172 174 TO 179 181 TO 184 198 -
270 350 TO 355 START MP 0.99
1 TO 7 9 11 TO 14 38 52 82 103 132 135 138 140 142 146 149 152 154 156 171 -
172 174 TO 179 181 TO 184 196 266 END MP 0.99
39 53 83 104 139 153 START FX MY MZ
************************************************************
MEMBER OFFSET
321 TO 326 START 0 -0.25 0
321 TO 326 END 0 -0.25 0
341 TO 345 START 0 -0.3 0
341 TO 345 END 0 -0.3 0
346 TO 349 START 0 -0.2 0
346 TO 349 END 0 -0.2 0
LOAD 1 DL
SELFWEIGHT Y -1.1 LIST 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 -
131 TO 135 137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 -
196 TO 214 221 TO 241 251 TO 280 350 TO 355
```

```
SELFWEIGHT Y -1 LIST 301 TO 314 321 TO 326 341 TO 349
*WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM & HANDRAIL LOAD @15KG/M
*WALKWAY WIDTH 1000MM & BELT WIDTHG 1400MM
MEMBER LOAD
71 TO 78 80 TO 99 101 TO 112 UNI GY -0.315
31 TO 58 UNI GY -0.465
*CABLE TRAY LOAD @112KG/TRAY/M RUN * 2 NOS OF TRAY = 224KG/M RUN
*CONCENTRATED FORCE = 224*3=6.72kN & MOMENT= 6.72*0.4 = 2.69kNM
178 179 181 TO 184 255 265 275 CON GY -6.72 0.395
178 179 181 TO 184 255 265 275 CMOM GX 2.69
*LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
178 179 181 TO 184 255 265 275 CON GY -1 0.595
171 172 174 TO 177 251 261 271 CON GY -1 0.2
*PIPE DEAD LOAD
178 179 181 TO 184 255 265 275 CON GY -2.03 0.2
171 172 174 TO 177 251 261 271 CON GY -3.57 0.595
*CONVEYOR DEAD LOAD VERTICAL
44 58 CON GY -6 2.13
43 57 CON GY -6 1.41
42 56 CON GY -6 0.71
41 55 CON GY -6 0.01
39 53 CON GY -6 1.77
38 52 CON GY -6 1
37 51 CON GY -6 0.3
35 49 CON GY -6 1.64
34 48 CON GY -6 0.81
32 46 CON GY -6 1.745
31 45 CON GY -6 1.51
*CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
44 58 CON GX 1.5 2.13
43 57 CON GX 1.5 1.41
42 56 CON GX 1.5 0.71
41 55 CON GX 1.5 0.01
39 53 CON GX 1.5 1.77
38 52 CON GX 1.5 1
37 51 CON GX 1.5 0.3
35 49 CON GX 1.5 1.64
34 48 CON GX 1.5 0.81
32 46 CON GX 1.5 1.745
31 45 CON GX 1.5 1.51
*CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
44 58 CON GZ 0.8 2.13
43 57 CON GZ 0.8 1.41
42 56 CON GZ 0.8 0.71
41 55 CON GZ 0.8 0.01
39 53 CON GZ 0.8 1.77
38 52 CON GZ 0.8 1
37 51 CON GZ 0.8 0.3
35 49 CON GZ 0.8 1.64
34 48 CON GZ 0.8 0.81
32 46 CON GZ 0.8 1.745
31 45 CON GZ 0.8 1.51
************************************************************
LOAD 2 LL
```

```
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
71 TO 78 80 TO 99 101 TO 112 UNI GY -2
31 TO 58 UNI GY -2
************************************************************
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
44 58 CON GY -6.5 2.13
43 57 CON GY -6.5 1.41
42 56 CON GY -6.5 0.71
41 55 CON GY -6.5 0.01
39 53 CON GY -6.5 1.77
38 52 CON GY -6.5 1
37 51 CON GY -6.5 0.3
35 49 CON GY -6.5 1.64
34 48 CON GY -6.5 0.81
32 46 CON GY -6.5 1.745
31 45 CON GY -6.5 1.51
** Spillage Load @75kg/ Sq. m on seal plate
31 TO 58 UNI GY -0.64
************************************************************
LOAD 4 WIND LOAD IN +Z DIRN
MEMBER LOAD
44 58 CON GZ 2.58 2.13
43 57 CON GZ 2.58 1.41
42 56 CON GZ 2.58 0.71
41 55 CON GZ 2.58 0.01
39 53 CON GZ 2.58 1.77
38 52 CON GZ 2.58 1
37 51 CON GZ 2.58 0.3
35 49 CON GZ 2.58 1.64
34 48 CON GZ 2.58 0.81
32 46 CON GZ 2.58 1.745
31 45 CON GZ 2.58 1.51
*****
44 58 CMOM GX 2.66 2.13
43 57 CMOM GX 2.66 1.41
42 56 CMOM GX 2.66 0.71
41 55 CMOM GX 2.66 0.01
39 53 CMOM GX 2.66 1.77
38 52 CMOM GX 2.66 1
37 51 CMOM GX 2.66 0.3
35 49 CMOM GX 2.66 1.64
34 48 CMOM GX 2.66 0.81
32 46 CMOM GX 2.66 1.745
31 45 CMOM GX 2.66 1.51
*31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX 1.38
*TRESTLE BODY WIND
1 TO 14 UNI GZ 1.2
*WIND ON CABLE TRAY SUPPORT
178 179 181 TO 184 255 265 275 CON GZ 1.11 0.395
178 179 181 TO 184 255 265 275 CMOM GX 0.81
************************************************************
```

```
LOAD 5 WIND LOAD IN -Z DIRN
MEMBER LOAD
44 58 CON GZ -2.58 2.13
43 57 CON GZ -2.58 1.41
42 56 CON GZ -2.58 0.71
41 55 CON GZ -2.58 0.01
39 53 CON GZ -2.58 1.77
38 52 CON GZ -2.58 1
37 51 CON GZ -2.58 0.3
35 49 CON GZ -2.58 1.64
34 48 CON GZ -2.58 0.81
32 46 CON GZ -2.58 1.745
31 45 CON GZ -2.58 1.51
*****
44 58 CMOM GX -2.66 2.13
43 57 CMOM GX -2.66 1.41
42 56 CMOM GX -2.66 0.71
41 55 CMOM GX -2.66 0.01
39 53 CMOM GX -2.66 1.77
38 52 CMOM GX -2.66 1
37 51 CMOM GX -2.66 0.3
35 49 CMOM GX -2.66 1.64
34 48 CMOM GX -2.66 0.81
32 46 CMOM GX -2.66 1.745
31 45 CMOM GX -2.66 1.51
*31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX -1.38
*WIND ON TRESTLE BODY
1 TO 14 UNI GZ -1.2
*WIND ON CABLE TRAY SUPPORT
178 179 181 TO 184 255 265 275 CON GZ -1.11 0.395
178 179 181 TO 184 255 265 275 CMOM GX -0.81
************************************************************
LOAD 6 TL1
TEMPERATURE LOAD
1 TO 14 31 TO 58 71 TO 74 76 78 81 83 85 TO 87 89 91 TO 95 97 99 102 104 106 -
107 TO 108 110 112 131 133 134 139 TO 141 145 147 149 153 TO 155 191 TO 194 -
196 TO 214 221 TO 241 251 TO 280 TEMP 25
************************************************************
LOAD 7 TL2
TEMPERATURE LOAD
1 TO 14 31 TO 58 71 TO 74 76 78 81 83 85 TO 87 89 91 TO 95 97 99 102 104 106 -
107 TO 108 110 112 131 133 134 139 TO 141 145 147 149 153 TO 155 191 TO 194 -
196 TO 214 221 TO 241 251 TO 280 TEMP -25
************************************************************
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
************************************************************
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
```

```
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 6 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 7 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 6 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 7 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
************************************************************
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
*137 TO 141 145 TO 149 151 TO 156 429 TO 458
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
*137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
```

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*FU 410000 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*TRACK 0 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*BEAM 1 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
*137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*TMAIN 300 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*MAIN 200 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*NSF 0.8 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
*KZ 0.8 MEMB 201 TO 214
*KY 0.8 MEMB 201 TO 214
*UNT 1.77 MEMB 429 TO 458
*LY 1.77 MEMB 429 TO 458
*LX 1.77 MEMB 429 TO 458
*LZ 5.5 MEMB 429 TO 458
*LX 5.5 MEMB 196 TO 198
*LY 5.5 MEMB 196 TO 198
*LZ 3.5 MEMB 196 TO 198
*LZ 4.75 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
*152 TO 158
*LY 3 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
*152 TO 158
*UNT 3 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
*152 TO 158
*CHECK CODE MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
*138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
*221 TO 241 429 TO 458
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 301 TO 314 321 TO 326 341 TO 349
FYMAIN 420000 MEMB 301 TO 314 321 TO 326 341 TO 349
TRACK 0 MEMB 301 TO 314 321 TO 326 341 TO 349
CLB 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
CLT 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
CLS 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
MAXMAIN 25 MEMB 301 TO 314 321 TO 326 341 TO 349
MINMAIN 12 MEMB 301 TO 314 321 TO 326 341 TO 349
MINSEC 8 MEMB 301 TO 314 321 TO 326 341 TO 349
REINF 0 MEMB 301 TO 314
DESIGN COLUMN 301 TO 314
DESIGN BEAM 321 TO 326 341 TO 349
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  1, 2020                *
*           Time=    12:50:52                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-001\Staad\Civil\3110-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 30-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART GROUND PORTAL
    9. JOB REF CONVEYOR 3110-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 0.35 -1.865; 2 0 0.35 3.635; 3 4.3 0.35 -1.865; 4 4.3 0.35 3.635
   18. 5 8.6 0.35 -1.865; 6 8.6 0.35 3.635; 7 13.015 0.35 -1.865; 8 13.015 0.35 3.635
   19. 9 17.765 0.35 -1.865; 10 17.765 0.35 3.635; 11 22.365 0.1 -1.865
   20. 12 22.365 0.1 3.635; 13 26.915 0.1 -1.865; 14 26.915 0.35 -1.07
   21. 15 26.915 0.35 0; 16 26.915 0.35 1.77; 17 26.915 0.35 2.84
   22. 18 26.915 0.1 3.635; 19 -3.5 3.7 -1.07; 20 -3.5 3.7 0; 21 -3.5 3.7 1.77
   23. 22 -3.5 3.7 2.84; 31 0 3.215 -1.865; 32 0 3.215 -1.07; 33 0 3.215 0
   24. 34 0 3.215 1.77; 35 0 3.215 2.84; 36 0 3.215 3.635; 37 4.3 2.611 -1.865
   25. 38 4.3 2.611 -1.07; 39 4.3 2.611 0; 40 4.3 2.611 1.77; 41 4.3 2.611 2.84
   26. 42 4.3 2.611 3.635; 43 8.6 2.006 -1.865; 44 8.6 2.006 -1.07; 45 8.6 2.006 0
   27. 46 8.6 2.006 1.77; 47 8.6 2.006 2.84; 48 8.6 2.006 3.635
   28. 49 13.015 1.386 -1.865; 50 13.015 1.386 -1.07; 51 13.015 1.386 0
   29. 52 13.015 1.386 1.77; 53 13.015 1.386 2.84; 54 13.015 1.386 3.635
   30. 55 17.765 1.035 -1.865; 56 17.765 1.035 -1.07; 57 17.765 1.035 0
   31. 58 17.765 1.035 1.77; 59 17.765 1.035 2.84; 60 17.765 1.035 3.635
   32. 61 22.365 0.715 -1.865; 62 22.365 0.715 -1.07; 63 22.365 0.715 0
   33. 64 22.365 0.715 1.77; 65 22.365 0.715 2.84; 66 22.365 0.715 3.635
   34. 67 13.015 1.261 -1.865; 68 13.015 1.261 -0.865; 69 13.015 1.261 2.635
   35. 70 13.015 1.261 3.635; 101 -1.75 3.4575 -1.07; 102 -1.75 3.4575 0
   36. 103 -1.75 3.4575 1.77; 104 -1.75 3.4575 2.84; 105 2.15 2.913 -1.07
   37. 106 2.15 2.913 0; 107 2.15 2.913 1.77; 108 2.15 2.913 2.84
   38. 109 2.65395 2.84221 -1.865; 110 2.65702 2.84178 -1.07
```

39. 111 2.65702 2.84178 2.84; 112 2.65395 2.84221 3.635; 113 5.62693 2.4243 -1.865
40. 114 5.62693 2.4243 -1.07; 115 5.62693 2.4243 2.84; 116 5.62693 2.4243 3.635
41. 117 6.45 2.3085 -1.865; 118 6.45 2.3085 -1.07; 119 6.45 2.3085 0
42. 120 6.45 2.3085 1.77; 121 6.45 2.3085 2.84; 122 6.45 2.3085 3.635
43. 127 10.8075 1.696 -1.07; 128 10.8075 1.696 0; 129 10.8075 1.696 1.77
44. 130 10.8075 1.696 2.84; 131 11.7879 1.55832 -1.865; 132 11.7879 1.55832 -1.07
45. 133 11.7879 1.55832 2.84; 134 11.7879 1.55832 3.635; 135 14.7722 1.2563 -1.865
46. 136 14.7722 1.2563 -1.07; 137 14.7722 1.2563 2.84; 138 14.7722 1.2563 3.635
47. 139 15.39 1.2105 -1.07; 140 15.39 1.2105 0; 141 15.39 1.2105 1.77
48. 142 15.39 1.2105 2.84; 143 20.065 0.875 -1.07; 144 20.065 0.875 0
49. 145 20.065 0.875 1.77; 146 20.065 0.875 2.84; 147 20.9229 0.815318 -1.865
50. 148 20.9229 0.815318 -1.07; 149 20.9229 0.815318 2.84
51. 150 20.9229 0.815318 3.635; 151 23.9212 0.605896 -1.865
52. 152 23.9212 0.605896 -1.07; 153 23.9212 0.605896 2.84
53. 154 23.9212 0.605896 3.635; 155 24.64 0.5555 -1.07; 156 24.64 0.5555 0
54. 157 24.64 0.5555 1.77; 158 24.64 0.5555 2.84; 201 0 -1 -1.865; 202 0 -1 3.635
55. 203 4.3 -1 -1.865; 204 4.3 -1 3.635; 205 8.6 -1 -1.865; 206 8.6 -1 3.635
56. 207 13.015 -1 -1.865; 208 13.015 -1 3.635; 209 17.765 -1 -1.865
57. 210 17.765 -1 3.635; 211 22.365 -1 -1.865; 212 22.365 -1 3.635
58. 213 26.915 -1 -1.865; 214 26.915 -1 3.635; 215 24.64 0.1 3.635
59. 216 24.64 0.1 -1.865; 219 24.64 0.544456 -1.865; 220 24.64 0.544456 3.635
60. 221 26.915 0.1 -1.07; 222 26.915 0.1 0; 223 26.915 0.1 1.77
61. 224 26.915 0.1 2.84
62. MEMBER INCIDENCES
63. 1 31 1; 2 36 2; 3 37 3; 4 42 4; 5 43 5; 6 48 6; 7 67 7; 8 49 67; 9 70 8
64. 10 54 70; 11 55 9; 12 60 10; 13 61 11; 14 66 12; 31 20 102; 32 102 33
65. 33 33 106; 34 106 39; 35 39 119; 36 119 45; 37 45 128; 38 128 51; 39 51 140
66. 40 140 57; 41 57 144; 42 144 63; 43 63 156; 44 156 15; 45 21 103; 46 103 34
67. 47 34 107; 48 107 40; 49 40 120; 50 120 46; 51 46 129; 52 129 52; 53 52 141
68. 54 141 58; 55 58 145; 56 145 64; 57 64 157; 58 157 16; 71 19 101; 72 101 32
69. 73 32 105; 74 105 110; 75 110 38; 76 38 114; 77 114 118; 78 118 44; 80 44 127
70. 81 127 132; 82 132 50; 83 50 136; 84 136 139; 85 139 56; 86 56 143; 87 143 148
71. 88 148 62; 89 62 152; 90 152 155; 91 155 14; 92 22 104; 93 104 35; 94 35 108
72. 95 108 111; 96 111 41; 97 41 115; 98 115 121; 99 121 47; 101 47 130
73. 102 130 133; 103 133 53; 104 53 137; 105 137 142; 106 142 59; 107 59 146
74. 108 146 149; 109 149 65; 110 65 153; 111 153 158; 112 158 17; 131 31 109
75. 132 109 37; 133 37 113; 134 113 117; 135 117 43; 137 43 131; 138 131 49
76. 139 49 135; 140 135 55; 141 55 147; 142 147 61; 143 61 151; 144 151 219
77. 145 36 112; 146 112 42; 147 42 116; 148 116 122; 149 122 48; 151 48 134
78. 152 134 54; 153 54 138; 154 138 60; 155 60 150; 156 150 66; 157 66 154
79. 158 154 220; 171 110 109; 172 114 113; 174 132 131; 175 136 135; 176 148 147
80. 177 152 151; 178 112 111; 179 116 115; 181 134 133; 182 138 137; 183 150 149
81. 184 154 153; 191 3 117; 192 5 117; 193 4 122; 194 6 122; 196 68 67; 197 69 68
82. 198 70 69; 199 7 68; 200 8 69; 201 103 20; 202 103 33; 203 107 33; 204 107 39
83. 205 120 39; 206 120 45; 207 129 45; 208 129 51; 209 141 51; 210 141 57
84. 211 145 57; 212 145 63; 213 157 63; 214 157 15; 221 102 101; 222 103 102
85. 223 104 103; 224 106 105; 225 107 106; 226 108 107; 227 119 118; 228 120 119
86. 229 121 120; 230 128 127; 231 129 128; 232 130 129; 233 140 139; 234 141 140
87. 235 142 141; 236 144 143; 237 145 144; 238 146 145; 239 156 155; 240 157 156
88. 241 158 157; 251 32 31; 252 33 32; 253 34 33; 254 35 34; 255 36 35; 256 38 37
89. 257 39 38; 258 40 39; 259 41 40; 260 42 41; 261 44 43; 262 45 44; 263 46 45
90. 264 47 46; 265 48 47; 266 50 49; 267 51 50; 268 52 51; 269 53 52; 270 54 53
91. 271 56 55; 272 57 56; 273 58 57; 274 59 58; 275 60 59; 276 62 61; 277 63 62
92. 278 64 63; 279 65 64; 280 66 65; 301 201 1; 302 202 2; 303 203 3; 304 204 4
93. 305 205 5; 306 206 6; 307 207 7; 308 208 8; 309 209 9; 310 210 10; 311 211 11
94. 312 212 12; 313 213 13; 314 214 18; 321 1 2; 322 3 4; 323 5 6; 324 7 8

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 95. 325 9 10; 326 11 12; 341 13 221; 342 221 222; 343 222 223; 344 223 224
 96. 345 224 18; 346 12 215; 347 11 216; 348 215 18; 349 216 13; 350 216 219
 97. 351 215 220; 352 221 14; 353 222 15; 354 223 16; 355 224 17
 98. DEFINE MATERIAL START
 99. ISOTROPIC STEEL
100. E 2.05E+008
101. POISSON 0.3
102. DENSITY 76.8195
103. ALPHA 1.2E-005
104. DAMP 0.03
105. TYPE STEEL
106. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
107. ISOTROPIC CONCRETE
108. E 2.17185E+007
109. POISSON 0.17
110. DENSITY 23.5616
111. ALPHA 1E-005
112. DAMP 0.05
113. TYPE CONCRETE
114. STRENGTH FCU 27579
115. END DEFINE MATERIAL
116. MEMBER PROPERTY EUROPEAN
117. 1 TO 6 11 TO 14 251 TO 280 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
118. 7 TO 10 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
119. 31 TO 58 353 354 TABLE ST UPN200
120. 71 TO 78 80 TO 99 101 TO 112 352 355 TABLE ST UPN140
121. 131 TO 135 137 TO 149 151 TO 158 350 351 TABLE ST UPN140
122. 171 172 174 TO 179 181 TO 184 TABLE ST UPN140
123. 196 TO 198 TABLE LD L75X75X8 SP 0.01
124. 191 TO 194 TABLE LD L65X65X6 SP 0.01
125. 201 TO 214 TABLE ST L65X65X6
126. 221 TO 241 TABLE ST L50X50X6
127. 199 200 TABLE ST L50X50X6
128. MEMBER PROPERTY EUROPEAN
129. 341 TO 345 PRIS YD 0.6 ZD 0.35
130. 301 TO 314 PRIS YD 0.6 ZD 0.6
131. 321 TO 326 PRIS YD 0.5 ZD 0.3
132. MEMBER PROPERTY EUROPEAN
133. 346 TO 349 PRIS YD 0.5 ZD 0.3
134. CONSTANTS
135. BETA 90 MEMB 1 TO 6 11 TO 14
136. MATERIAL STEEL MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
137. 137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
138. 221 TO 241 251 TO 280 350 TO 355
139. MATERIAL CONCRETE MEMB 301 TO 314 321 TO 326 341 TO 349
140. SUPPORTS
141. 19 TO 22 FIXED BUT FX MX MY MZ
142. 201 TO 214 FIXED
143. MEMBER TRUSS
144. 191 TO 194 199 TO 214 221 TO 241
145. MEMBER RELEASE
146. 131 133 137 141 143 145 147 151 155 157 171 172 174 TO 179 181 TO 184 198 -
147. 270 350 TO 355 START MP 0.99
148. 1 TO 7 9 11 TO 14 38 52 82 103 132 135 138 140 142 146 149 152 154 156 171 -
149. 172 174 TO 179 181 TO 184 196 266 END MP 0.99
150. 39 53 83 104 139 153 START FX MY MZ
```

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151. ************************************************************
152. MEMBER OFFSET
153. 321 TO 326 START 0 -0.25 0
154. 321 TO 326 END 0 -0.25 0
155. 341 TO 345 START 0 -0.3 0
156. 341 TO 345 END 0 -0.3 0
157. 346 TO 349 START 0 -0.2 0
158. 346 TO 349 END 0 -0.2 0
159. LOAD 1 DL
160. SELFWEIGHT Y -1.1 LIST 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 -
161. 131 TO 135 137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 -
162. 196 TO 214 221 TO 241 251 TO 280 350 TO 355
163. SELFWEIGHT Y -1 LIST 301 TO 314 321 TO 326 341 TO 349
164. *WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM _HANDRAIL LOAD @1
165. *WALKWAY WIDTH 1000MM _BELT WIDTHG 1400MM
166. MEMBER LOAD
167. 71 TO 78 80 TO 99 101 TO 112 UNI GY -0.315
168. 31 TO 58 UNI GY -0.465
169. *CABLE TRAY LOAD @112KG/TRAY/M RUN * 2 NOS OF TRAY = 224KG/M RUN
170. *CONCENTRATED FORCE = 224*3=6.72KN _MOMENT= 6.72*0.4 = 2.69KNM
171. 178 179 181 TO 184 255 265 275 CON GY -6.72 0.395
172. 178 179 181 TO 184 255 265 275 CMOM GX 2.69
173. *LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
174. 178 179 181 TO 184 255 265 275 CON GY -1 0.595
175. 171 172 174 TO 177 251 261 271 CON GY -1 0.2
176. *PIPE DEAD LOAD
177. 178 179 181 TO 184 255 265 275 CON GY -2.03 0.2
178. 171 172 174 TO 177 251 261 271 CON GY -3.57 0.595
179. *CONVEYOR DEAD LOAD VERTICAL
180. 44 58 CON GY -6 2.13
181. 43 57 CON GY -6 1.41
182. 42 56 CON GY -6 0.71
183. 41 55 CON GY -6 0.01
184. 39 53 CON GY -6 1.77
185. 38 52 CON GY -6 1
186. 37 51 CON GY -6 0.3
187. 35 49 CON GY -6 1.64
188. 34 48 CON GY -6 0.81
189. 32 46 CON GY -6 1.745
190. 31 45 CON GY -6 1.51
191. *CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
192. 44 58 CON GX 1.5 2.13
193. 43 57 CON GX 1.5 1.41
194. 42 56 CON GX 1.5 0.71
195. 41 55 CON GX 1.5 0.01
196. 39 53 CON GX 1.5 1.77
197. 38 52 CON GX 1.5 1
198. 37 51 CON GX 1.5 0.3
199. 35 49 CON GX 1.5 1.64
200. 34 48 CON GX 1.5 0.81
201. 32 46 CON GX 1.5 1.745
202. 31 45 CON GX 1.5 1.51
203. *CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
204. 44 58 CON GZ 0.8 2.13
205. 43 57 CON GZ 0.8 1.41
206. 42 56 CON GZ 0.8 0.71
```

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207. 41 55 CON GZ 0.8 0.01
208. 39 53 CON GZ 0.8 1.77
209. 38 52 CON GZ 0.8 1
210. 37 51 CON GZ 0.8 0.3
211. 35 49 CON GZ 0.8 1.64
212. 34 48 CON GZ 0.8 0.81
213. 32 46 CON GZ 0.8 1.745
214. 31 45 CON GZ 0.8 1.51
215. ************************************************************
216. LOAD 2 LL
217. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
218. MEMBER LOAD
219. 71 TO 78 80 TO 99 101 TO 112 UNI GY -2
220. 31 TO 58 UNI GY -2
221. ************************************************************
222. LOAD 3 ML
223. *CONVEYOR MATERIAL LOAD
224. MEMBER LOAD
225. 44 58 CON GY -6.5 2.13
226. 43 57 CON GY -6.5 1.41
227. 42 56 CON GY -6.5 0.71
228. 41 55 CON GY -6.5 0.01
229. 39 53 CON GY -6.5 1.77
230. 38 52 CON GY -6.5 1
231. 37 51 CON GY -6.5 0.3
232. 35 49 CON GY -6.5 1.64
233. 34 48 CON GY -6.5 0.81
234. 32 46 CON GY -6.5 1.745
235. 31 45 CON GY -6.5 1.51
236. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
237. 31 TO 58 UNI GY -0.64
238. ************************************************************
239. LOAD 4 WIND LOAD IN +Z DIRN
240. MEMBER LOAD
241. 44 58 CON GZ 2.58 2.13
242. 43 57 CON GZ 2.58 1.41
243. 42 56 CON GZ 2.58 0.71
244. 41 55 CON GZ 2.58 0.01
245. 39 53 CON GZ 2.58 1.77
246. 38 52 CON GZ 2.58 1
247. 37 51 CON GZ 2.58 0.3
248. 35 49 CON GZ 2.58 1.64
249. 34 48 CON GZ 2.58 0.81
250. 32 46 CON GZ 2.58 1.745
251. 31 45 CON GZ 2.58 1.51
252. *****
253. 44 58 CMOM GX 2.66 2.13
254. 43 57 CMOM GX 2.66 1.41
255. 42 56 CMOM GX 2.66 0.71
256. 41 55 CMOM GX 2.66 0.01
257. 39 53 CMOM GX 2.66 1.77
258. 38 52 CMOM GX 2.66 1
259. 37 51 CMOM GX 2.66 0.3
260. 35 49 CMOM GX 2.66 1.64
261. 34 48 CMOM GX 2.66 0.81
262. 32 46 CMOM GX 2.66 1.745
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263. 31 45 CMOM GX 2.66 1.51
264. *31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX 1.38
265. *TRESTLE BODY WIND
266. 1 TO 14 UNI GZ 1.2
267. *WIND ON CABLE TRAY SUPPORT
268. 178 179 181 TO 184 255 265 275 CON GZ 1.11 0.395
269. 178 179 181 TO 184 255 265 275 CMOM GX 0.81
270. ************************************************************
271. LOAD 5 WIND LOAD IN -Z DIRN
272. MEMBER LOAD
273. 44 58 CON GZ -2.58 2.13
274. 43 57 CON GZ -2.58 1.41
275. 42 56 CON GZ -2.58 0.71
276. 41 55 CON GZ -2.58 0.01
277. 39 53 CON GZ -2.58 1.77
278. 38 52 CON GZ -2.58 1
279. 37 51 CON GZ -2.58 0.3
280. 35 49 CON GZ -2.58 1.64
281. 34 48 CON GZ -2.58 0.81
282. 32 46 CON GZ -2.58 1.745
283. 31 45 CON GZ -2.58 1.51
284. *****
285. 44 58 CMOM GX -2.66 2.13
286. 43 57 CMOM GX -2.66 1.41
287. 42 56 CMOM GX -2.66 0.71
288. 41 55 CMOM GX -2.66 0.01
289. 39 53 CMOM GX -2.66 1.77
290. 38 52 CMOM GX -2.66 1
291. 37 51 CMOM GX -2.66 0.3
292. 35 49 CMOM GX -2.66 1.64
293. 34 48 CMOM GX -2.66 0.81
294. 32 46 CMOM GX -2.66 1.745
295. 31 45 CMOM GX -2.66 1.51
296. *31 32 34 35 37 TO 39 41 TO 46 48 49 51 TO 53 55 TO 58 CMOM GX -1.38
297. *WIND ON TRESTLE BODY
298. 1 TO 14 UNI GZ -1.2
299. *WIND ON CABLE TRAY SUPPORT
300. 178 179 181 TO 184 255 265 275 CON GZ -1.11 0.395
301. 178 179 181 TO 184 255 265 275 CMOM GX -0.81
302. ************************************************************
303. LOAD 6 TL1
304. TEMPERATURE LOAD
305. 1 TO 14 31 TO 58 71 TO 74 76 78 81 83 85 TO 87 89 91 TO 95 97 99 102 104 106 -
306. 107 TO 108 110 112 131 133 134 139 TO 141 145 147 149 153 TO 155 191 TO 194 -
307. 196 TO 214 221 TO 241 251 TO 280 TEMP 25
308. ************************************************************
309. LOAD 7 TL2
310. TEMPERATURE LOAD
311. 1 TO 14 31 TO 58 71 TO 74 76 78 81 83 85 TO 87 89 91 TO 95 97 99 102 104 106 -
312. 107 TO 108 110 112 131 133 134 139 TO 141 145 147 149 153 TO 155 191 TO 194 -
313. 196 TO 214 221 TO 241 251 TO 280 TEMP -25
314. ************************************************************
315. LOAD 11 G
316. REPEAT LOAD
317. 1 1.0
318. LOAD 12 G + QN + MN
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319. REPEAT LOAD
320. 1 1.0 2 1.0 3 1.0
321. ************************************************************
322. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
323. *LOAD COMB 201 1.4 DL
324. *1 1.4
325. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
326. *1 1.2 6 1.2 2 1.6 3 1.6
327. *LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
328. *1 1.2 7 1.2 2 1.6 3 1.6
329. *LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
330. *1 1.2 4 1.6 2 0.5 3 0.5
331. *LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
332. *1 1.2 5 1.6 2 0.5 3 0.5
333. *LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
334. *1 0.9 4 1.6
335. *LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
336. *1 0.9 5 1.6
337. ************************************************************
338. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
339. LOAD COMB 301 1.0 DL
340. 1 1.0
341. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
342. 1 1.0 6 1.0 2 1.0 3 1.0
343. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
344. 1 1.0 7 1.0 2 1.0 3 1.0
345. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
346. 1 1.0 4 0.75 2 0.75 3 0.75
347. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
348. 1 1.0 5 0.75 2 0.75 3 0.75
349. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
350. 1 0.6 4 1.0
351. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
352. 1 0.6 5 1.0
353. ************************************************************
354. ****LOAD COMBINATION FOR FOUNDATION DESIGN
355. LOAD COMB 401 1.4 DL
356. 1 1.4
357. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
358. 1 1.4 6 1.4 2 1.7 3 1.7
359. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
360. 1 1.4 7 1.4 2 1.7 3 1.7
361. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
362. 1 1.2 4 1.6 2 0.5 3 0.5
363. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
364. 1 1.2 5 1.6 2 0.5 3 0.5
365. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
366. 1 0.9 4 1.6
367. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
368. 1 0.9 5 1.6
369. ************************************************************
370. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**138**	**NUMBER OF MEMBERS**	**229**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**18**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 122/ 13/ 78 DOF
TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 736
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 58 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.1/ 50221.4 MB

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371. *STEEL TAKE OFF LIST 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
372. *137 TO 141 145 TO 149 151 TO 156 429 TO 458
373. *LOAD LIST 201 TO 207
374. *PARAMETER 1
375. *CODE LRFD
376. *FYLD 275000 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
377. *137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
378. *221 TO 241 429 TO 458
379. *FU 410000 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
380. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
381. *221 TO 241 429 TO 458
382. *TRACK 0 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
383. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
384. *221 TO 241 429 TO 458
385. *BEAM 1 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 -
386. *137 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
387. *221 TO 241 429 TO 458
388. *TMAIN 300 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
389. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
390. *221 TO 241 429 TO 458
391. *MAIN 200 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
392. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
393. *221 TO 241 429 TO 458
394. *NSF 0.8 MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
395. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
396. *221 TO 241 429 TO 458
397. *KZ 0.8 MEMB 201 TO 214
398. *KY 0.8 MEMB 201 TO 214
399. *UNT 1.77 MEMB 429 TO 458
400. *LY 1.77 MEMB 429 TO 458
401. *LX 1.77 MEMB 429 TO 458
402. *LZ 5.5 MEMB 429 TO 458
403. *LX 5.5 MEMB 196 TO 198
404. *LY 5.5 MEMB 196 TO 198
405. *LZ 3.5 MEMB 196 TO 198
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406. *LZ 4.75 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
407. *152 TO 158
408. *LY 3 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
409. *152 TO 158
410. *UNT 3 MEMB 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 TO 149 151 -
411. *152 TO 158
412. *CHECK CODE MEMB 1 TO 14 31 TO 58 71 TO 78 80 TO 99 101 TO 112 131 TO 135 137 -
413. *138 TO 149 151 TO 158 171 172 174 TO 179 181 TO 184 191 TO 194 196 TO 214 -
414. *221 TO 241 429 TO 458
415. LOAD LIST 401 TO 407
416. START CONCRETE DESIGN
417. CODE ACI 2008
418. FC 35000 MEMB 301 TO 314 321 TO 326 341 TO 349
419. FYMAIN 420000 MEMB 301 TO 314 321 TO 326 341 TO 349
420. TRACK 0 MEMB 301 TO 314 321 TO 326 341 TO 349
421. CLB 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
422. CLT 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
423. CLS 0.05 MEMB 301 TO 314 321 TO 326 341 TO 349
424. MAXMAIN 25 MEMB 301 TO 314 321 TO 326 341 TO 349
425. MINMAIN 12 MEMB 301 TO 314 321 TO 326 341 TO 349
426. MINSEC 8 MEMB 301 TO 314 321 TO 326 341 TO 349
427. REINF 0 MEMB 301 TO 314
428. DESIGN COLUMN 301 TO 314
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COLUMN NO. 301 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 302 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 303 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 304 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 305 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 306 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 307 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 308 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 309 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 310 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 311 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 312 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 313 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

==========

COLUMN NO. 314 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

429. DESIGN BEAM 321 TO 326 341 TO 349

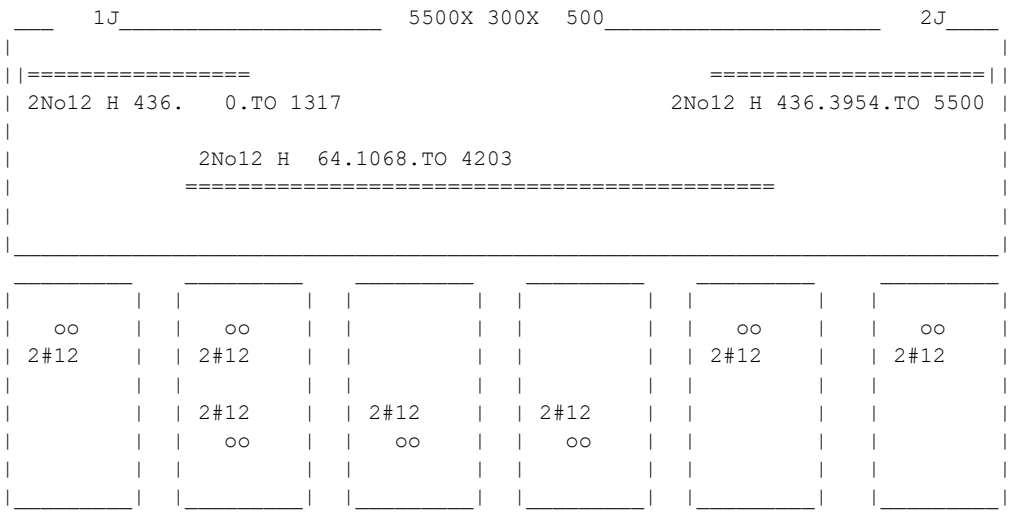
BEAM NO. 321 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	1068.	4203.	NO	NO
2	436.	2 - 12MM	0.	1317.	YES	NO
3	436.	2 - 12MM	3954.	5500.	NO	YES

B E A M N O. 321 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.43 KNS Vc= 122.92 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.49 KNS Vc= 122.90 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



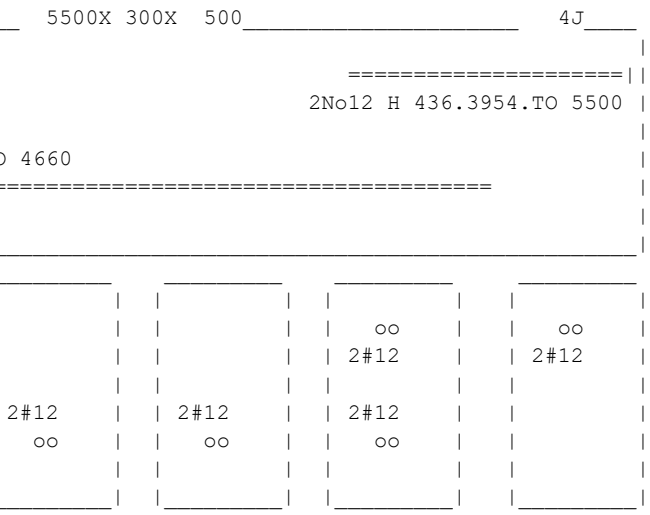
BEAM NO. 322 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	610.	4660.	NO	NO
2	436.	2 - 12MM	0.	1317.	YES	NO
3	436.	2 - 12MM	3954.	5500.	NO	YES

B E A M N O. 322 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.40 KNS Vc= 122.88 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.51 KNS Vc= 122.86 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



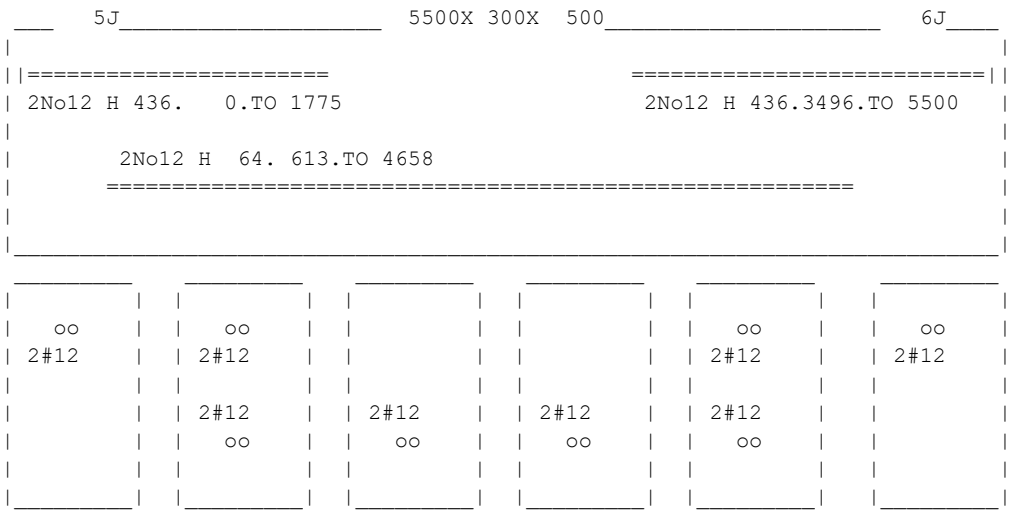
BEAM NO. 323 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	613.	4658.	NO	NO
2	436.	2 - 12MM	0.	1775.	YES	NO
3	436.	2 - 12MM	3496.	5500.	NO	YES

B E A M N O. 323 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.46 KNS Vc= 127.13 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.46 KNS Vc= 127.13 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



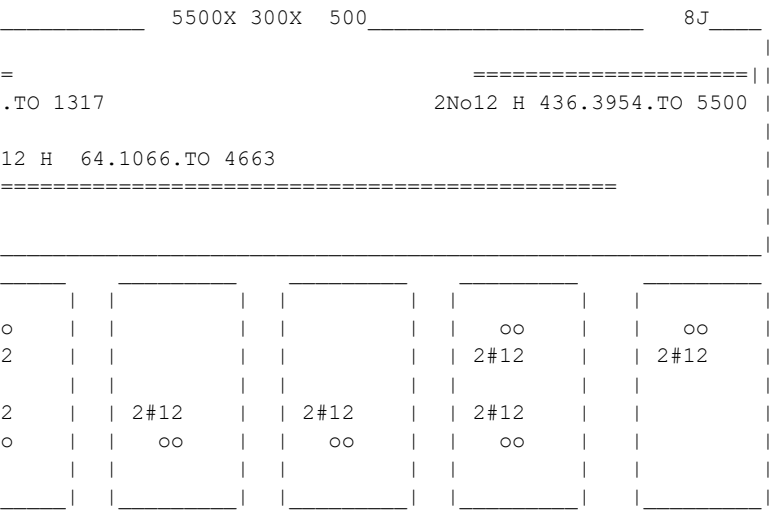
BEAM NO. 324 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	1066.	4663.	NO	NO
2	436.	2 - 12MM	0.	1317.	YES	NO
3	436.	2 - 12MM	3954.	5500.	NO	YES

B E A M N O. 324 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.56 KNS Vc= 123.12 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.36 KNS Vc= 123.20 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



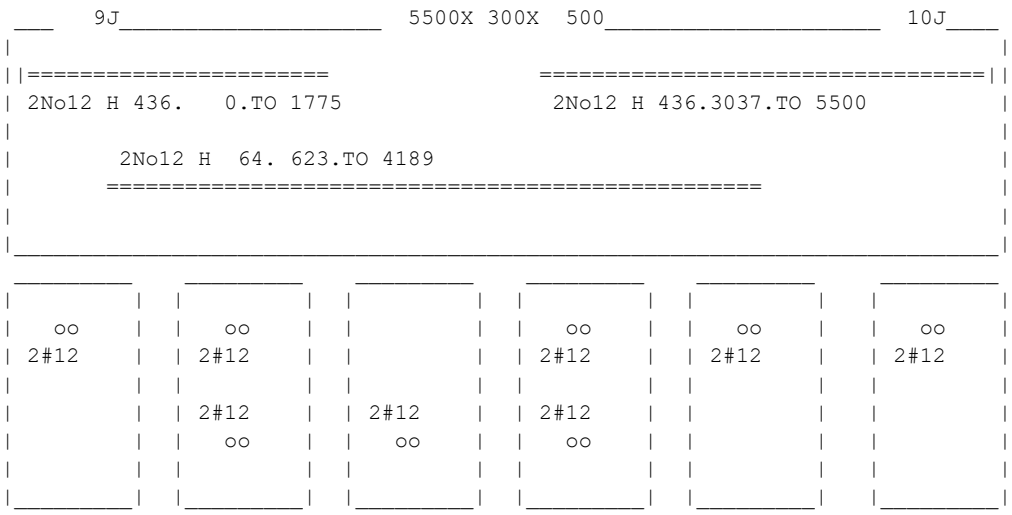
BEAM NO. 325 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	623.	4189.	NO	NO
2	436.	2 - 12MM	0.	1775.	YES	NO
3	436.	2 - 12MM	3037.	5500.	NO	YES

B E A M N O. 325 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.88 KNS Vc= 121.28 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.04 KNS Vc= 121.28 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



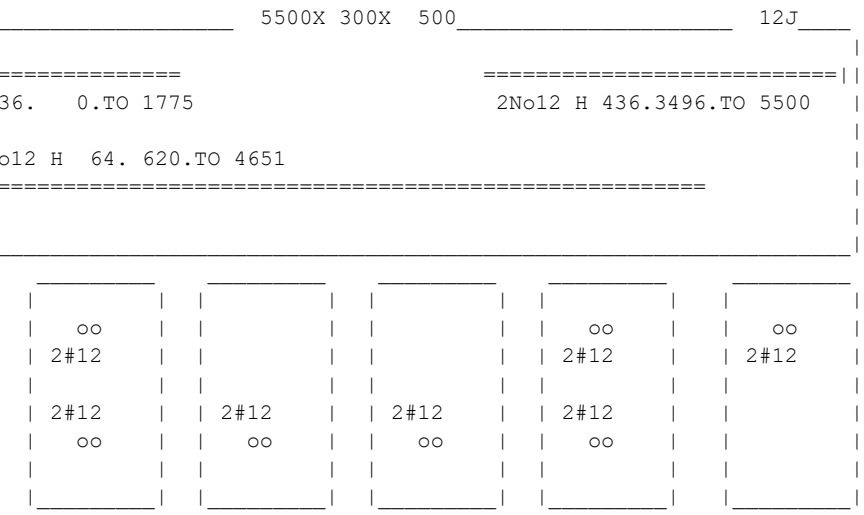
BEAM NO. 326 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	620.	4651.	NO	NO
2	436.	2 - 12MM	0.	1775.	YES	NO
3	436.	2 - 12MM	3496.	5500.	NO	YES

B E A M N O. 326 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.41 KNS Vc= 123.72 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.51 KNS Vc= 123.72 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 341 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

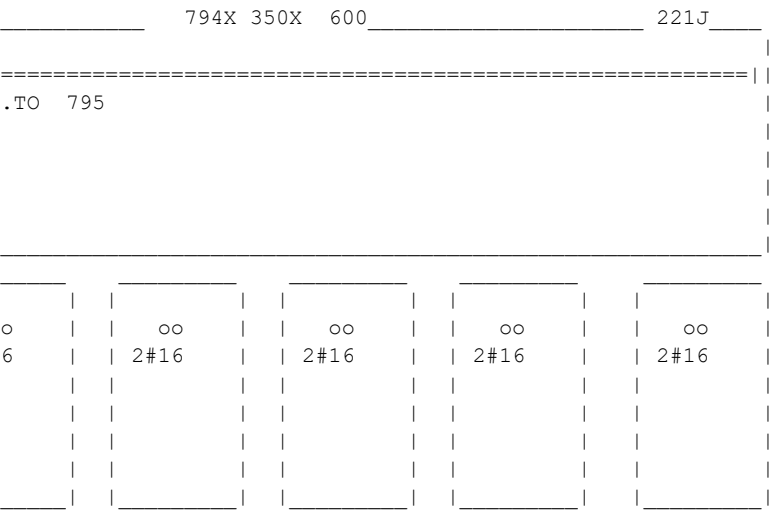
LEN - 795. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	2 - 16MM	0.	795.	YES	YES

B E A M N O. 341 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 341 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 341 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



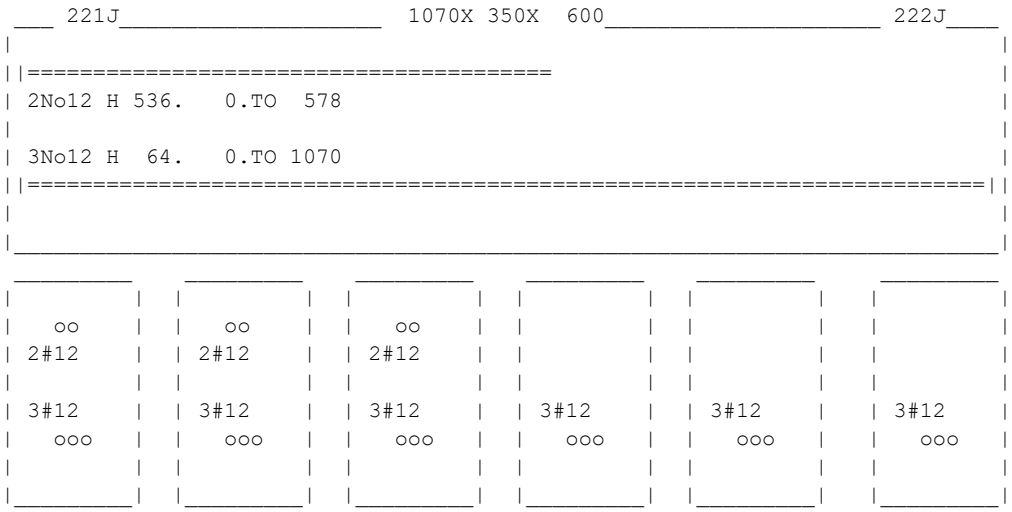
BEAM NO. 342 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1070.	YES	YES
2	536.	2 - 12MM	0.	578.	YES	NO

B E A M N O. 342 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.32 KNS Vc= 187.59 KNS Vs= 0.00 KNS
Tu= 3.79 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 23.31 KNS Vc= 187.45 KNS Vs= 0.00 KNS
Tu= 3.79 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



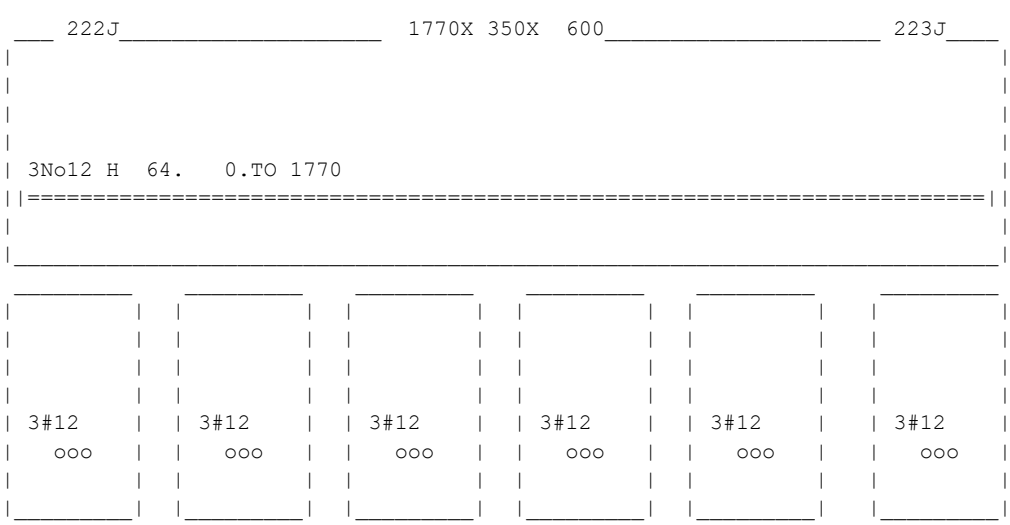
BEAM NO. 343 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1770. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1770.	YES	YES

B E A M N O. 343 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.48 KNS Vc= 175.45 KNS Vs= 0.00 KNS
Tu= 0.36 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.38 KNS Vc= 175.44 KNS Vs= 0.00 KNS
Tu= 0.36 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



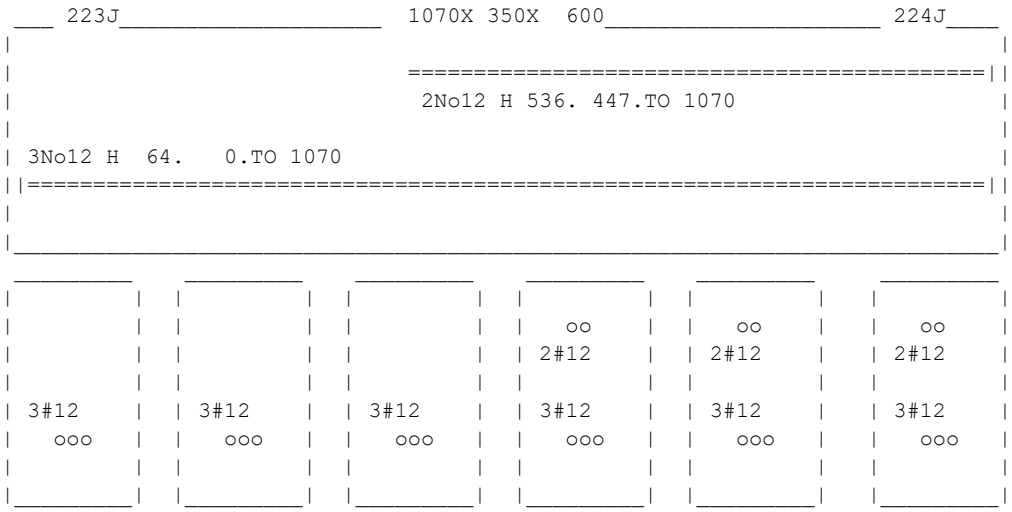
BEAM NO. 344 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	1070.	YES	YES
2	536.	2 - 12MM	447.	1070.	NO	YES

B E A M N O. 344 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.79 KNS Vc= 187.82 KNS Vs= 0.00 KNS
Tu= 4.42 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 23.80 KNS Vc= 187.96 KNS Vs= 0.00 KNS
Tu= 4.42 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 345 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

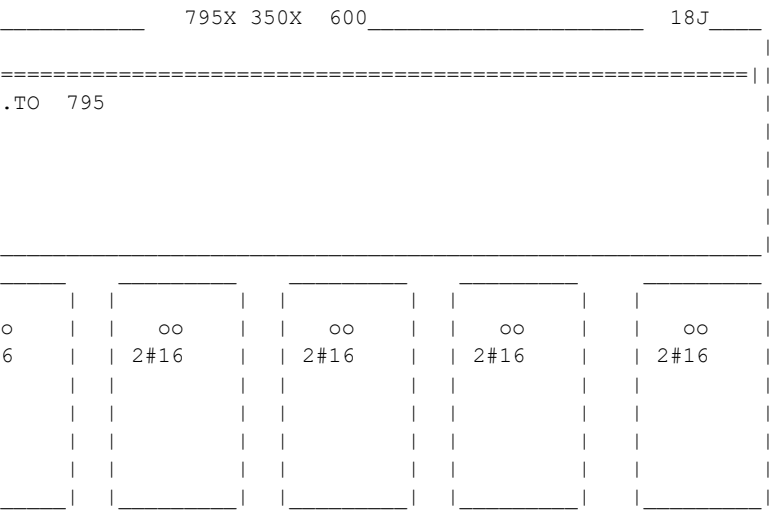
LEN - 795. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	2 - 16MM	0.	795.	YES	YES

B E A M N O. 345 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 345 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 345 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



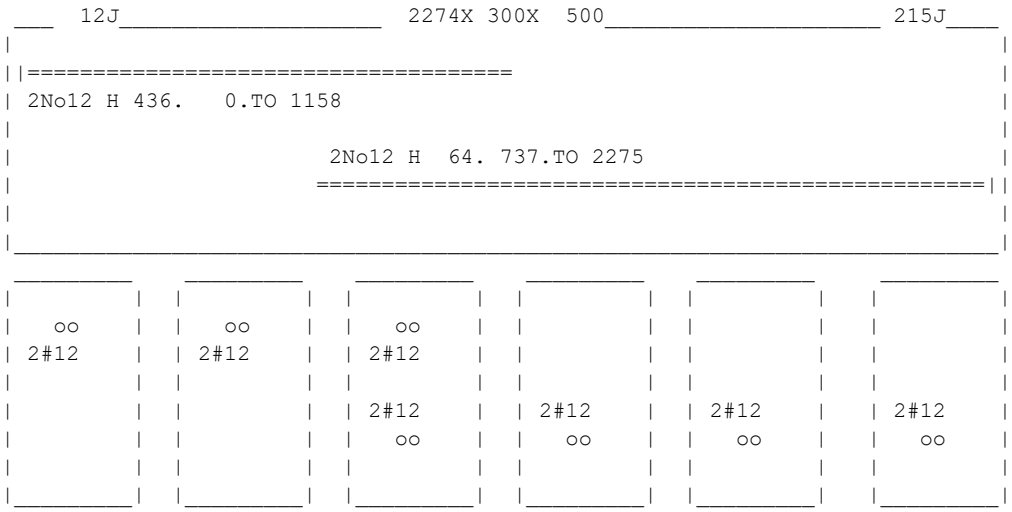
BEAM NO. 346 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2275. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	737.	2275.	NO	YES
2	436.	2 - 12MM	0.	1158.	YES	NO

B E A M N O. 346 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.03 KNS Vc= 123.79 KNS Vs= 0.00 KNS
Tu= 0.69 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.06 KNS Vc= 122.58 KNS Vs= 0.00 KNS
Tu= 0.69 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



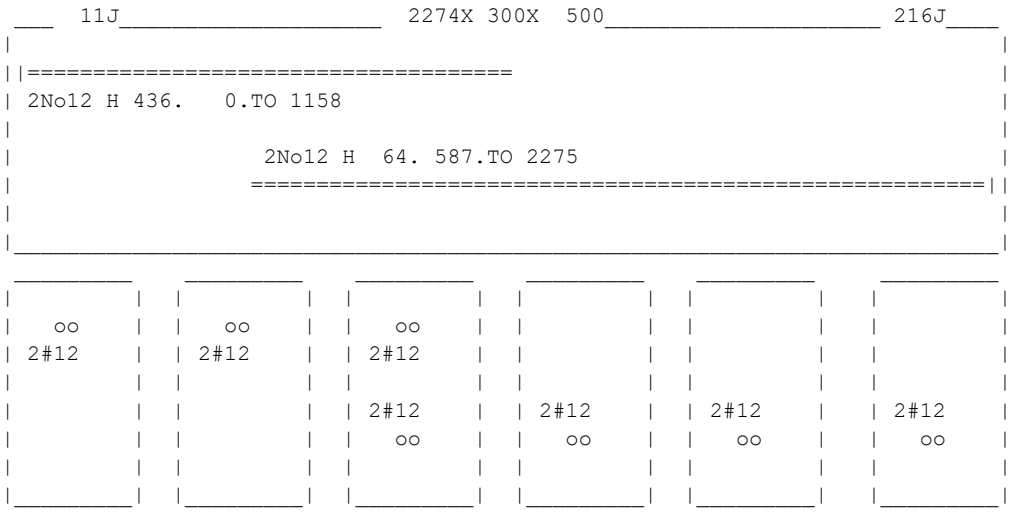
BEAM NO. 347 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2275. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	587.	2275.	NO	YES
2	436.	2 - 12MM	0.	1158.	YES	NO

B E A M N O. 347 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.22 KNS Vc= 123.37 KNS Vs= 0.00 KNS
Tu= 0.67 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.26 KNS Vc= 122.33 KNS Vs= 0.00 KNS
Tu= 0.67 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



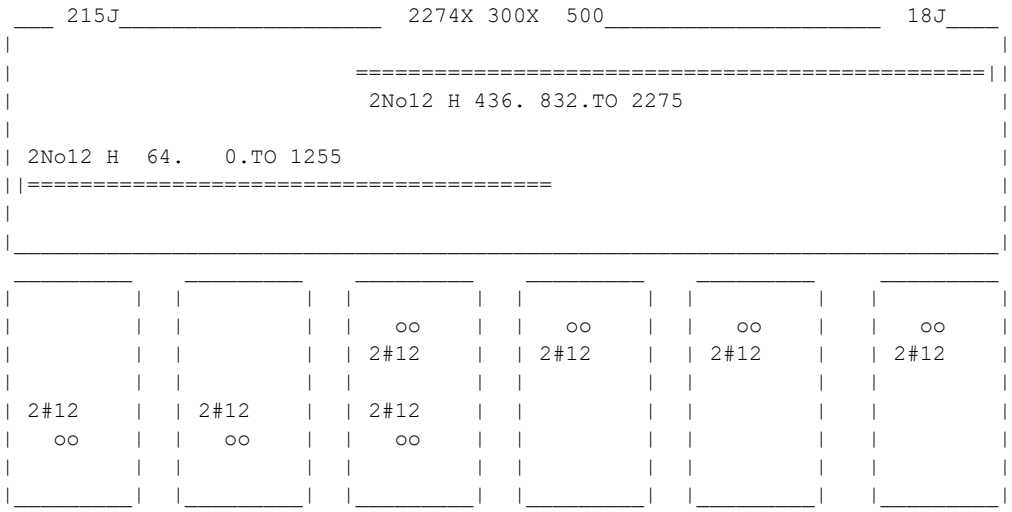
BEAM NO. 348 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2275. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1255.	YES	NO
2	436.	2 - 12MM	832.	2275.	NO	YES

B E A M N O. 348 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.47 KNS Vc= 122.77 KNS Vs= 0.00 KNS
Tu= 0.70 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.44 KNS Vc= 123.47 KNS Vs= 0.00 KNS
Tu= 0.70 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



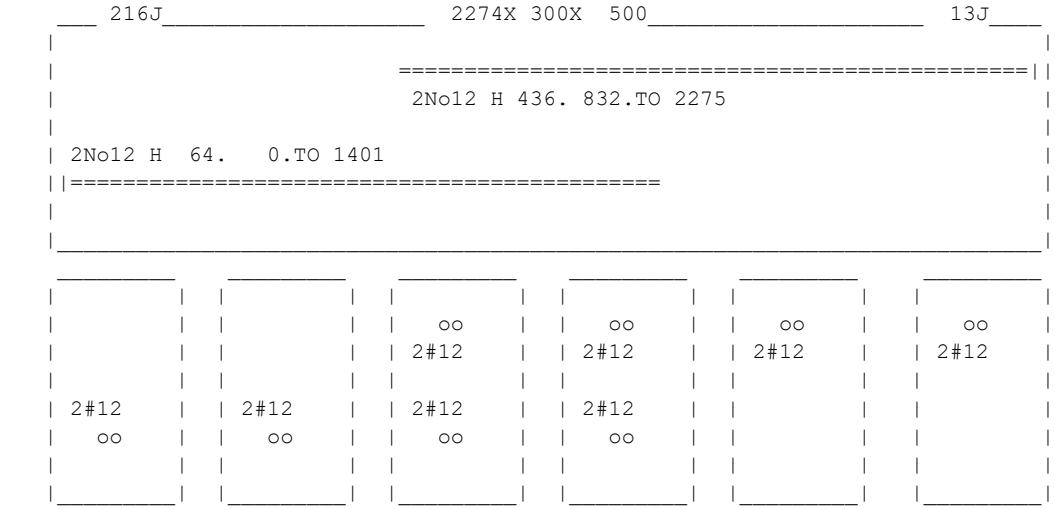
BEAM NO. 349 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2275. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	1401.	YES	NO
2	436.	2 - 12MM	832.	2275.	NO	YES

B E A M N O. 349 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.20 KNS Vc= 122.41 KNS Vs= 0.00 KNS
Tu= 0.68 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.16 KNS Vc= 122.93 KNS Vs= 0.00 KNS
Tu= 0.68 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

430. END CONCRETE DESIGN
431. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 1,2020 TIME= 12:50:57 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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